EWB AWS VPC PEERING PROJECT

Implementation, Validation & Evidence Report

Prepared for: Mocharla Varsha
Region: Asia Pacific (Sydney) — ap-southeast-2
Report date: 31 October 2026

This report documents the AWS VPC peering lab using the existing resources, configuration screenshots, terminal outputs, and the EWB student-project workbook. Screenshots are included in the evidence appendix.

1. Executive summary

The project connects two private Amazon VPC networks through an AWS VPC Peering connection. A Bastion host in VPC A provides controlled SSH administration to the private application instances. App-A and App-B communicate over private IPv4 addresses through the peering connection.

The peering connection is Active. Both VPC route tables contain the required peer routes. A Blackhole route was found in the VPC B private route table, replaced with a route targeting the active peering connection, and verified Active. Bidirectional ICMP tests then succeeded with zero packet loss.

A controlled security-group failure was also demonstrated: removing the App-A ICMP source rule from App-B caused 100% packet loss from App-A; restoring the rule restored connectivity. This documents both troubleshooting and recovery.

2. Architecture and addressing

Component	VPC / Subnet	CIDR / Private IP	Role
VPC A — ewb-vpc-a	vpc-024f7fcb61f158e51	10.10.0.0/16	Administration and App-A
Bastion — ewb-bastion	VPC A public subnet	10.10.1.10; public 54.252.166.57	SSH entry point
App-A — ewb-app-a	ewb-a-private	10.10.2.10	Private application host
VPC B — ewb-vpc-b	vpc-06ad3a55183d56581	10.20.0.0/16	Private application network
App-B — ewb-app-b	ewb-b-private	10.20.1.10	Private application host

The App-A and App-B EC2 instance summary screenshots show no Public IPv4 address. Both instances are t3.micro and use Amazon Linux 2023. The Bastion is the only instance with a public IPv4 address.

3. VPC peering and route tables

Peering connection: ewb-vpc-a-to-vpc-b — pcx-0ae44744490fc31e2 — Active. Requester VPC CIDR 10.10.0.0/16; accepter VPC CIDR 10.20.0.0/16. Both are in Sydney (ap-southeast-2).

Route table	Destination	Target	Observed status
ewb-rt-a-public	10.10.0.0/16	local	Active
ewb-rt-a-public	0.0.0.0/0	Internet Gateway	Active
ewb-rt-a-public	10.20.0.0/16	VPC peering	Active
ewb-rt-a-private	10.10.0.0/16	local	Active

Route table	Destination	Target	Observed status
ewb-rt-a-private	10.20.0.0/16	VPC peering	Active
ewb-rt-b-private	10.20.0.0/16	local	Active
ewb-rt-b-private	10.10.0.0/16	VPC peering	Active after repair

Subnet associations visible in the screenshots: ewb-rt-a-public → ewb-a-public (subnet-0a13f73a405ba0abc); ewb-rt-a-private → ewb-a-private (subnet-096ab28c22c23240e); ewb-rt-b-private → ewb-b-private (subnet-0a9cb81bce3a16d9d).

Troubleshooting note: the VPC B private route to 10.10.0.0/16 initially showed Blackhole even though the peering connection detail showed Active. The route was removed and recreated with the same peering connection as target. AWS then showed Active.

4. Security-group configuration

Security group	Inbound rule	Source
ewb-sg-bastion	SSH TCP 22	110.235.236.128/32 (public IP at test time)
ewb-sg-bastion	SSH TCP 22	EC2 Instance Connect managed prefix list
ewb-sg-app-a	SSH TCP 22	10.10.1.10/32 (Bastion)
ewb-sg-app-a	All ICMP - IPv4	10.20.1.10/32 (App-B)
ewb-sg-app-b	SSH TCP 22	10.10.1.10/32 (Bastion)
ewb-sg-app-b	All ICMP - IPv4	10.10.1.10/32 (Bastion)
ewb-sg-app-b	All ICMP - IPv4	10.10.2.10/32 (App-A)

The Bastion SSH source is restricted to the observed client public IPv4 /32 and the regional EC2 Instance Connect prefix list. No SSH rule is open to 0.0.0.0/0. App-A and App-B have no public IPv4 addresses.

5. Access method and validation results

The private key ewb-lab.pem remained on the Windows workstation. SSH agent forwarding was used from Windows PowerShell to Bastion (ssh -A), then from Bastion to App-A and App-B using their private IPs. The private key was not copied to either EC2 instance.

Test	Command / method	Observed result
Windows → Bastion	ssh -A ... ec2-user@54.252.166.57	Connected
Bastion → App-A	ssh ec2-user@10.10.2.10	Connected

Test	Command / method	Observed result
Bastion → App-B	ssh ec2-user@10.20.1.10	Connected
App-A → App-B	ping -c 4 10.20.1.10	4 received, 0% loss
App-B → App-A	ping -c 4 10.10.2.10	4 received, 0% loss after route repair
Controlled failure	Remove App-A ICMP rule from App-B SG; ping from App-A	4 transmitted, 0 received, 100% loss
Recovery	Restore App-A ICMP rule; ping from App-A	4 received, 0% loss

Measured reverse ping after route repair: 4 packets received, 0% loss; average RTT 0.258 ms. App-A to App-B recovery ping: 4 received, 0% loss; average RTT 0.282 ms.

6. Conclusion and remaining operational notes

The VPC peering lab is operational: both private instances communicate bidirectionally using private IP addresses, SSH administration is routed through the Bastion, and route/security-group troubleshooting was demonstrated. The final screenshots confirm all three security groups and both private instances' lack of public IPv4 addresses.

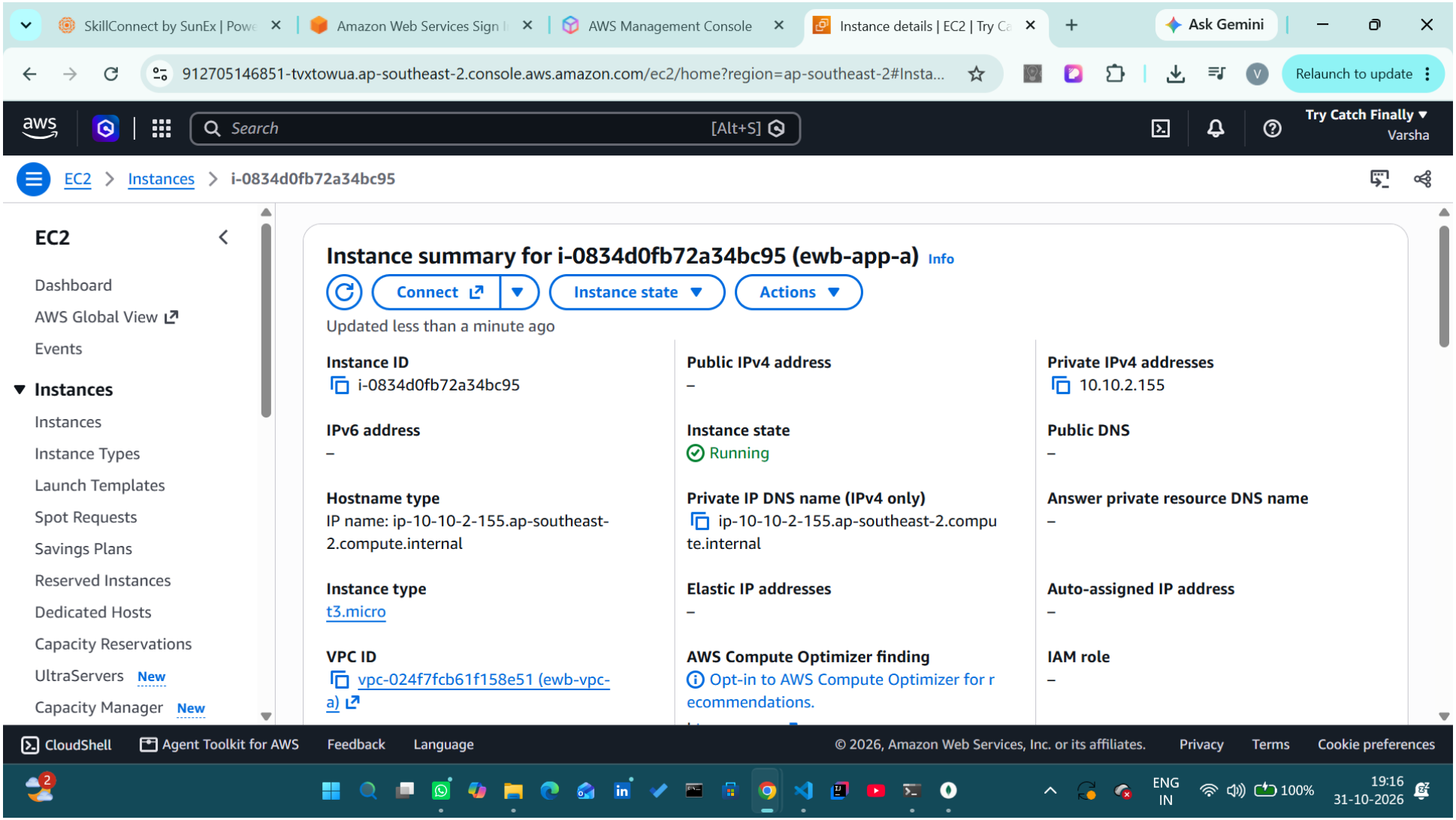
Cost and cleanup: the EC2 instances were intentionally left running at the time this report was prepared so evidence could be reviewed and the project retained. Stop or terminate resources only after the instructor accepts the evidence and the report is saved. If cleaning up, follow the workbook's prescribed order and ensure no required screenshots remain to be captured.

Security note: the Bastion SSH source shown is 110.235.236.128/32 as captured during the lab. If your public IP changes, update that rule to your current public IPv4 /32 before reconnecting.

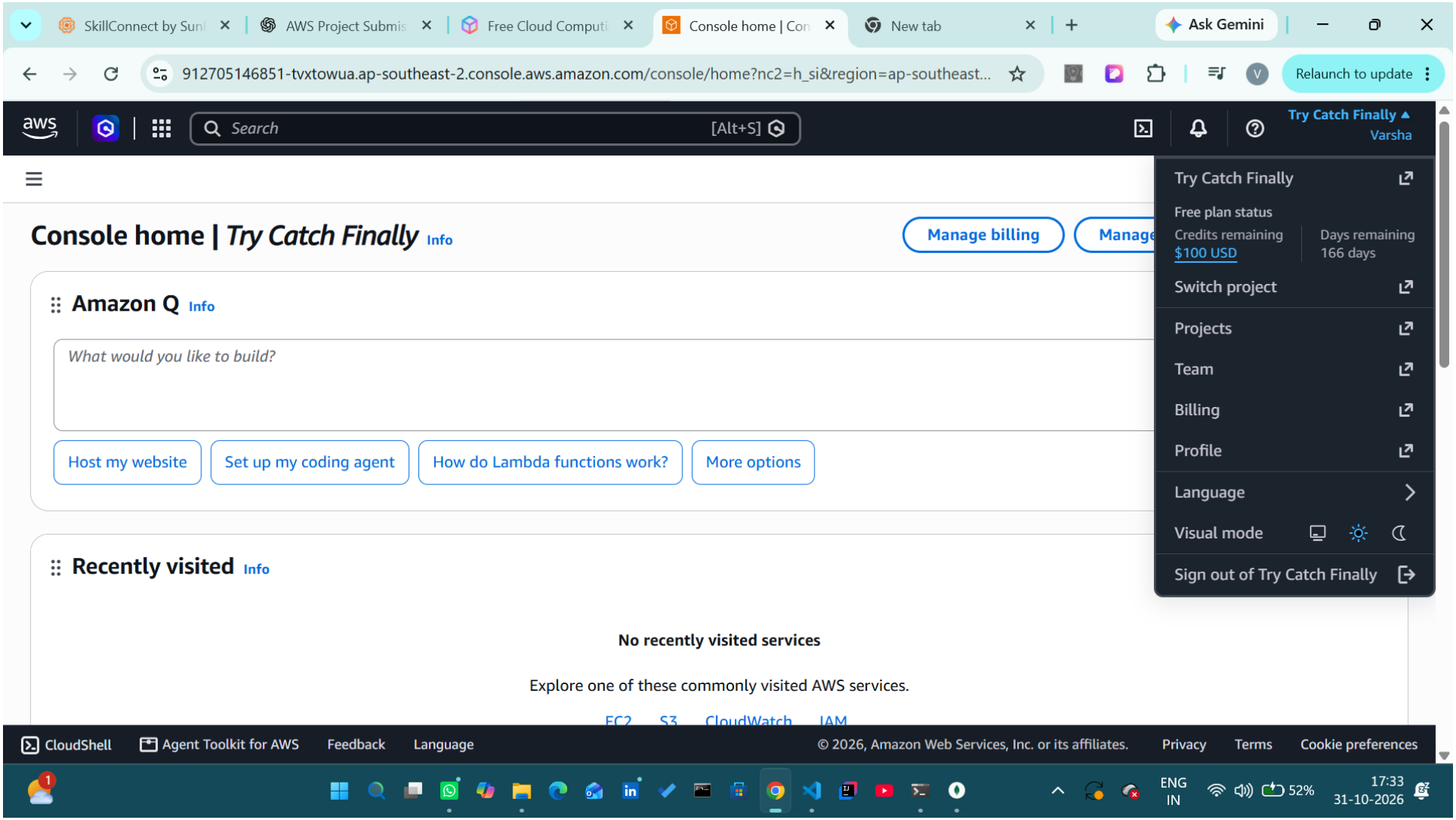
7. Evidence appendix — screenshots shared in this project conversation

The appendix contains 59 mounted screenshot files available from this project conversation. Each is included as received, with its filename and a short evidence label. Some screenshots show repeated views or intermediate states; the report narrative identifies the final verified configuration and distinguishes the temporary failure state from the recovered state.

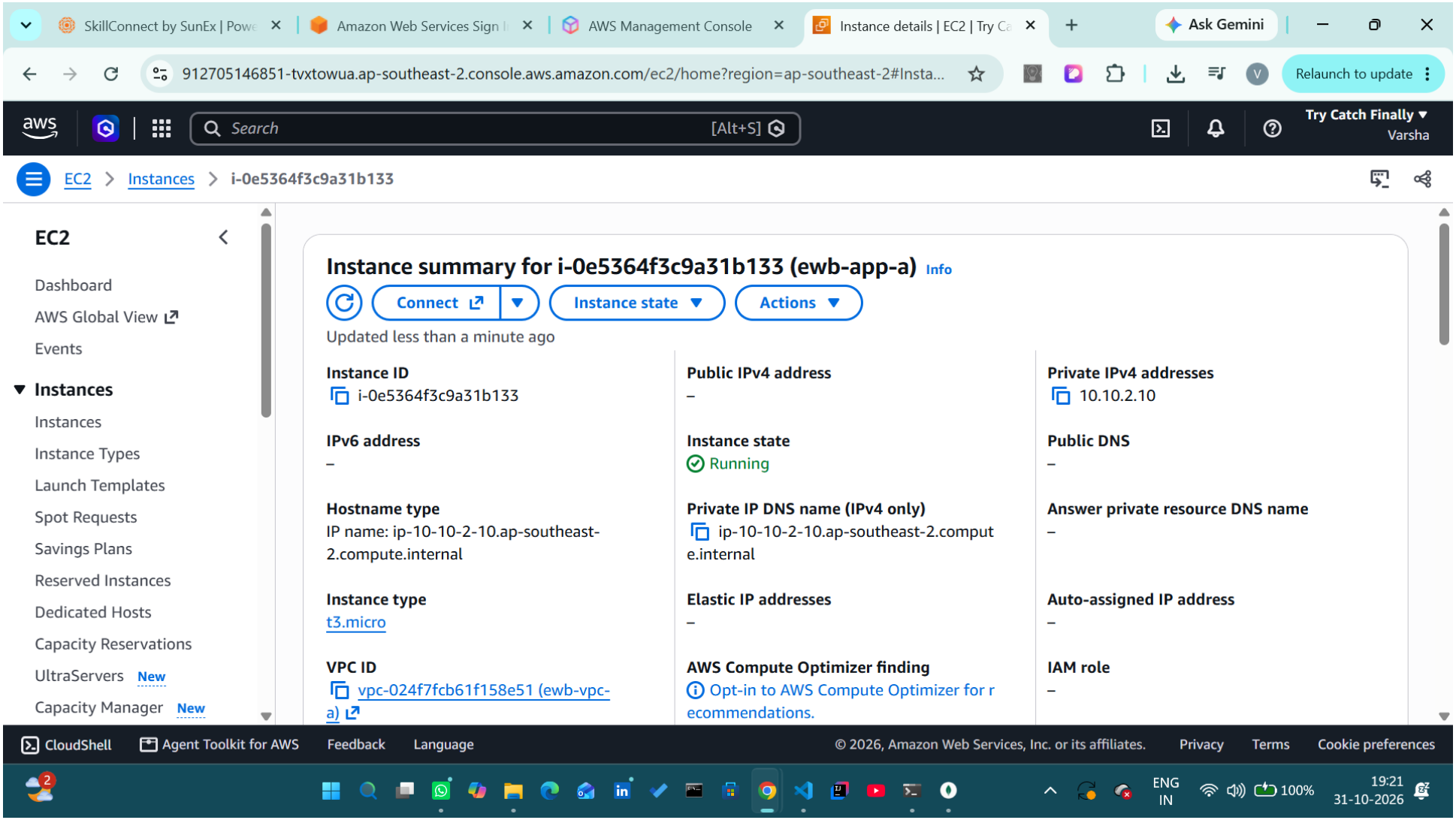
Evidence 01 — image(10).png



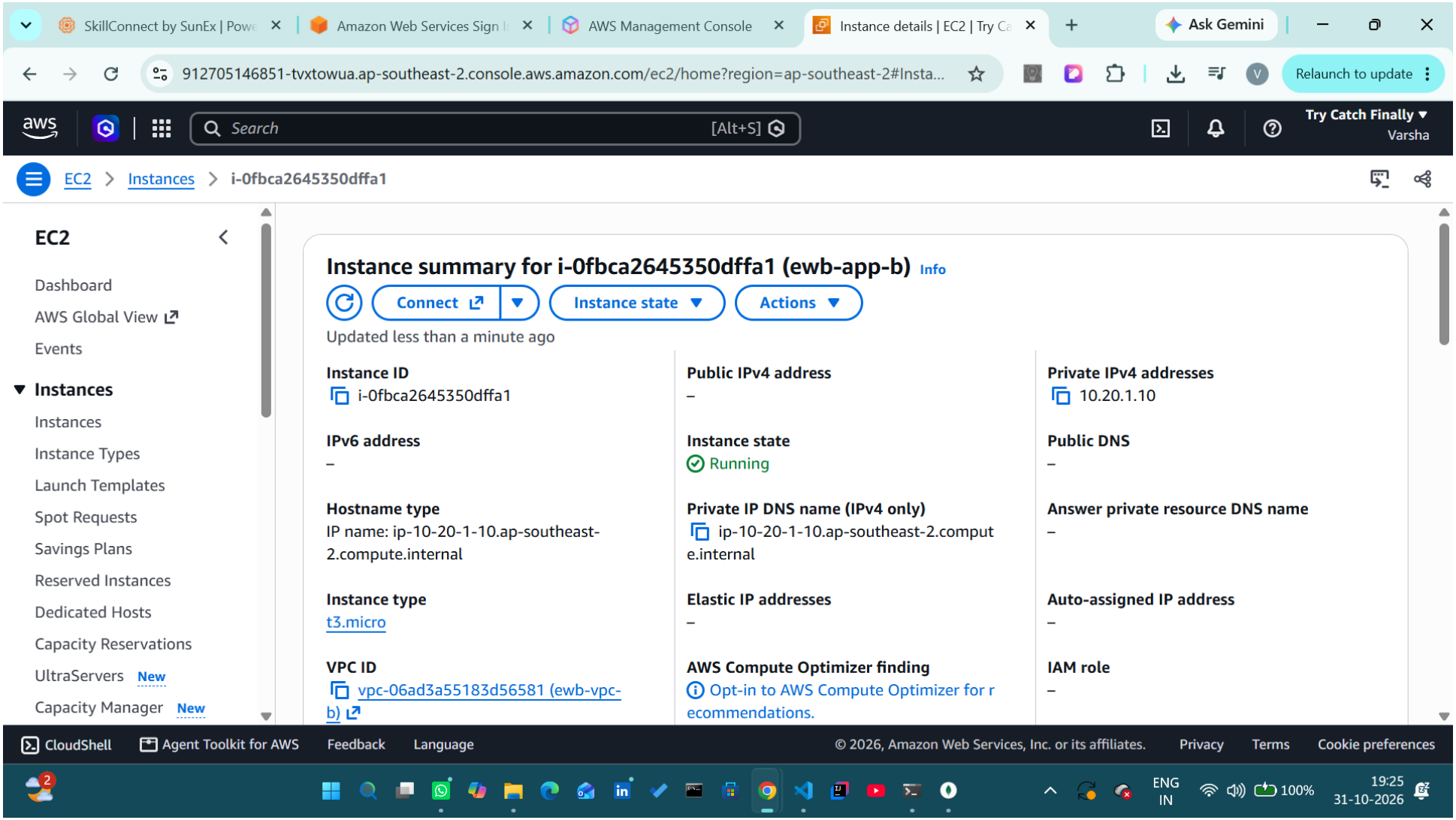
Evidence 02 — image(2).png



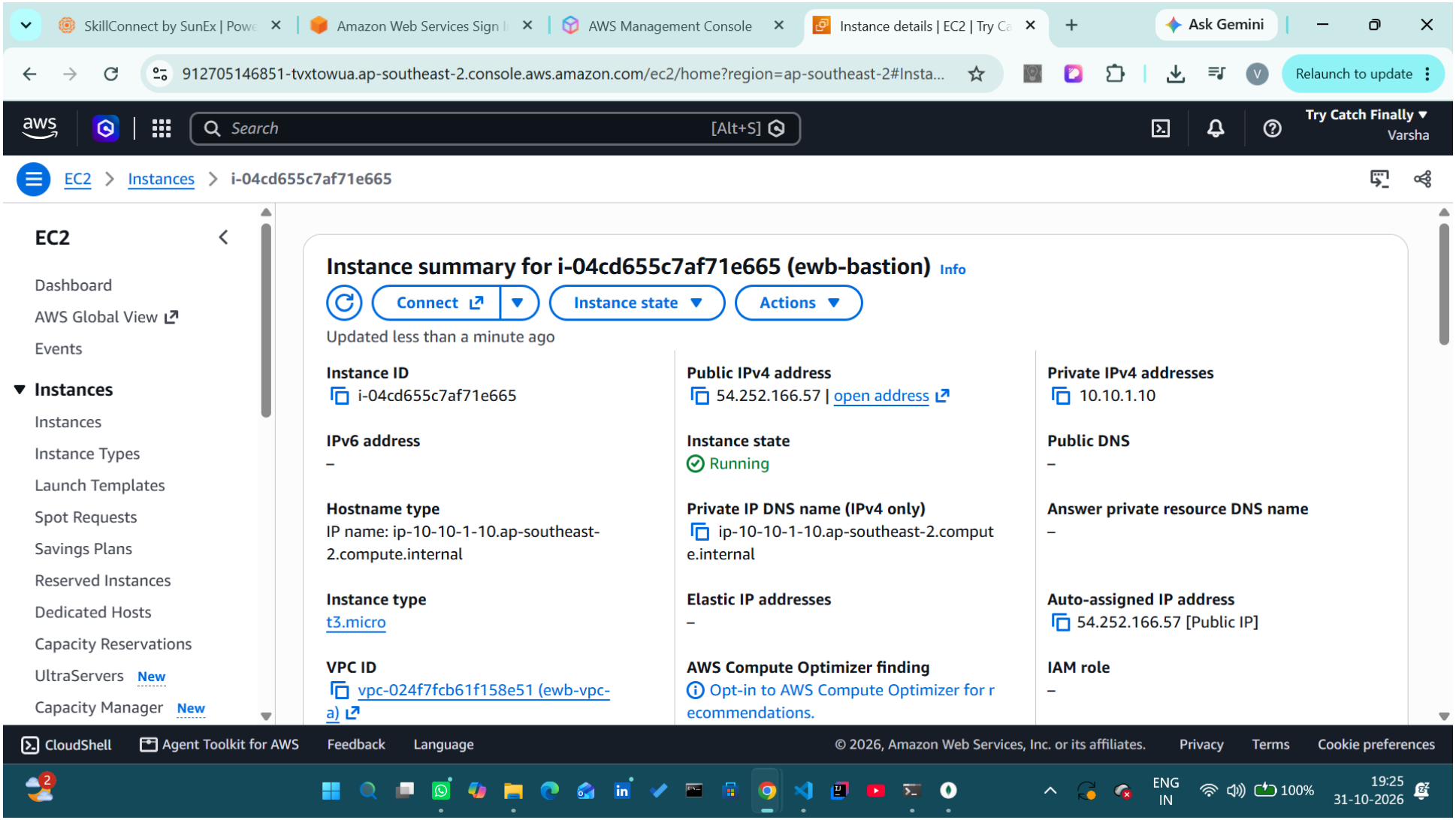
Evidence 03 — image(20261031-135131).png



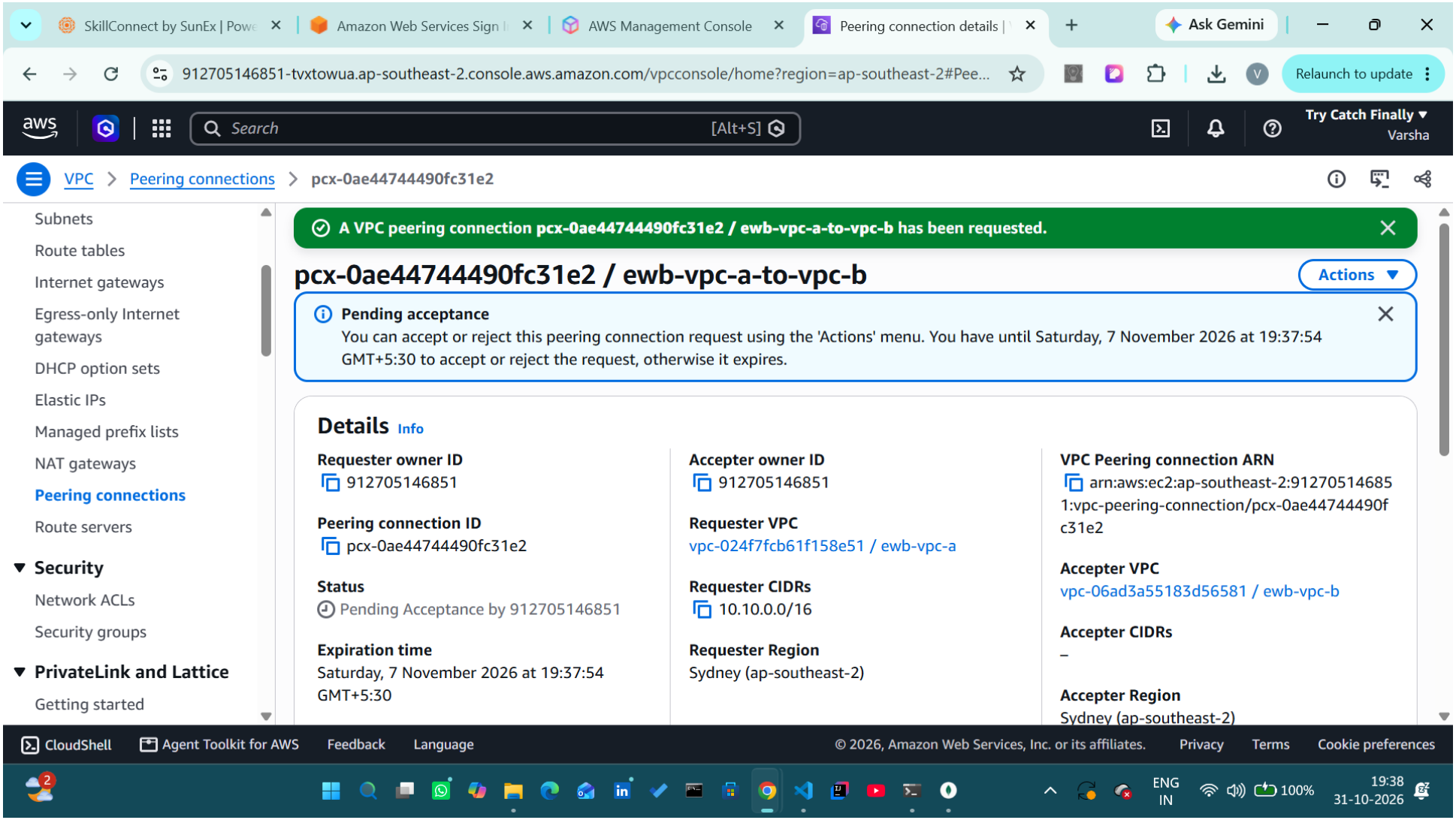
Evidence 04 — image(20261031-135513).png



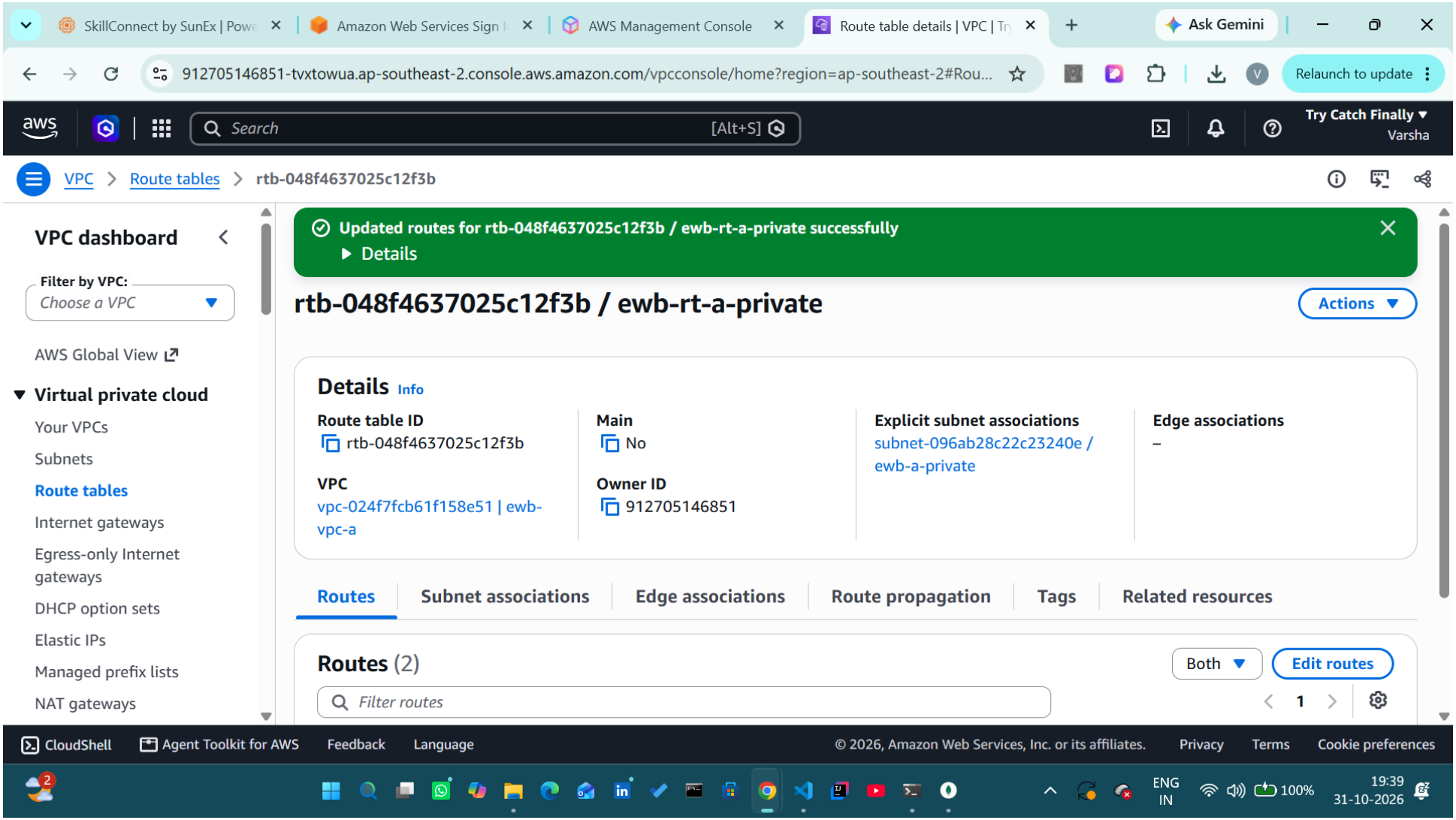
Evidence 05 — image(20261031-135531).png



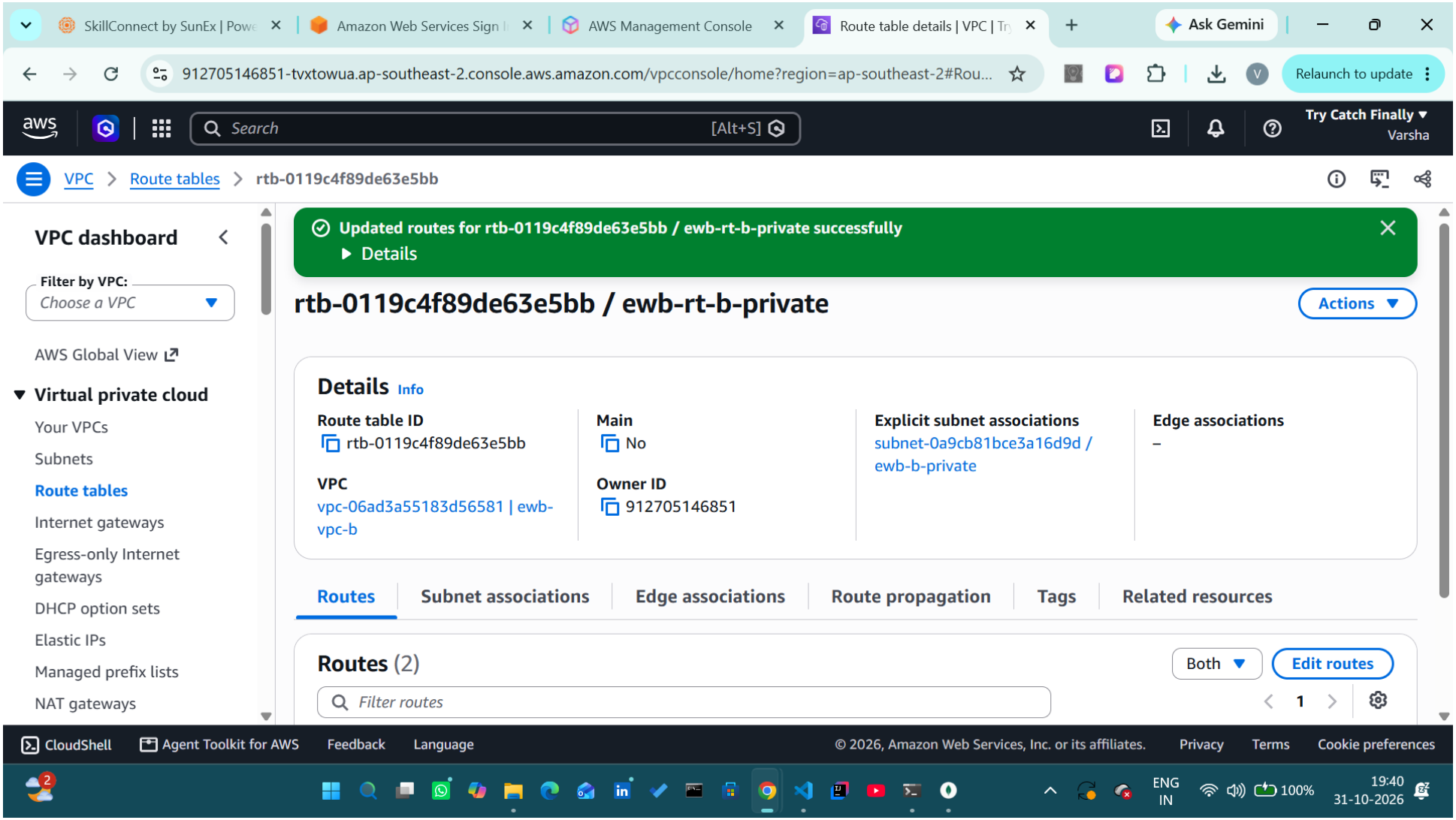
Evidence 06 — image(20261031-140813).png



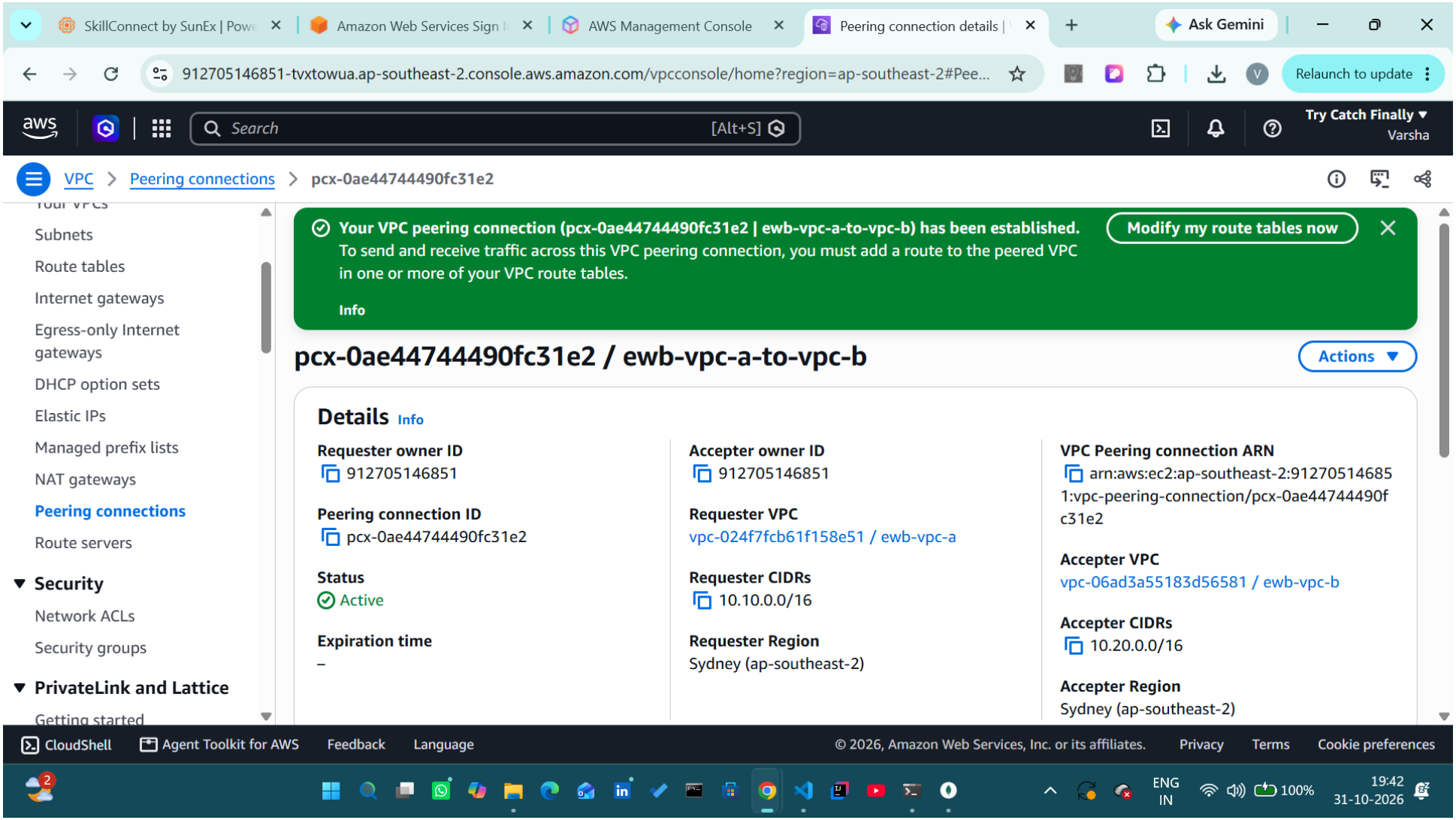
Evidence 07 — image(20261031-140927).png



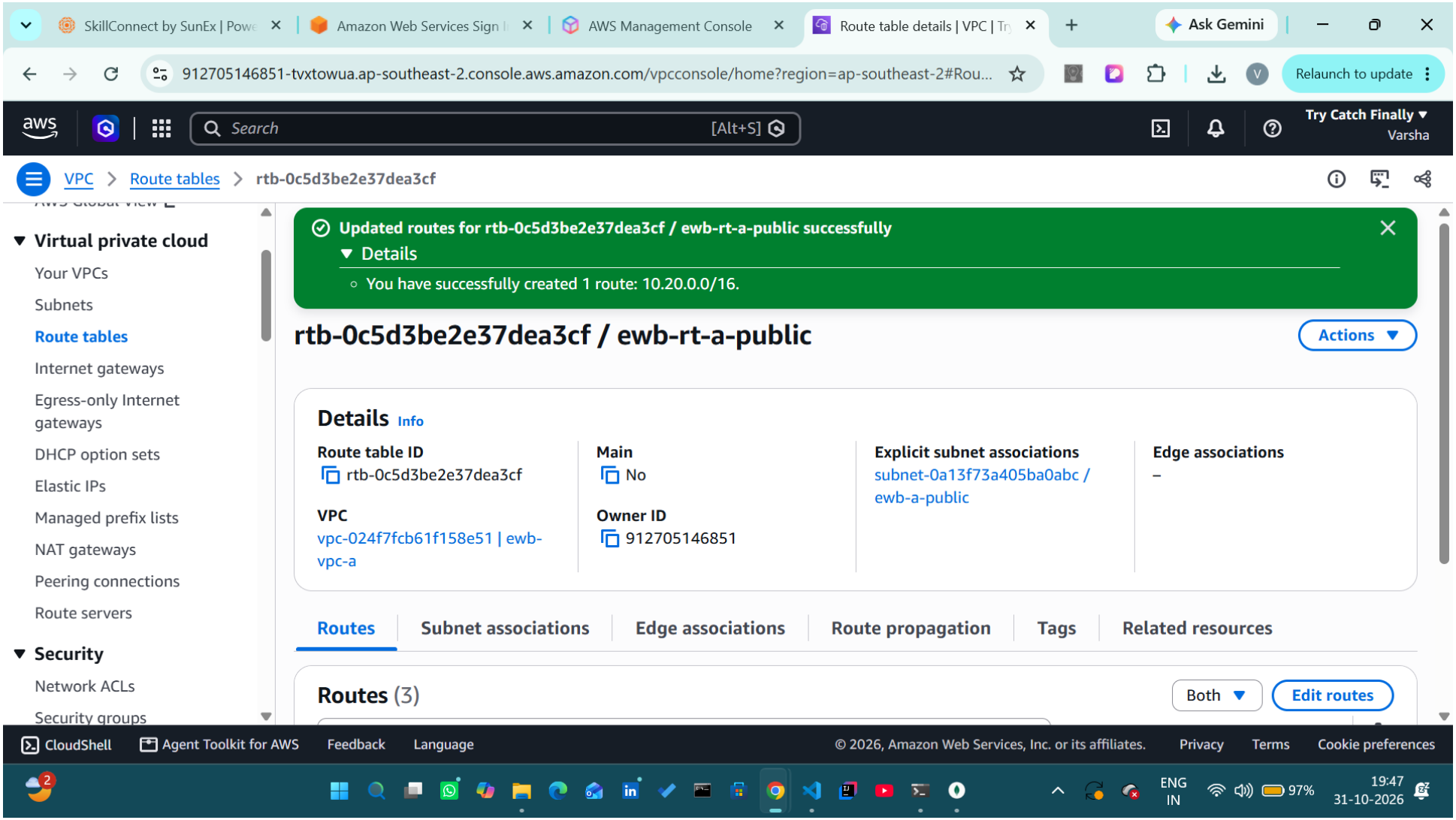
Evidence 08 — image(20261031-141103).png



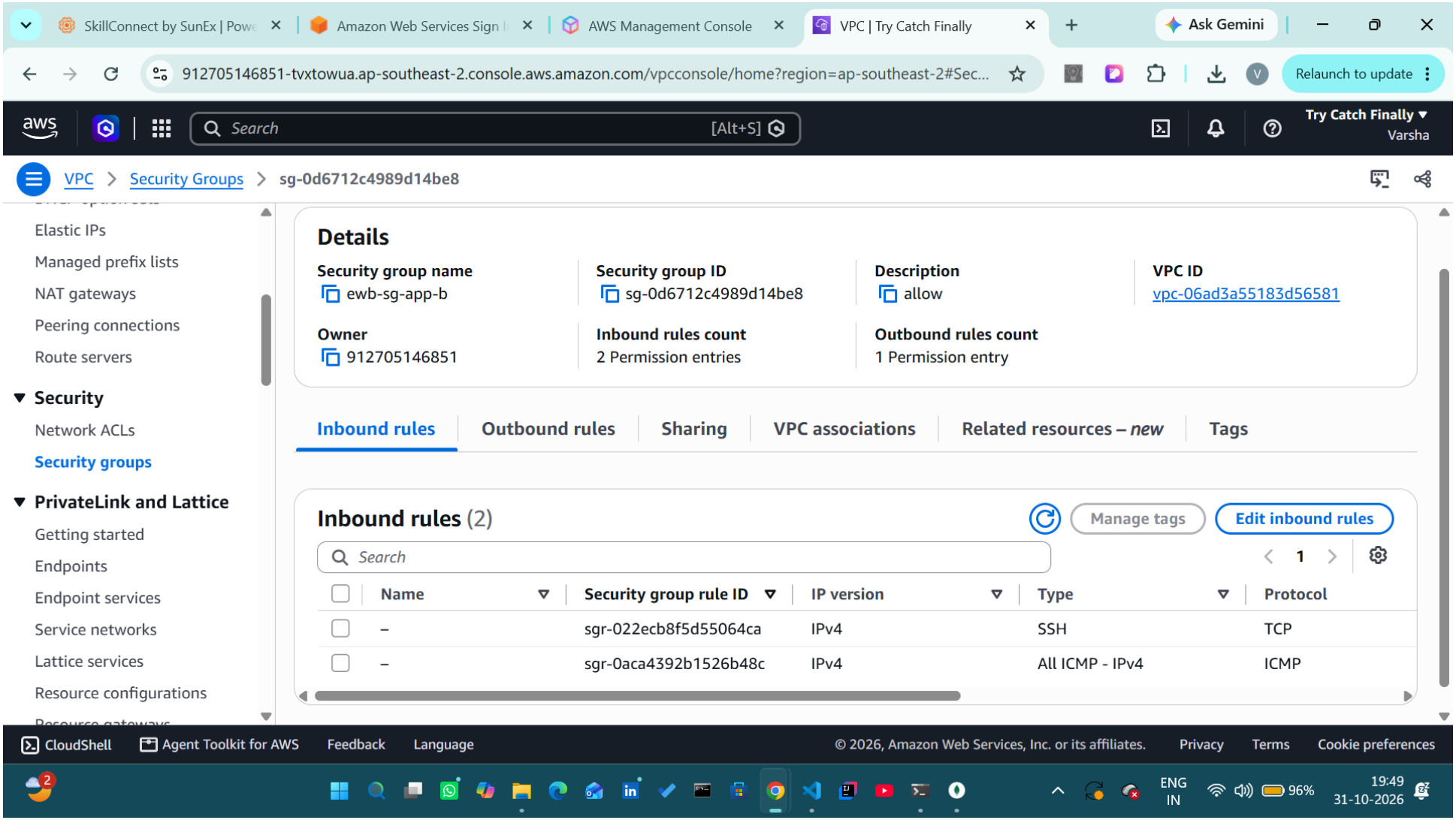
Evidence 09 — image(20261031-141215).png



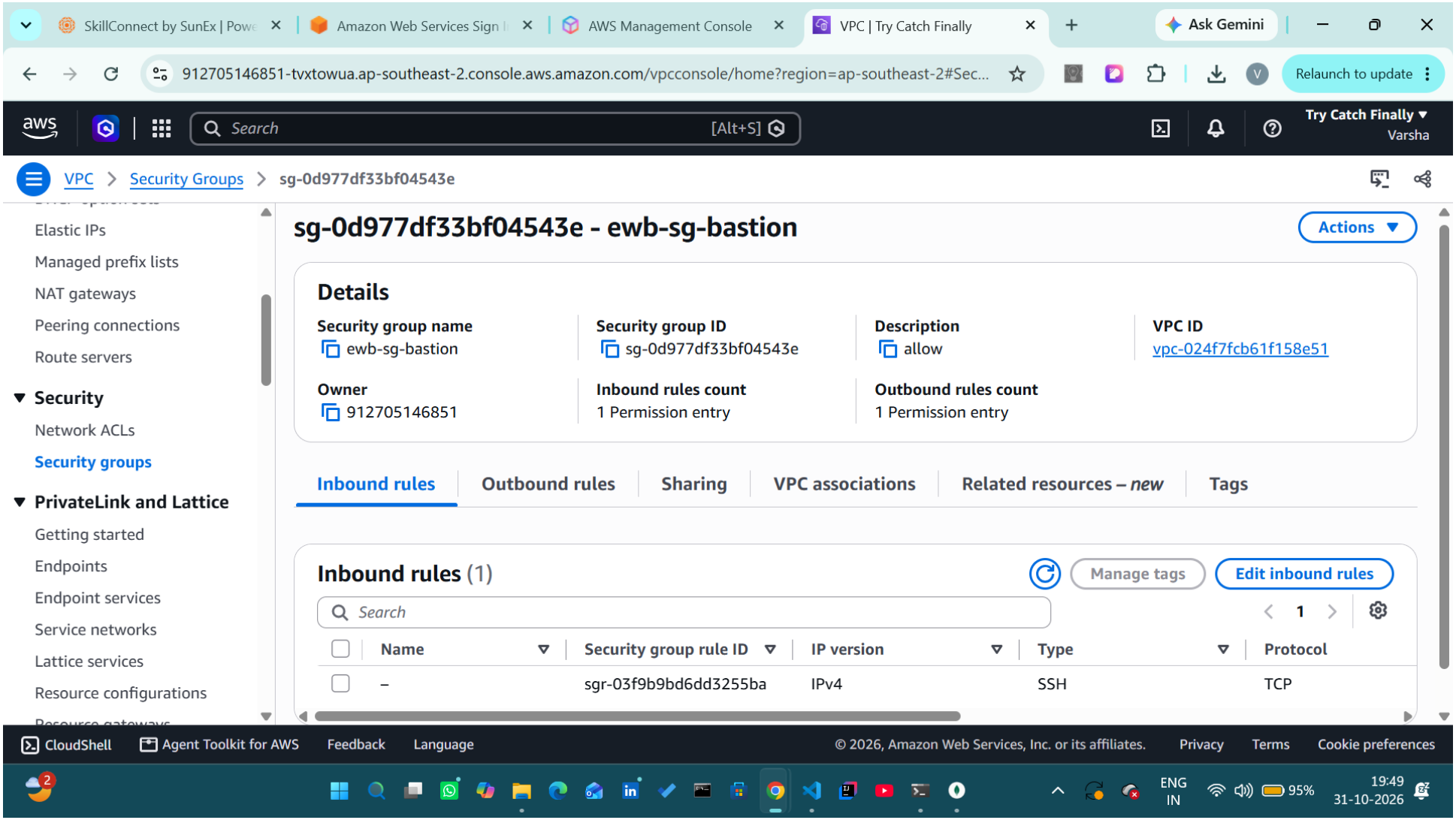
Evidence 10 — image(20261031-141729).png



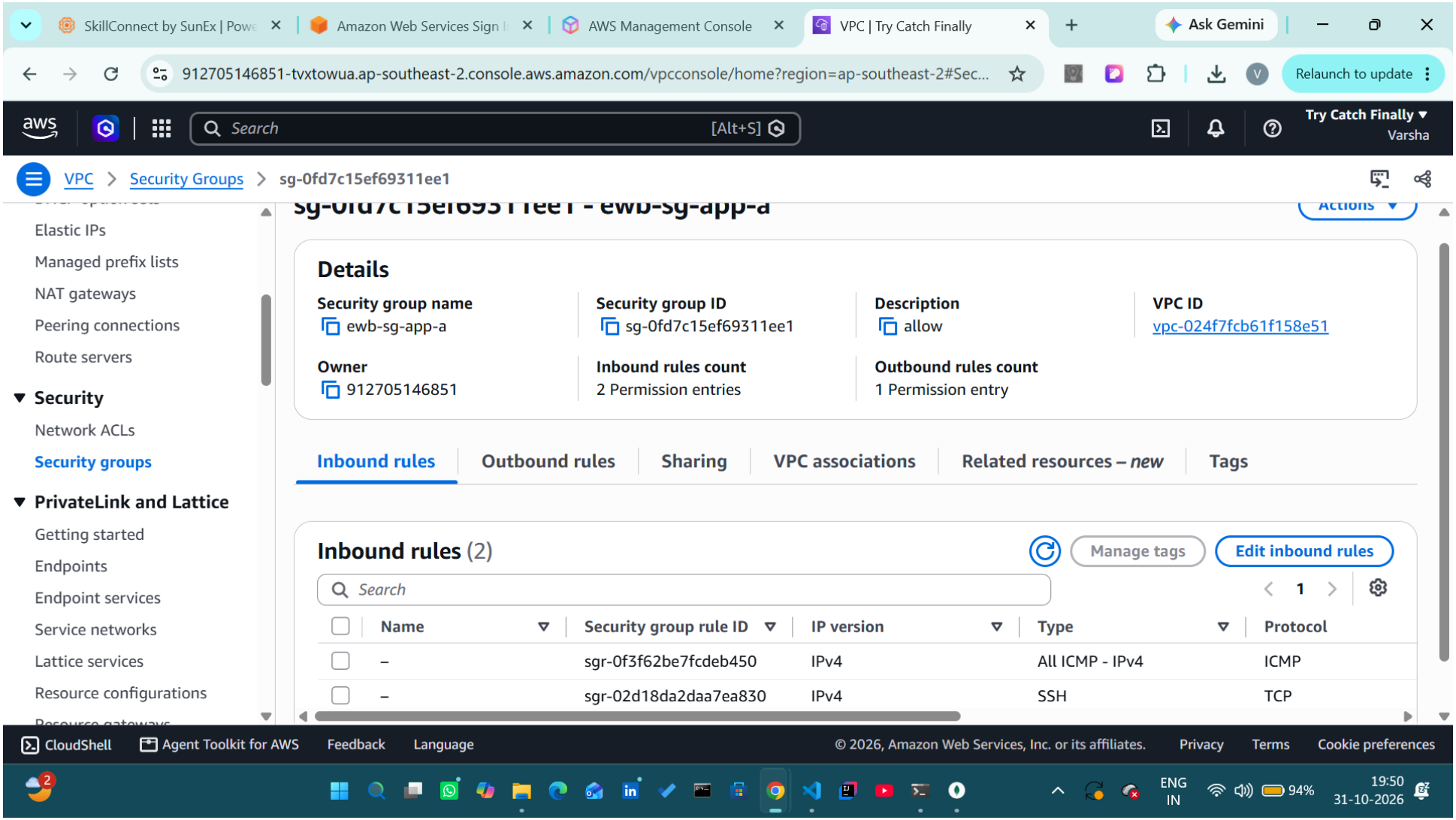
Evidence 11 — image(20261031-141930).png



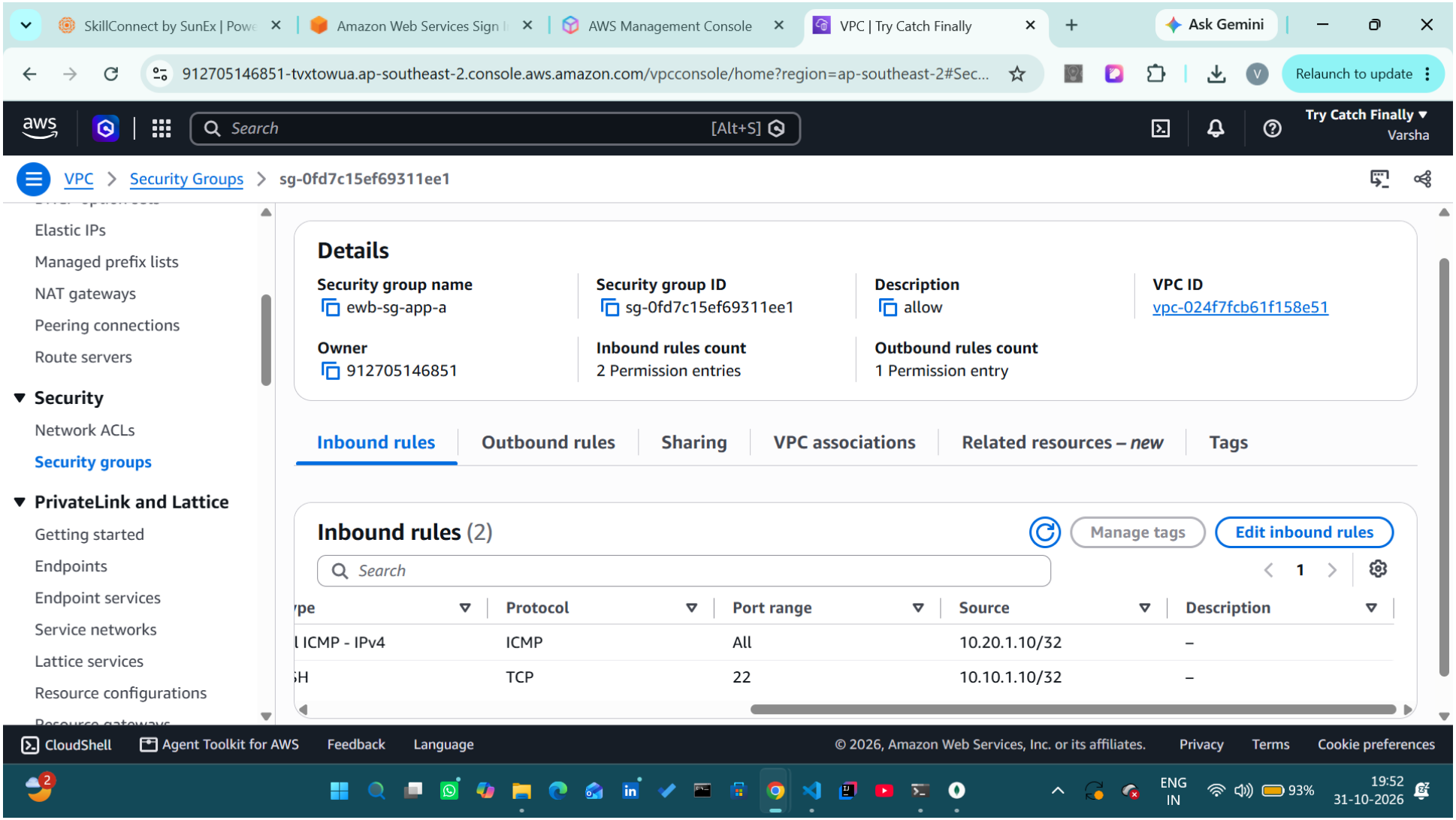
Evidence 12 — image(20261031-142008).png



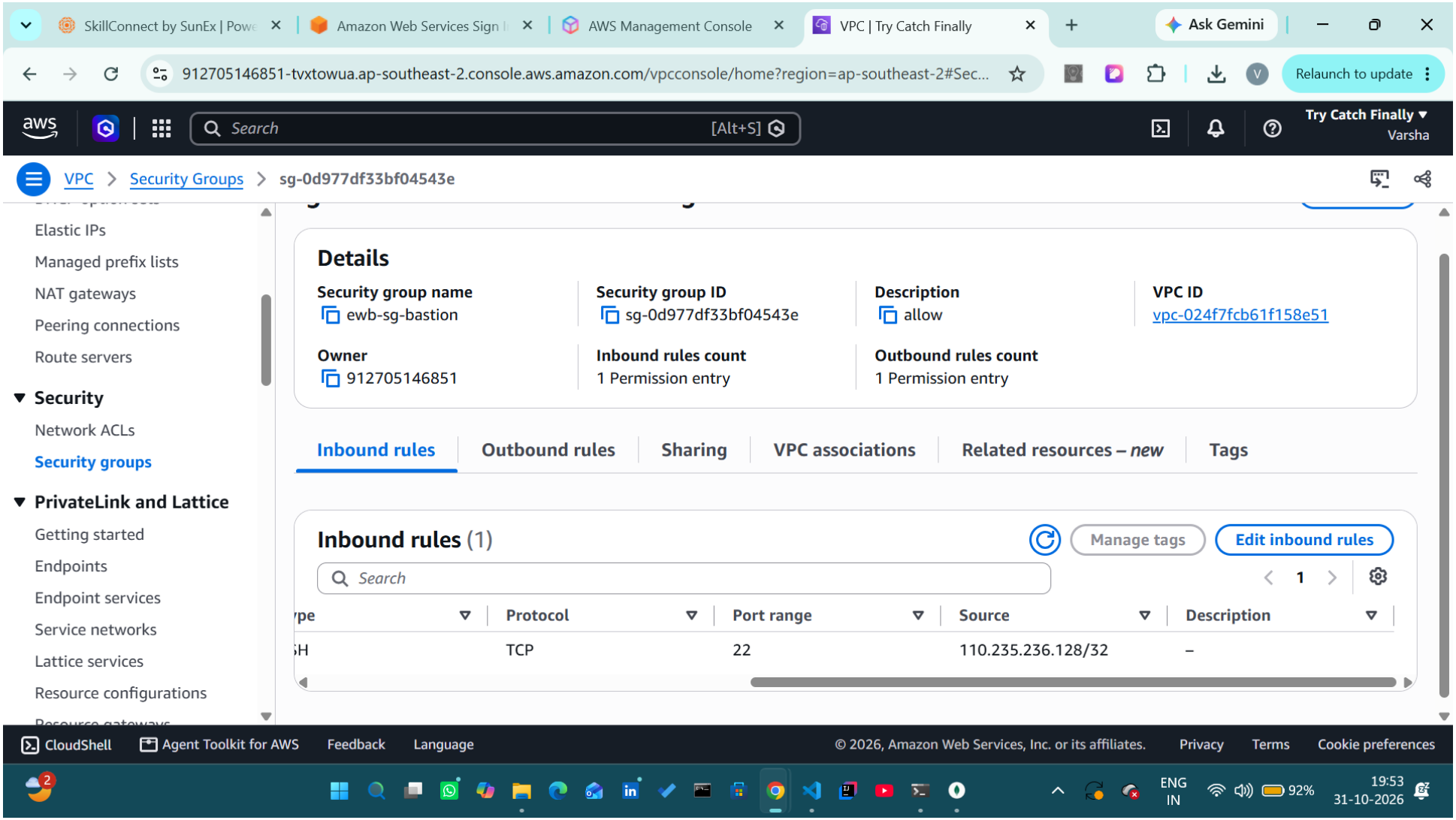
Evidence 13 — image(20261031-142104).png



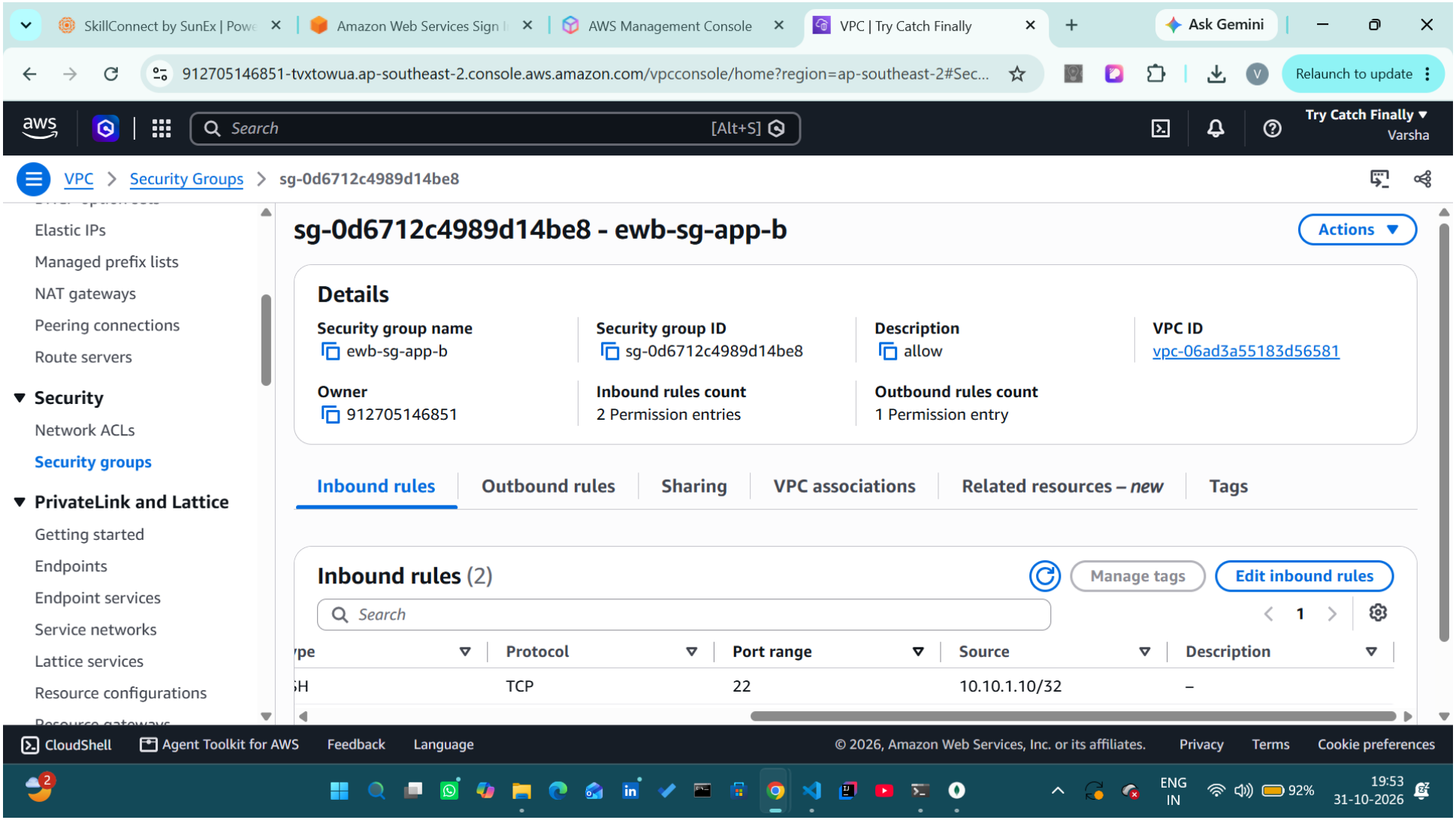
Evidence 14 — image(20261031-142232).png



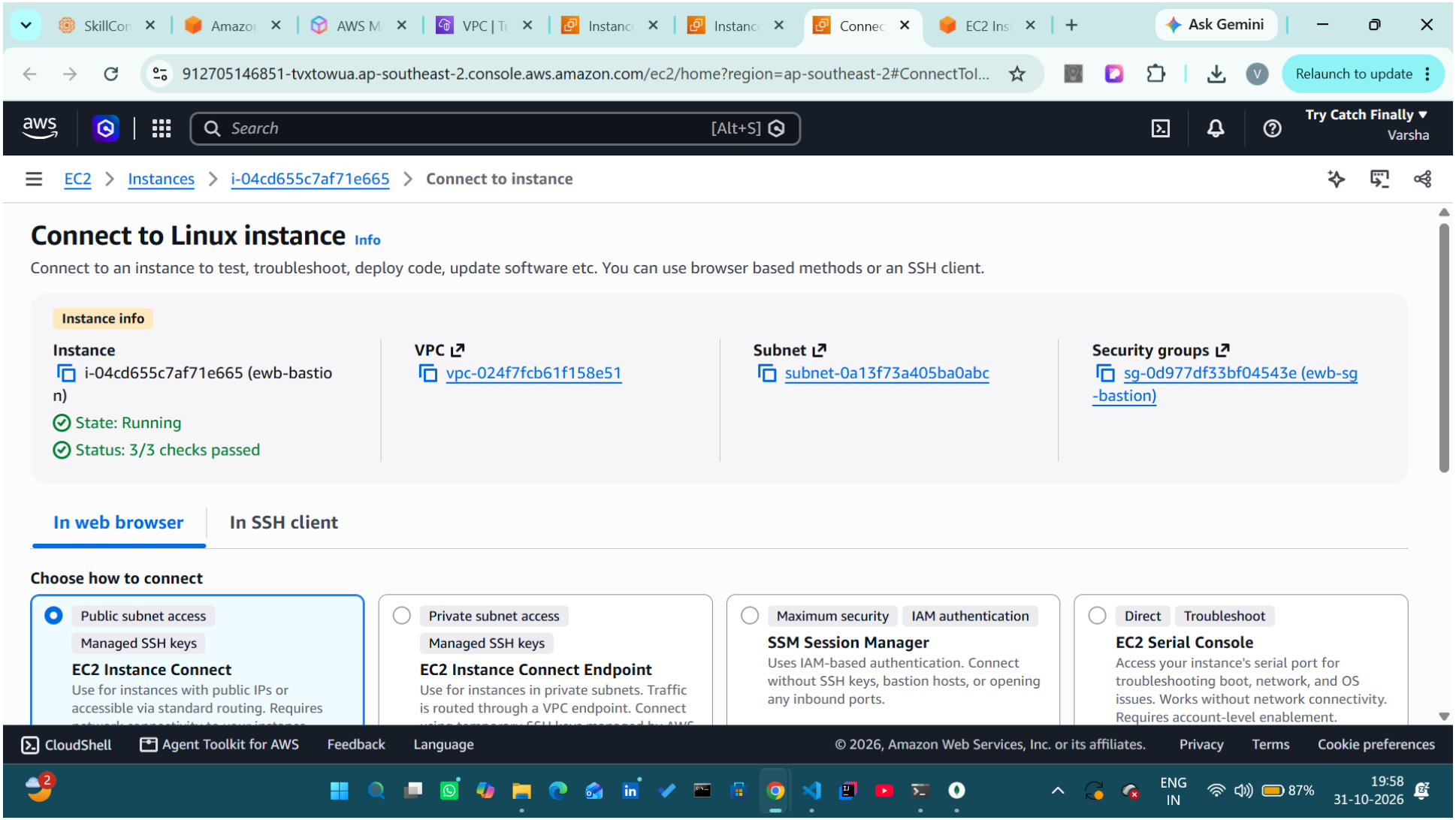
Evidence 15 — image(20261031-142317).png



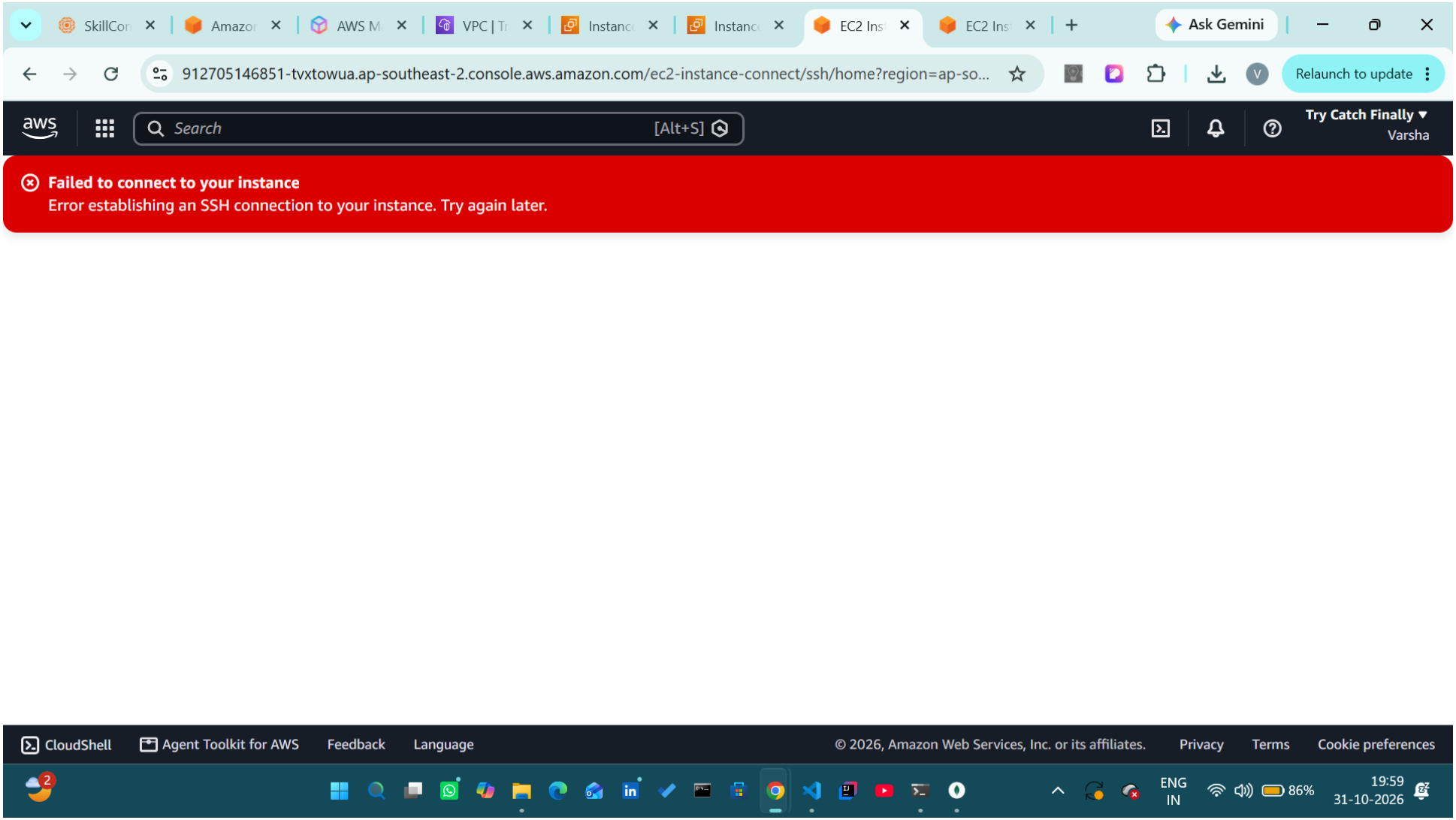
Evidence 16 — image(20261031-142336).png



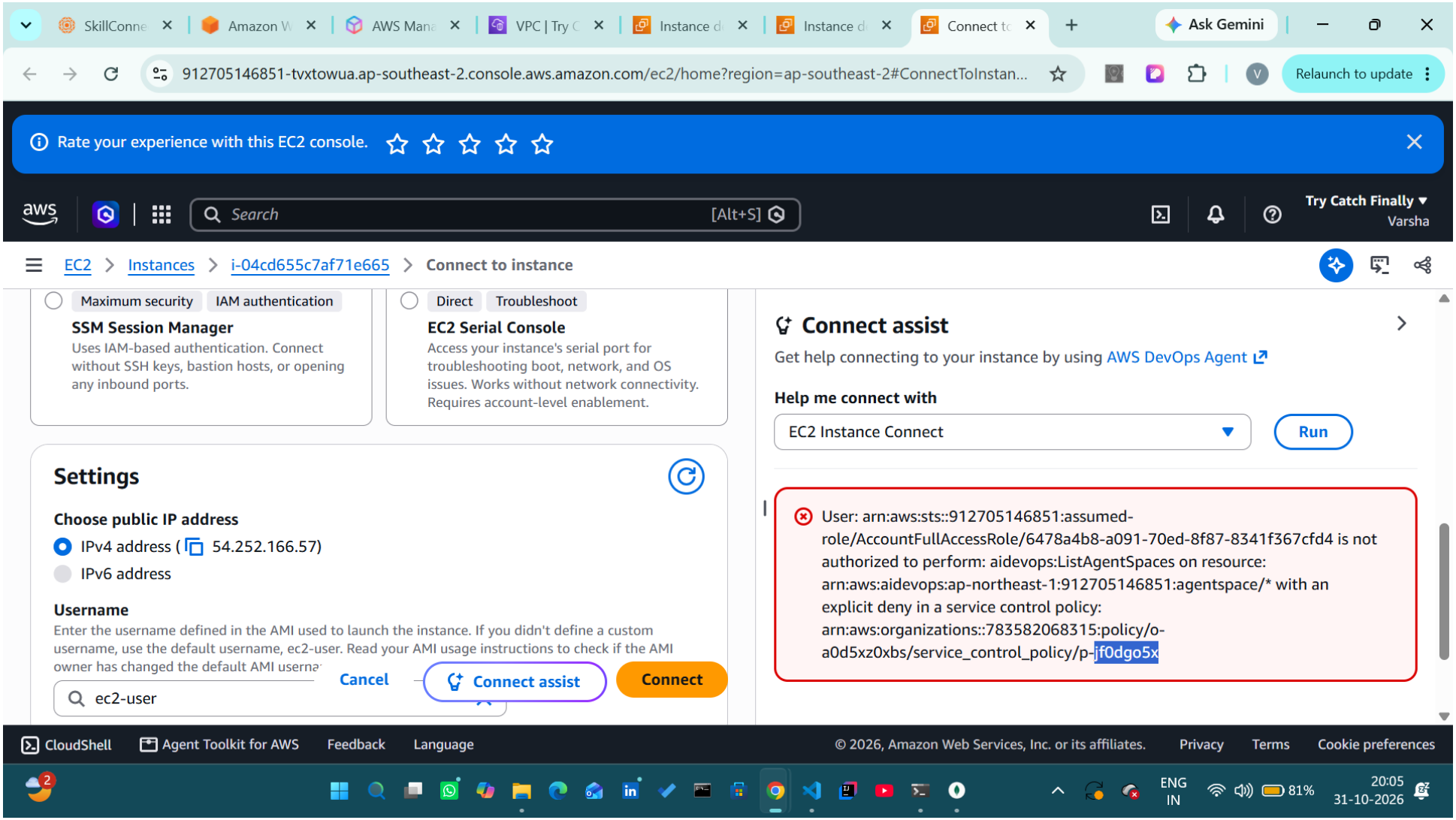
Evidence 17 — image(20261031-142830).png



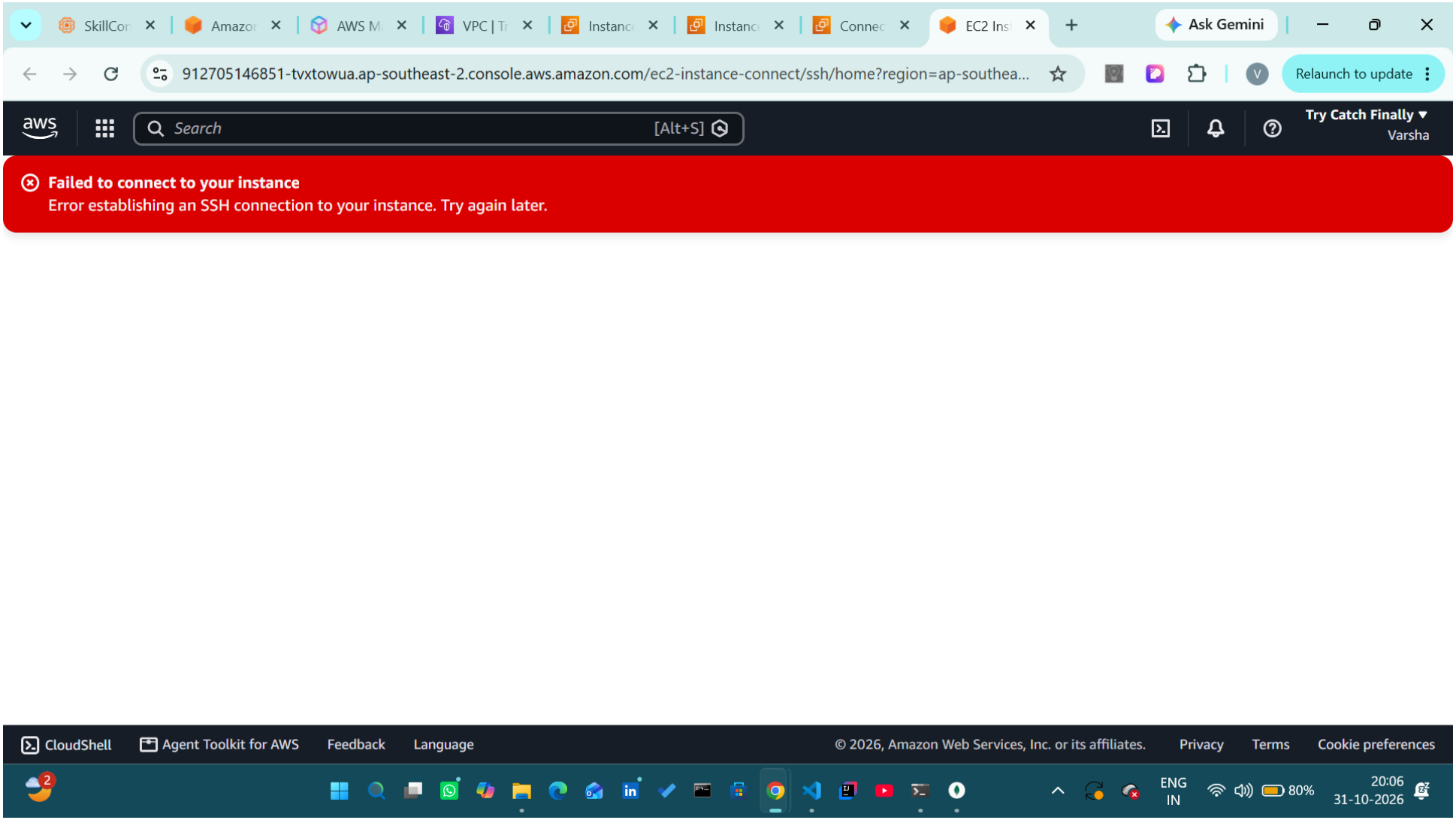
Evidence 18 — image(20261031-143007).png



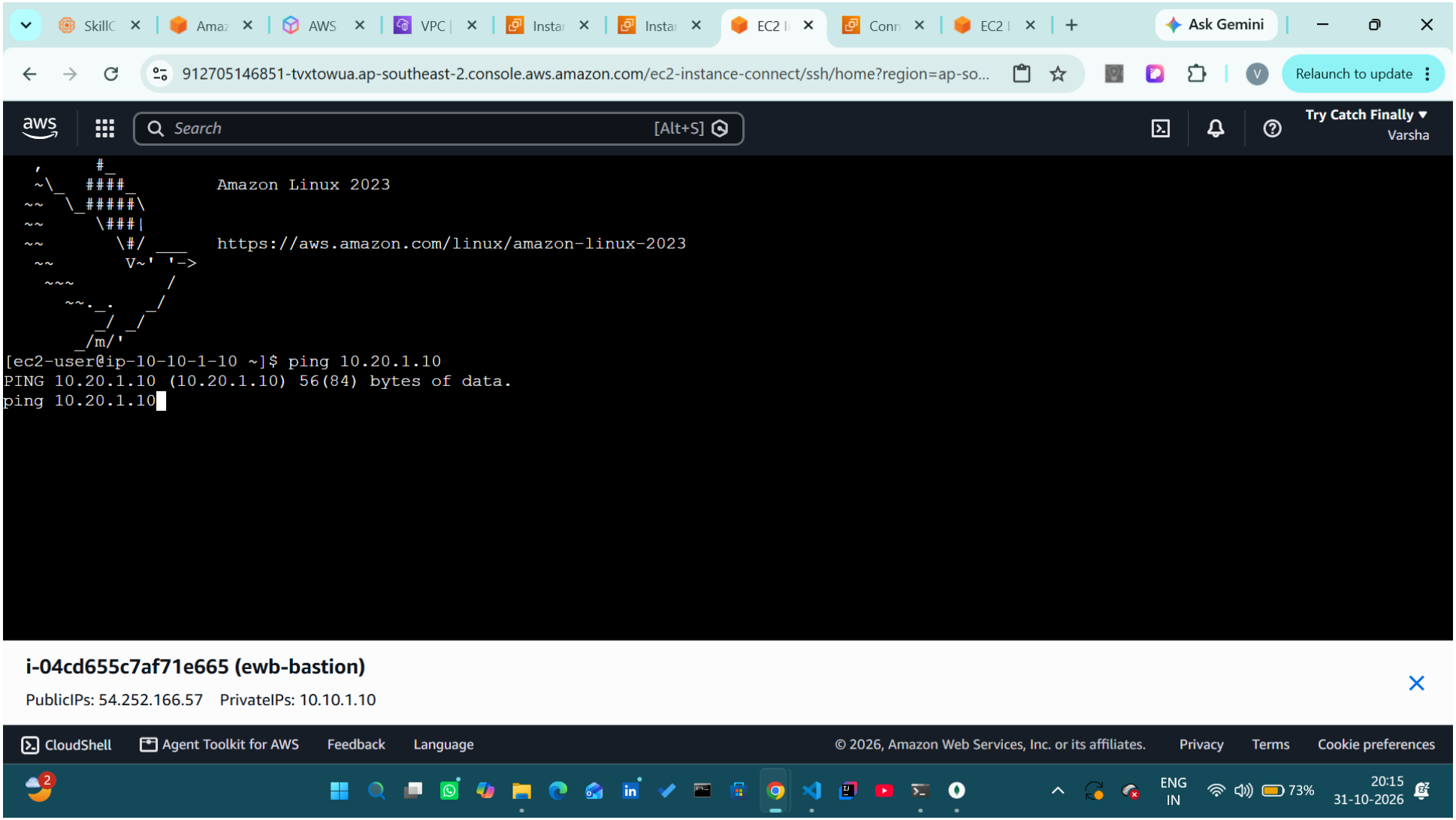
Evidence 19 — image(20261031-143529).png



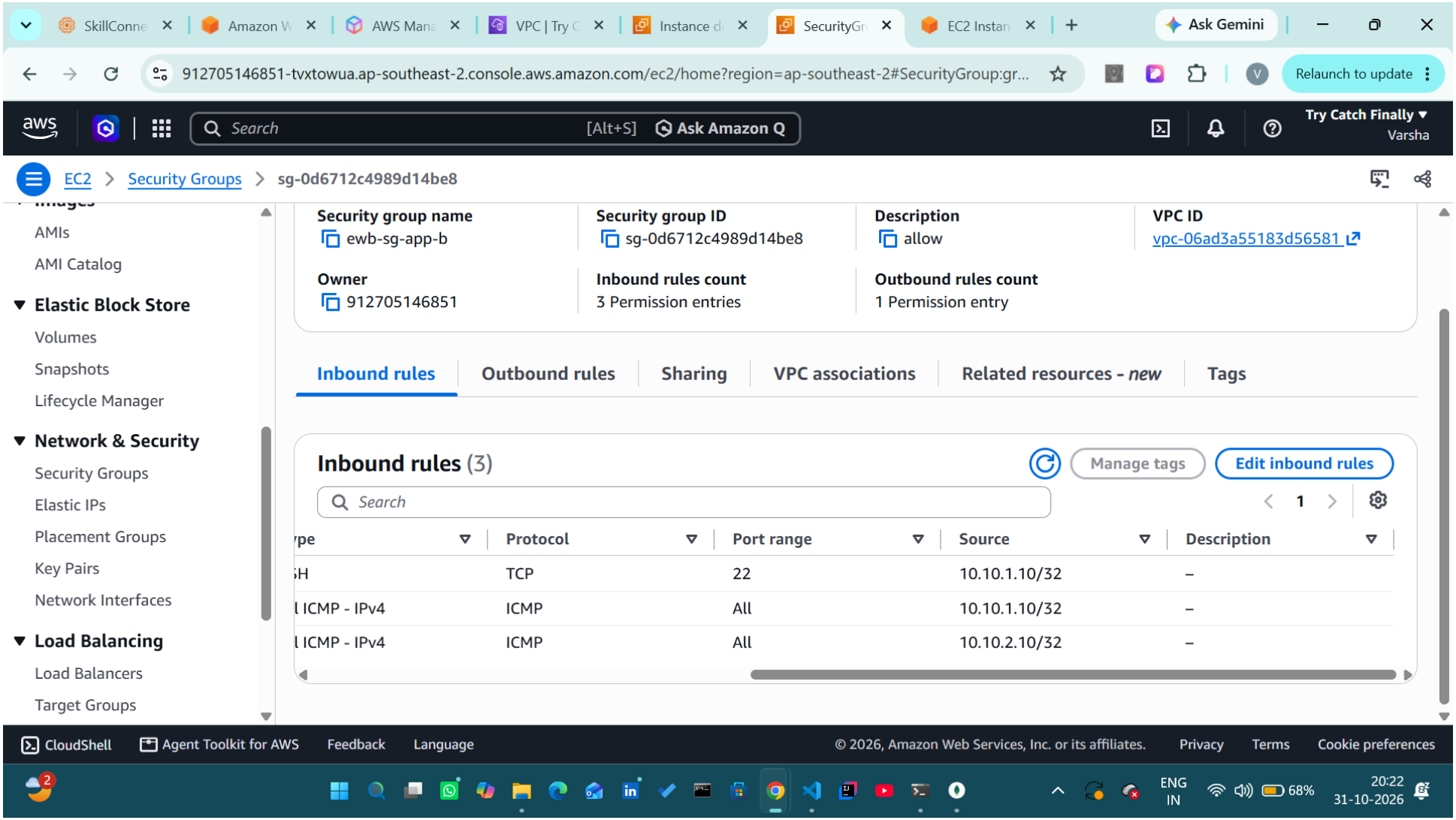
Evidence 20 — image(20261031-143647).png



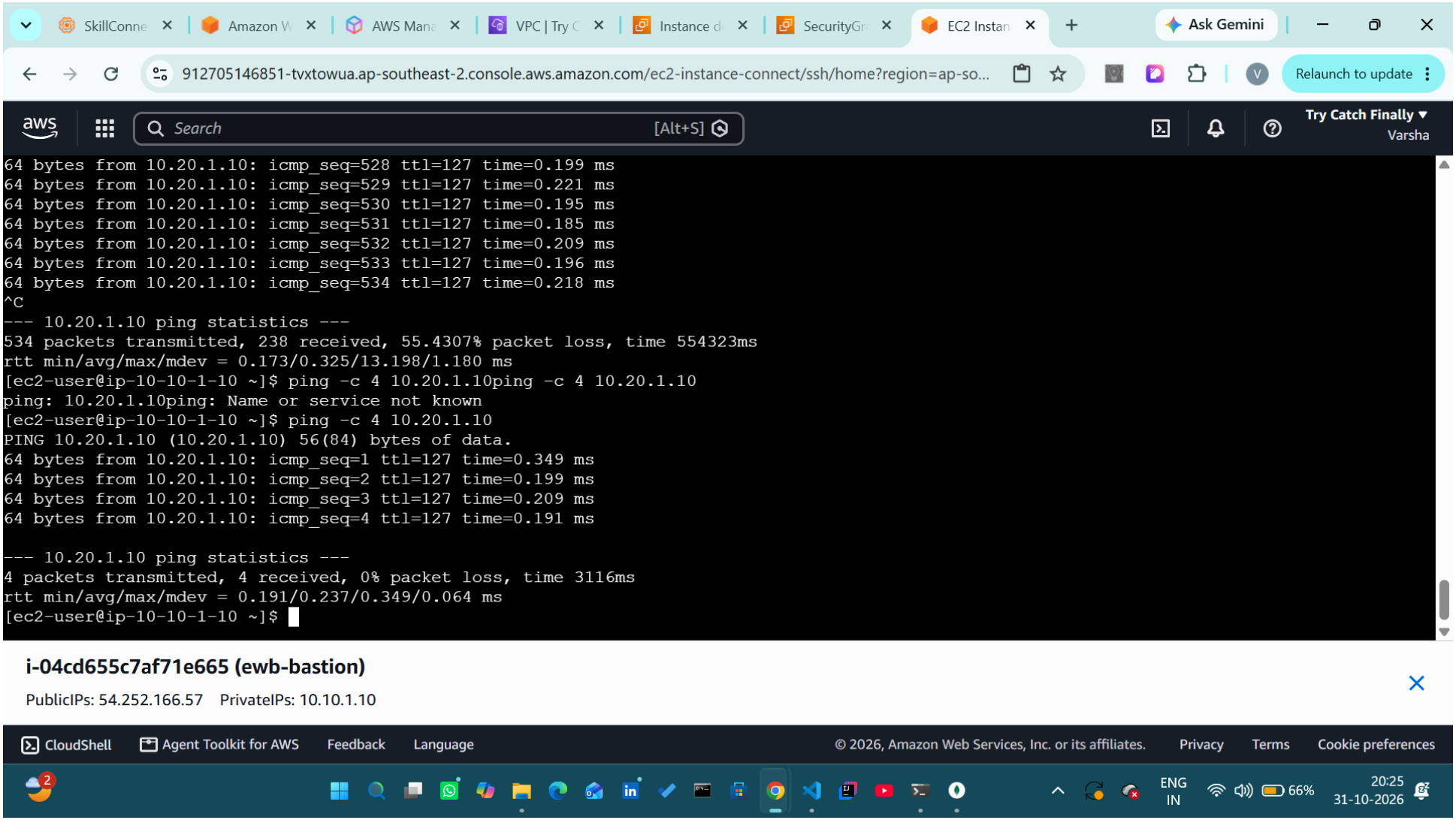
Evidence 21 — image(20261031-144510).png



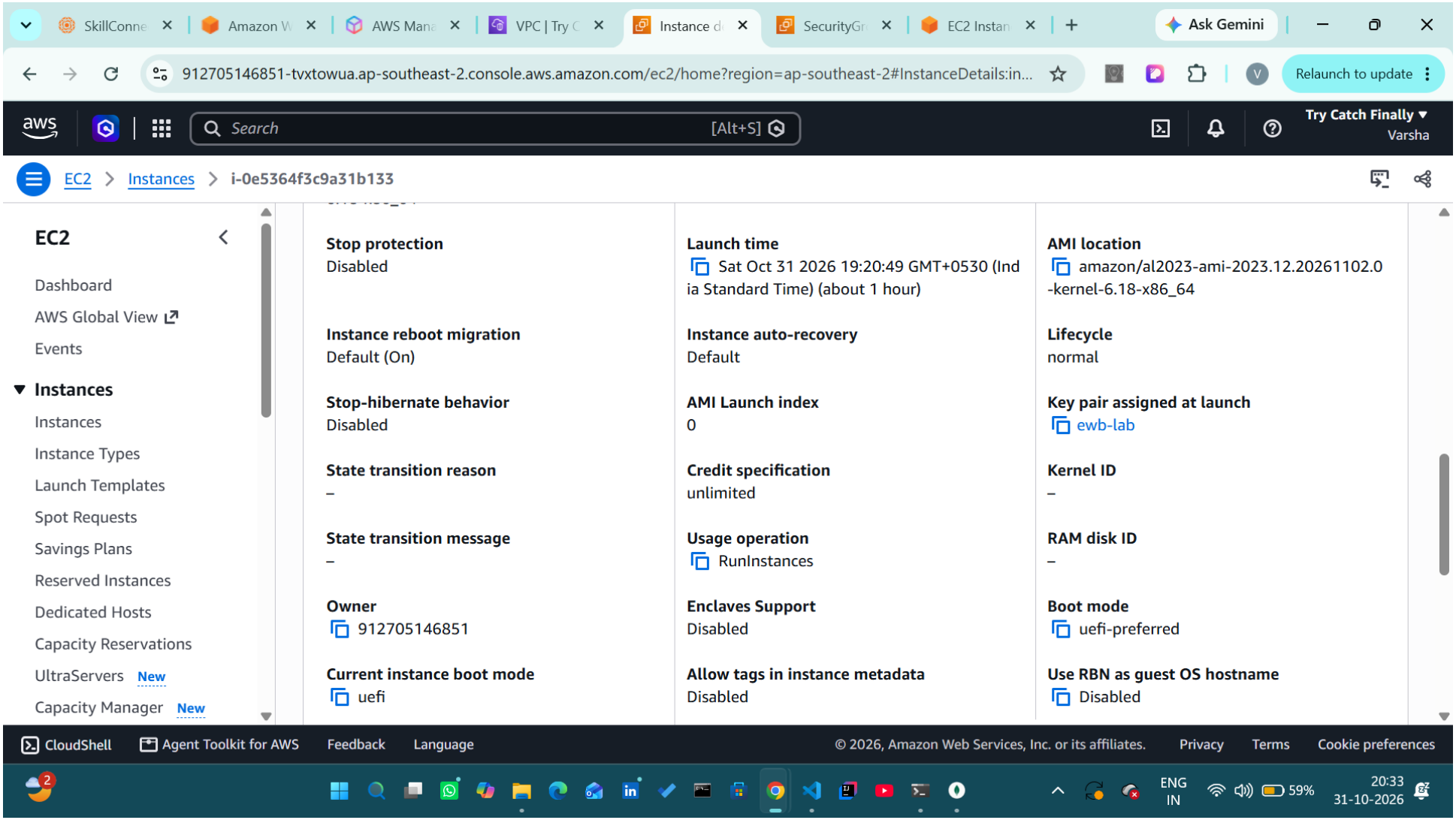
Evidence 22 — image(20261031-145223).png



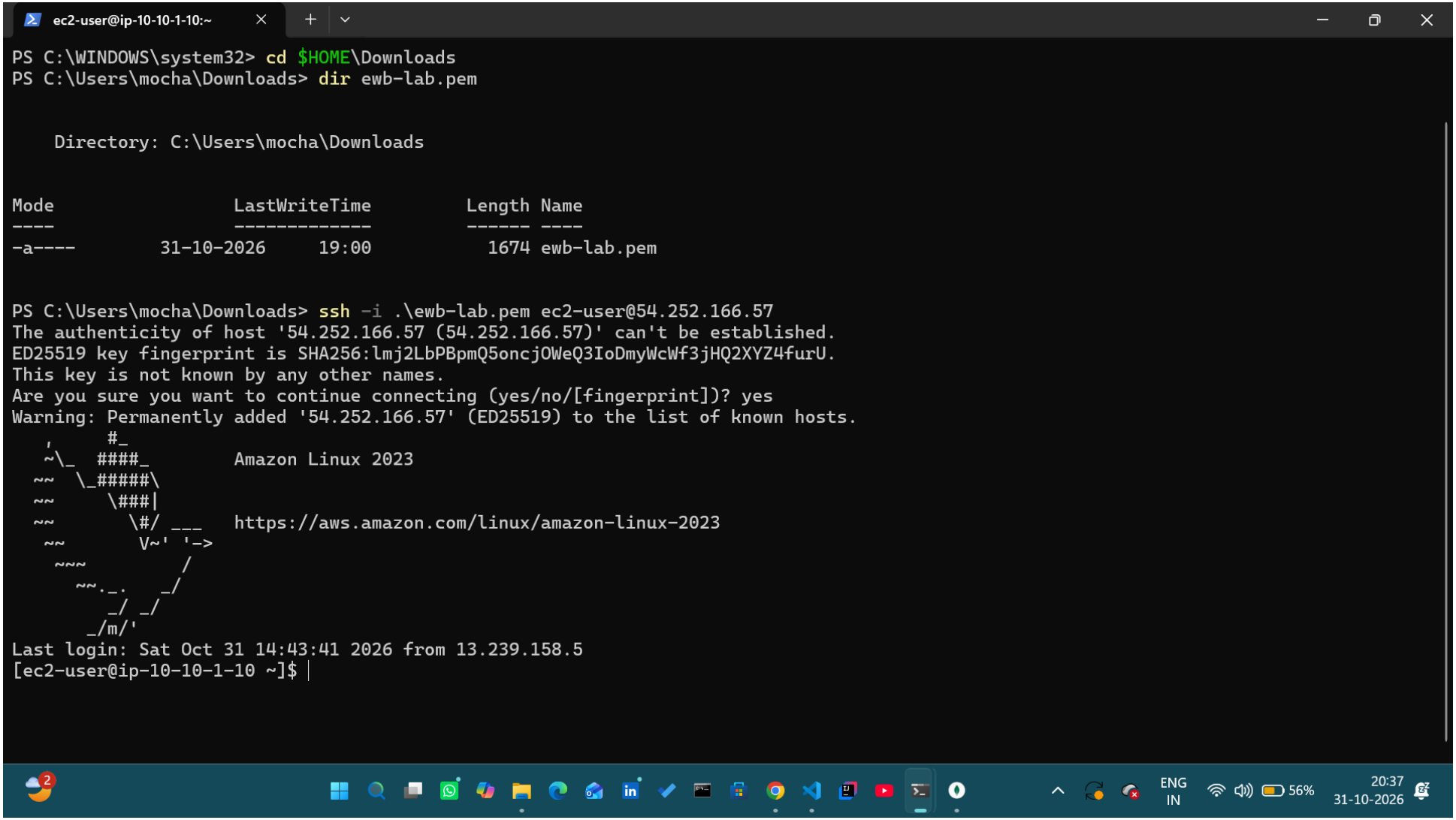
Evidence 23 — image(20261031-145526).png



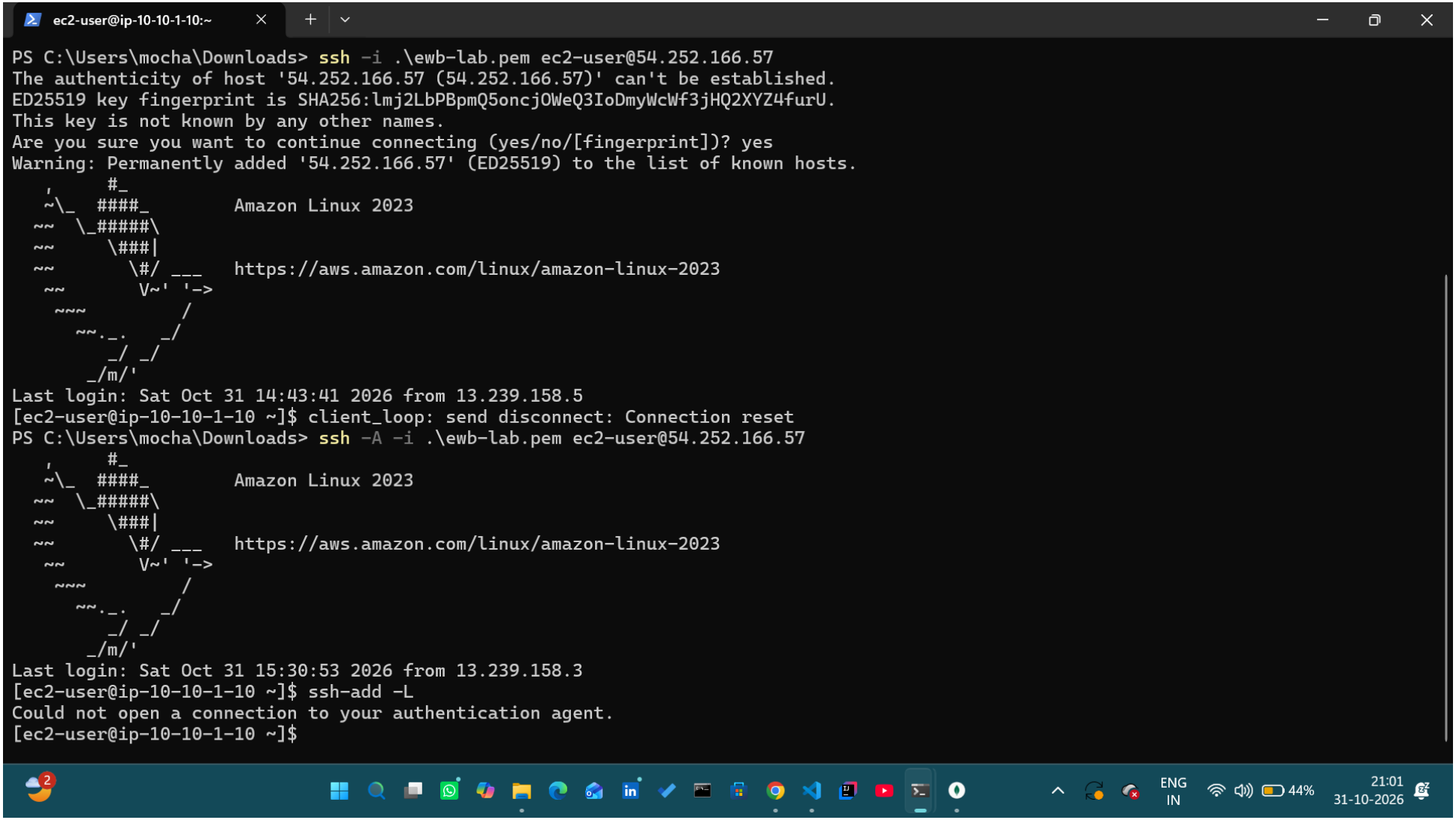
Evidence 24 — image(20261031-150403).png



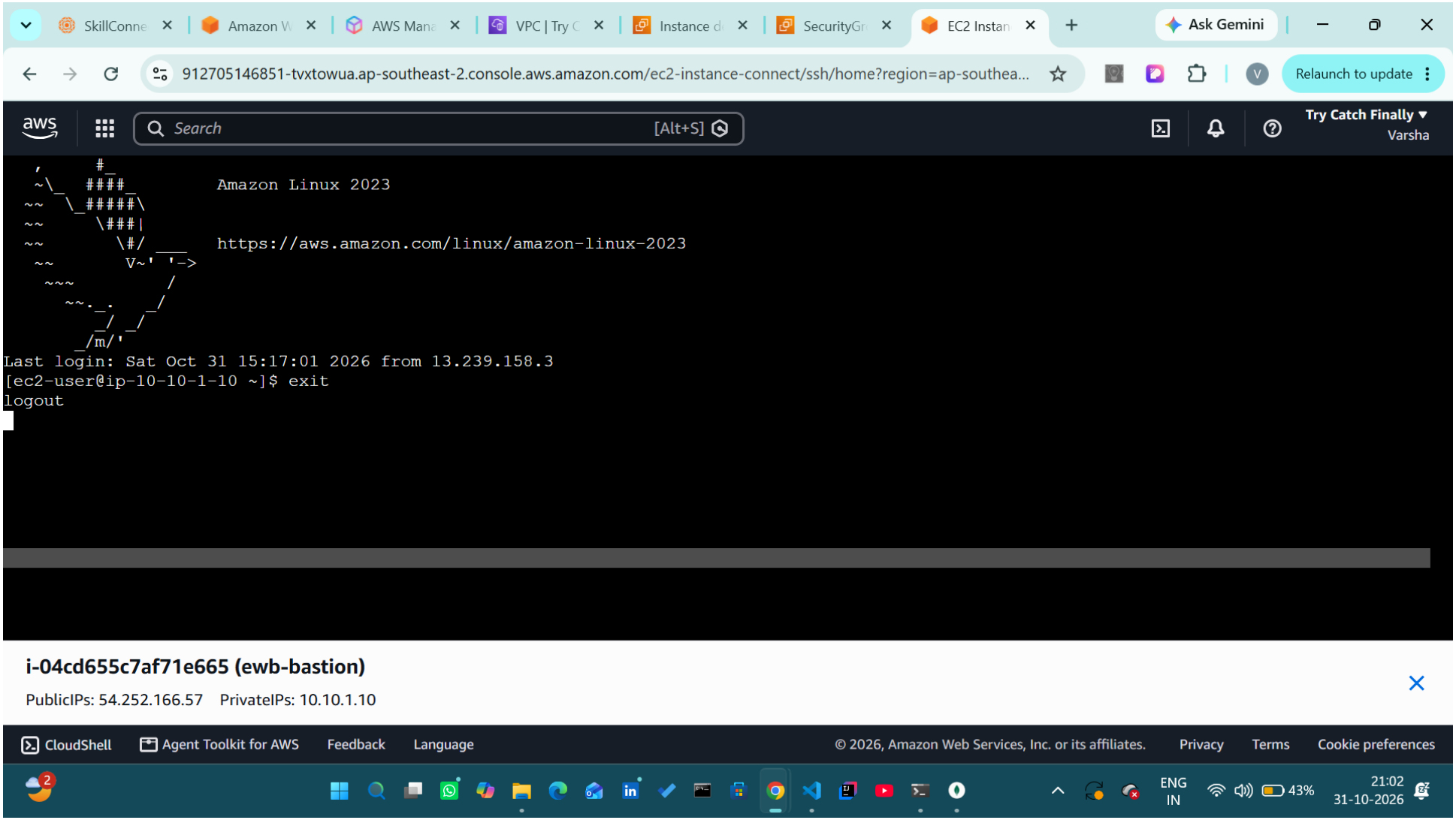
Evidence 25 — image(20261031-150804).png



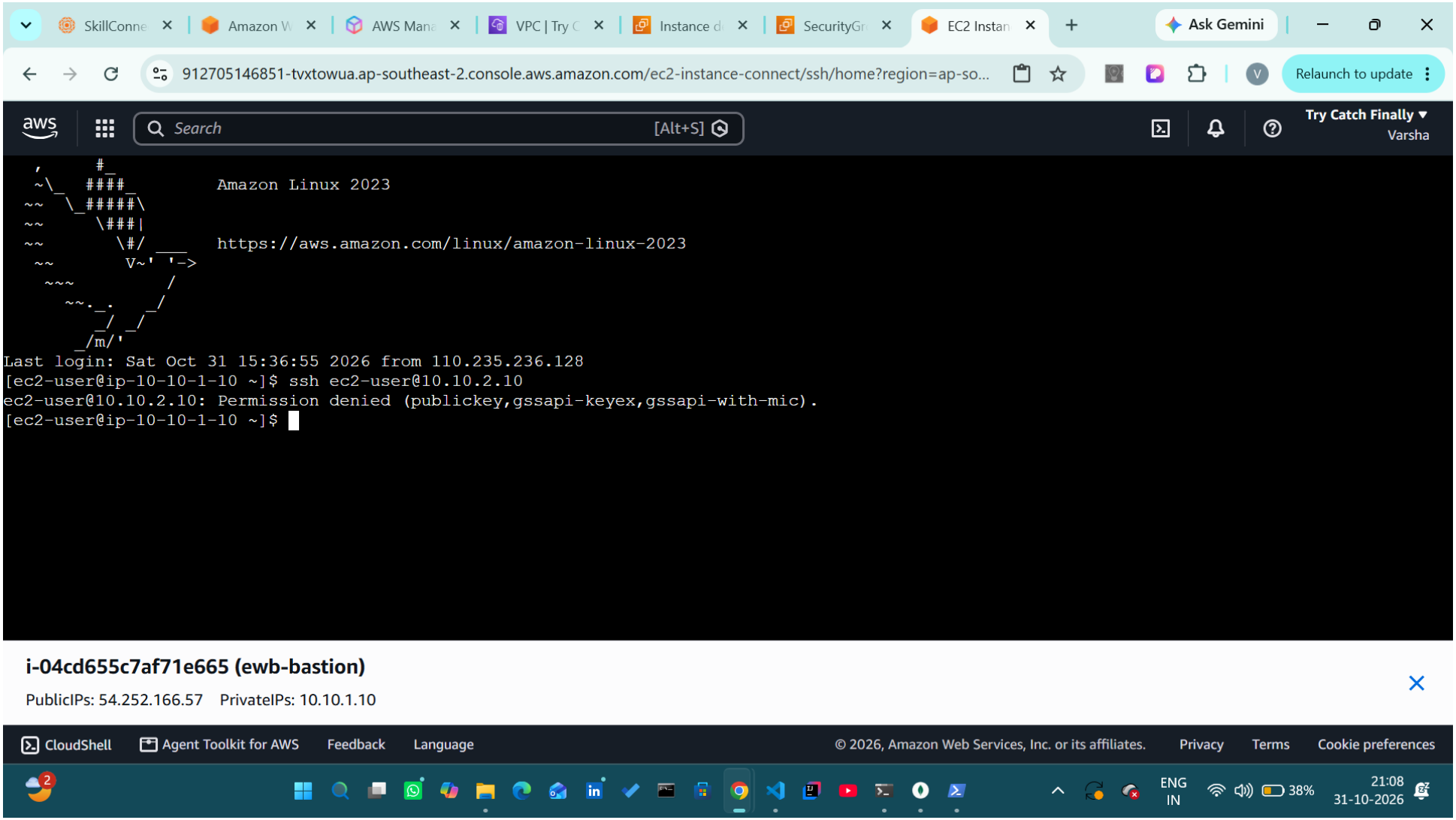
Evidence 26 — image(20261031-153144).png



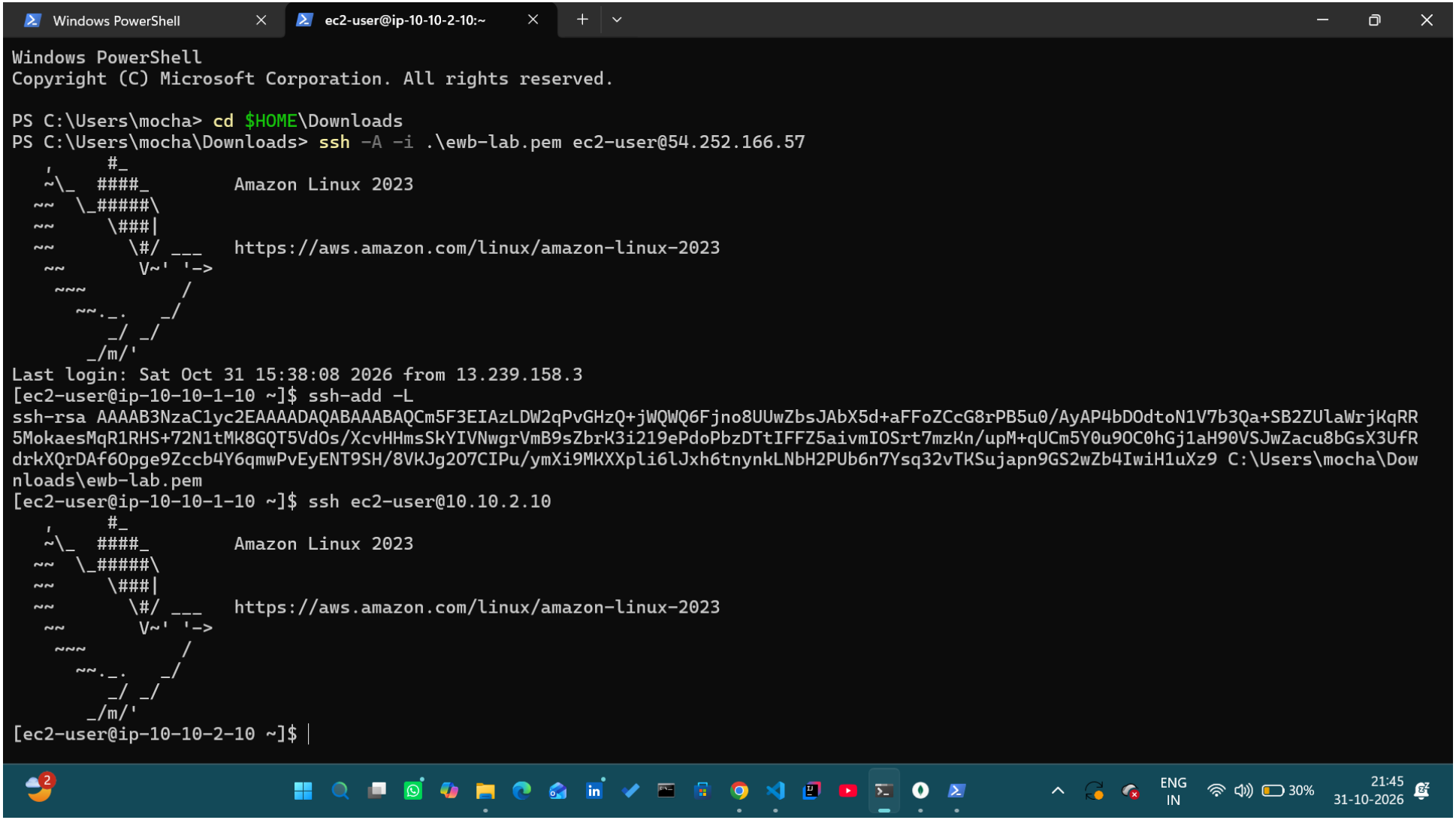
Evidence 27 — image(20261031-153231).png



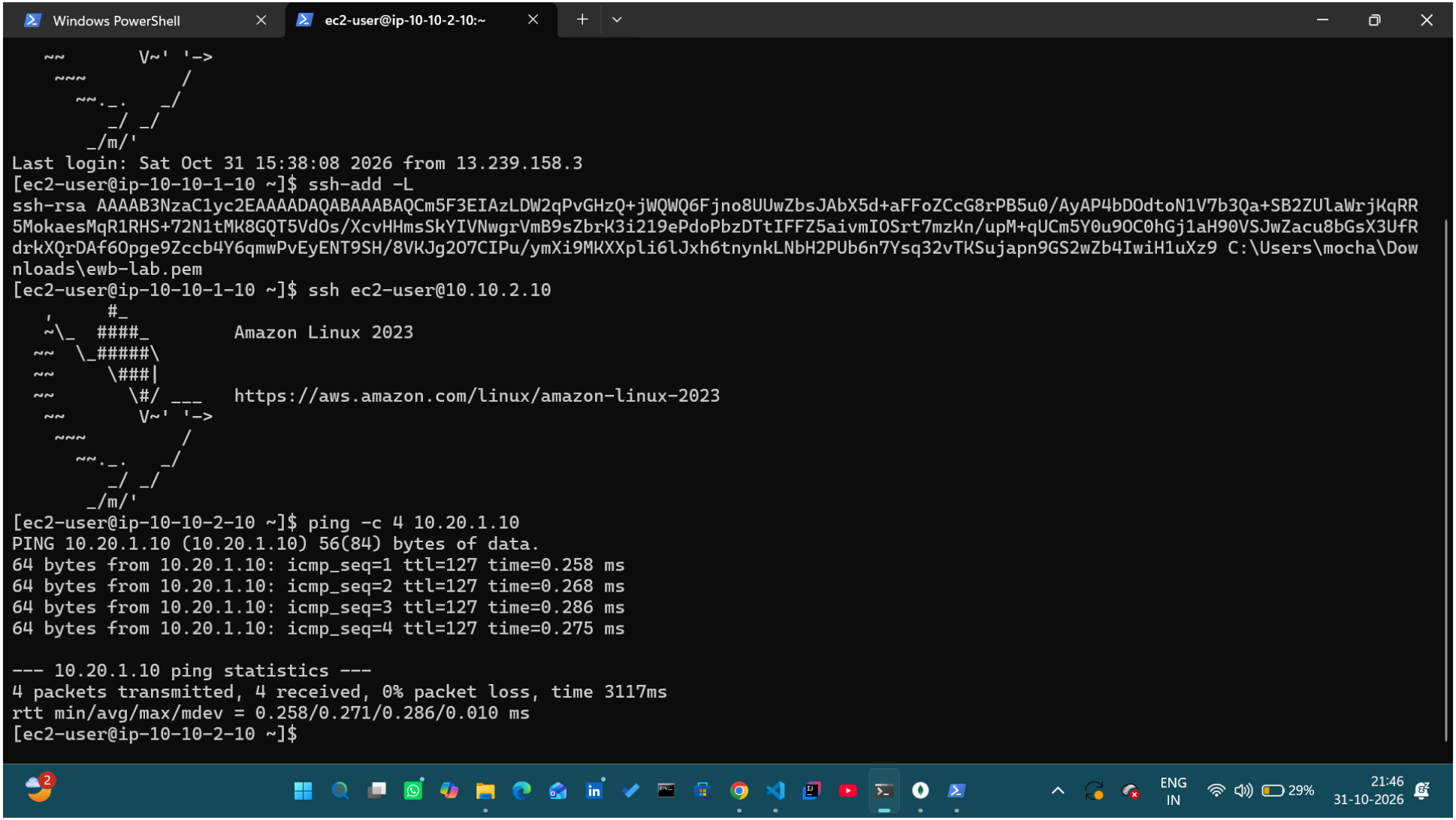
Evidence 28 — image(20261031-153848).png



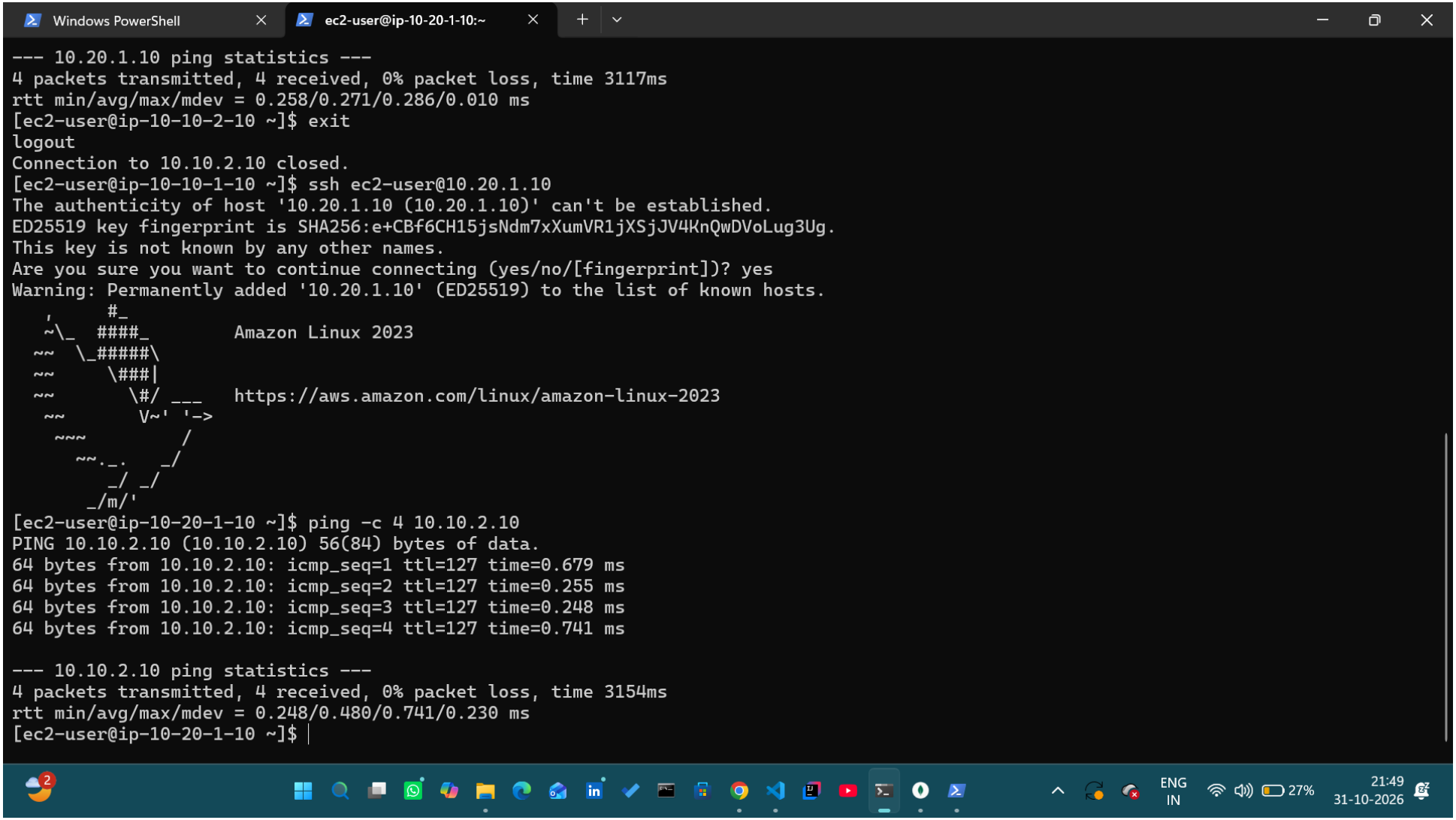
Evidence 29 — image(20261031-161540).png



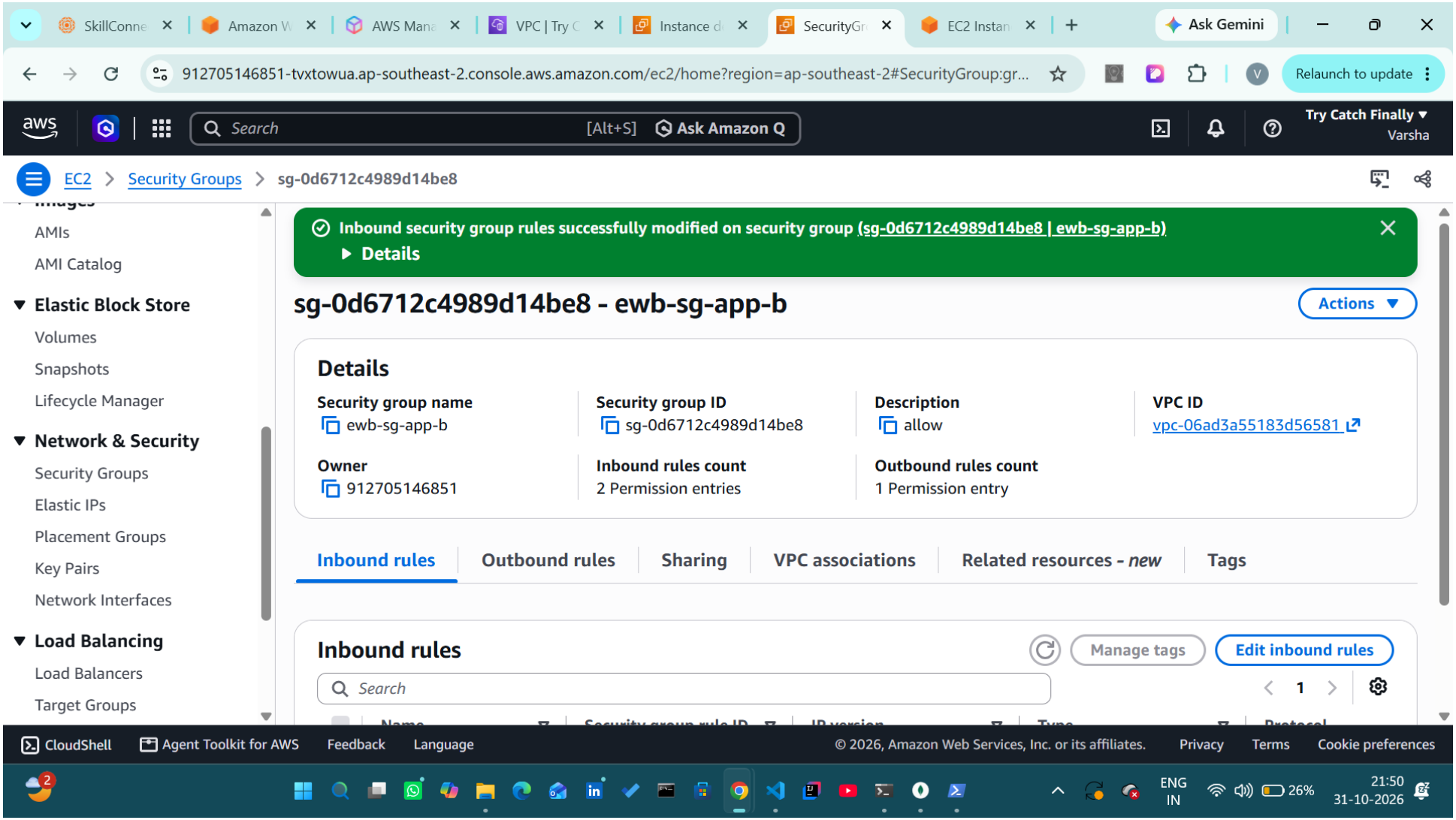
Evidence 30 — image(20261031-161641).png



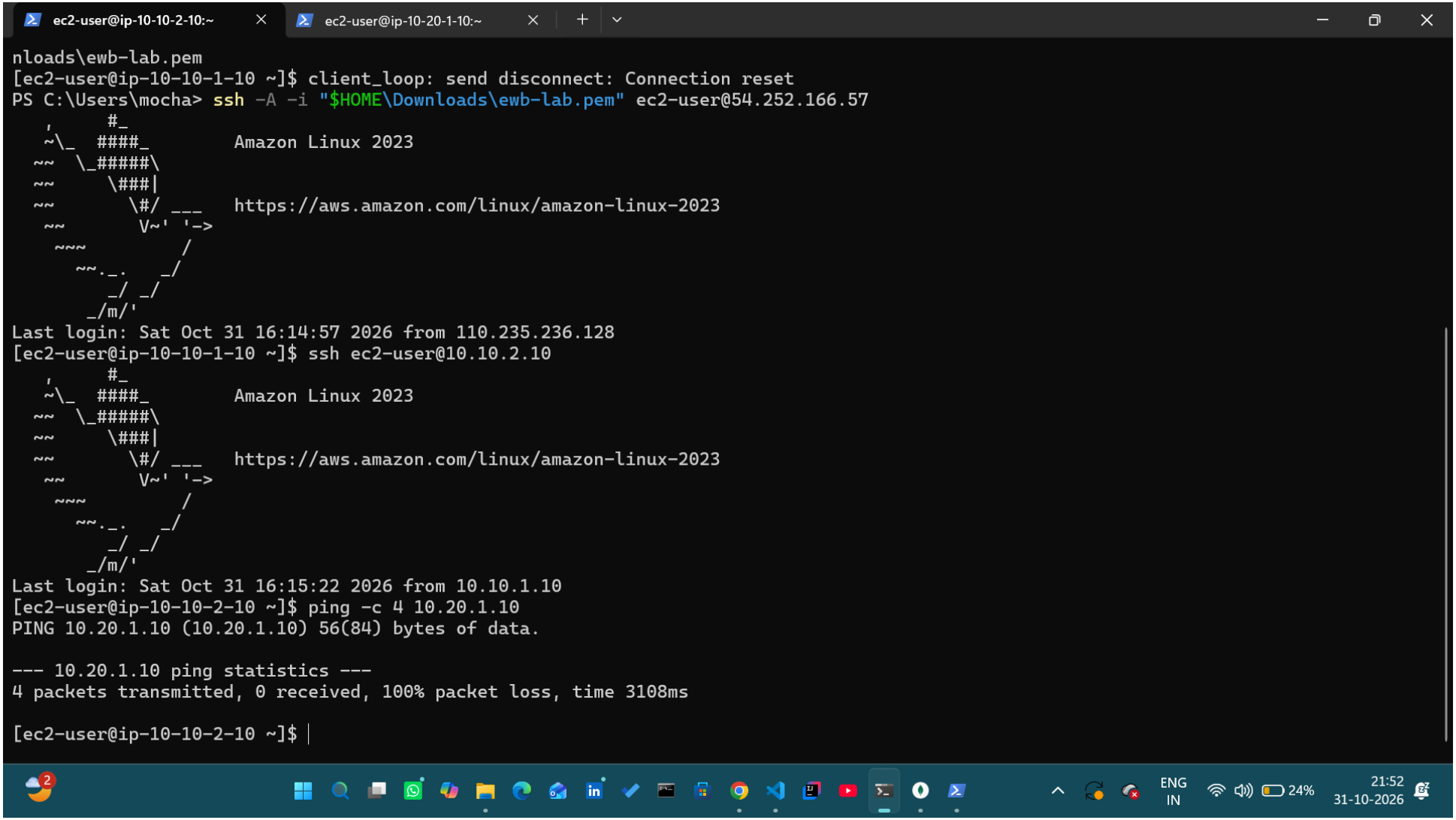
Evidence 31 — image(20261031-161915).png



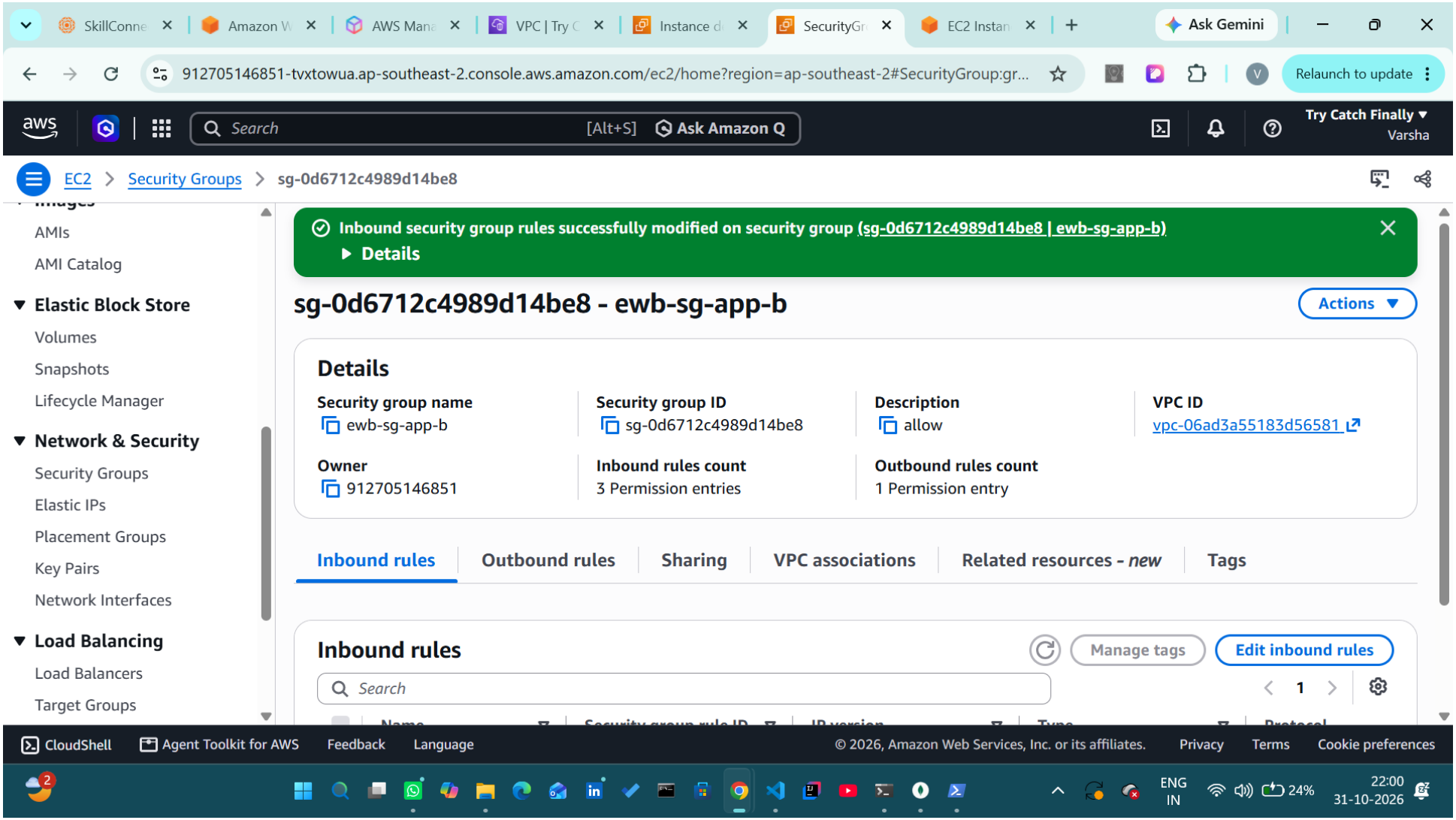
Evidence 32 — image(20261031-162049).png



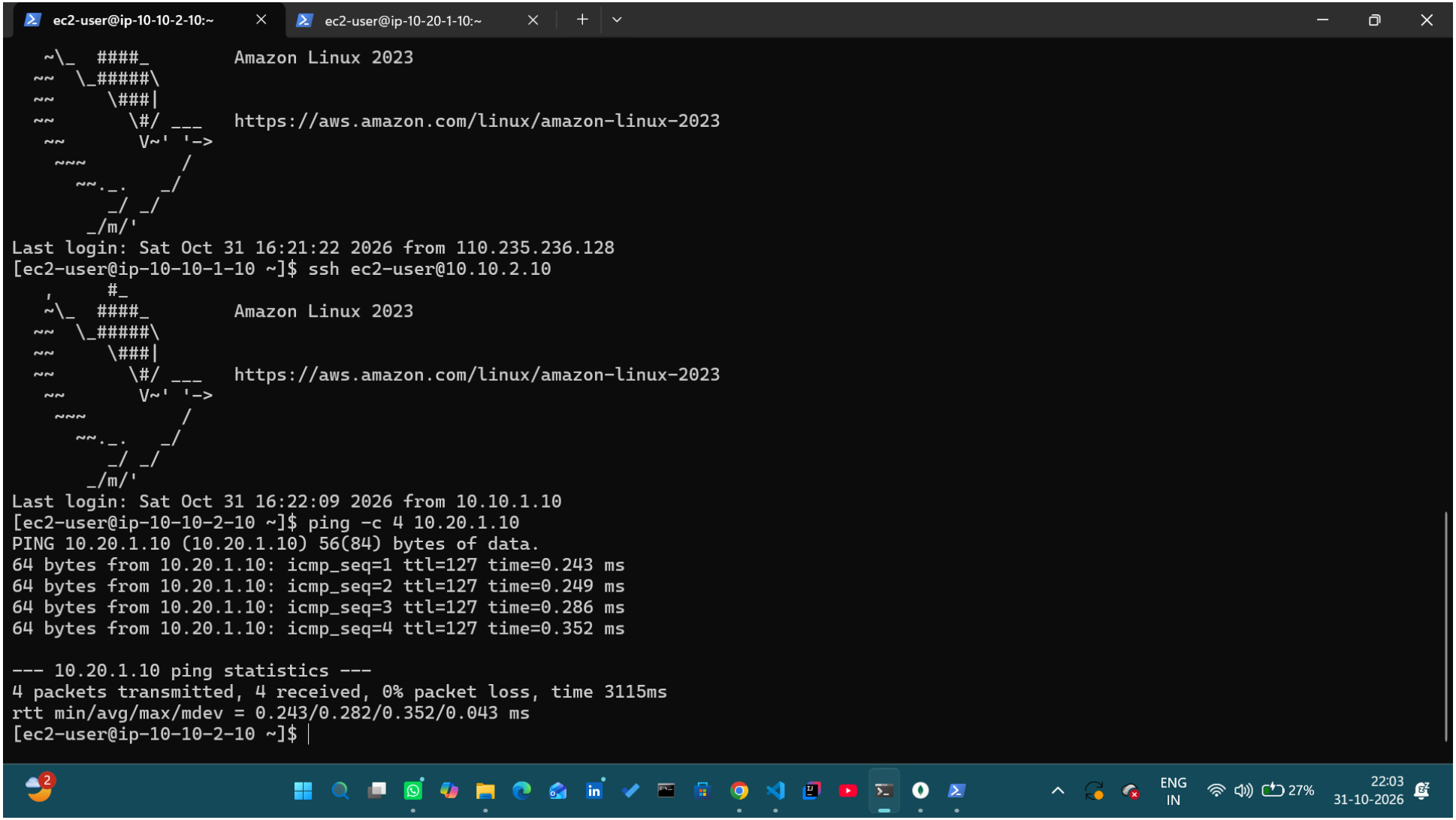
Evidence 33 — image(20261031-162313).png



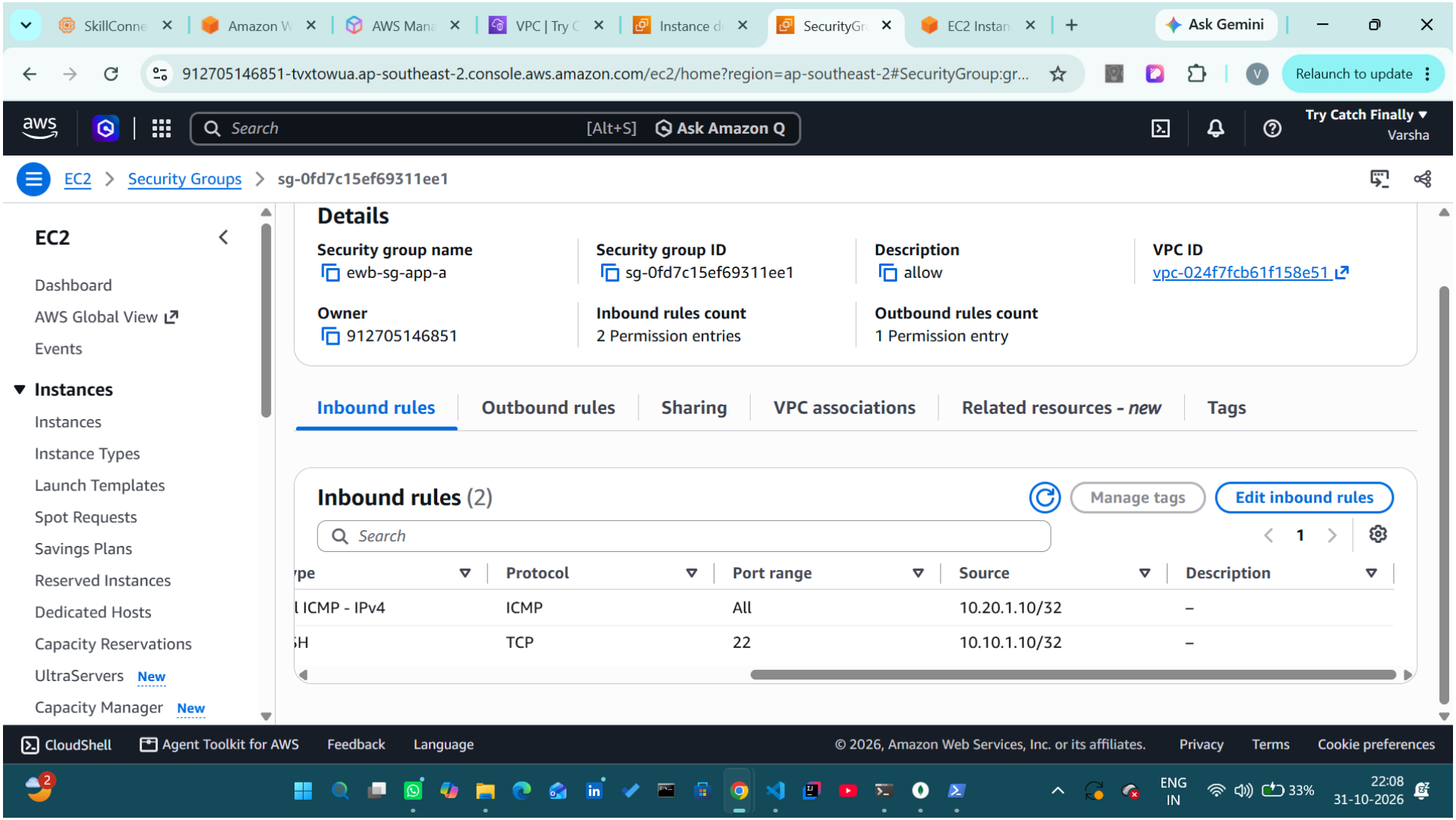
Evidence 34 — image(20261031-163020).png



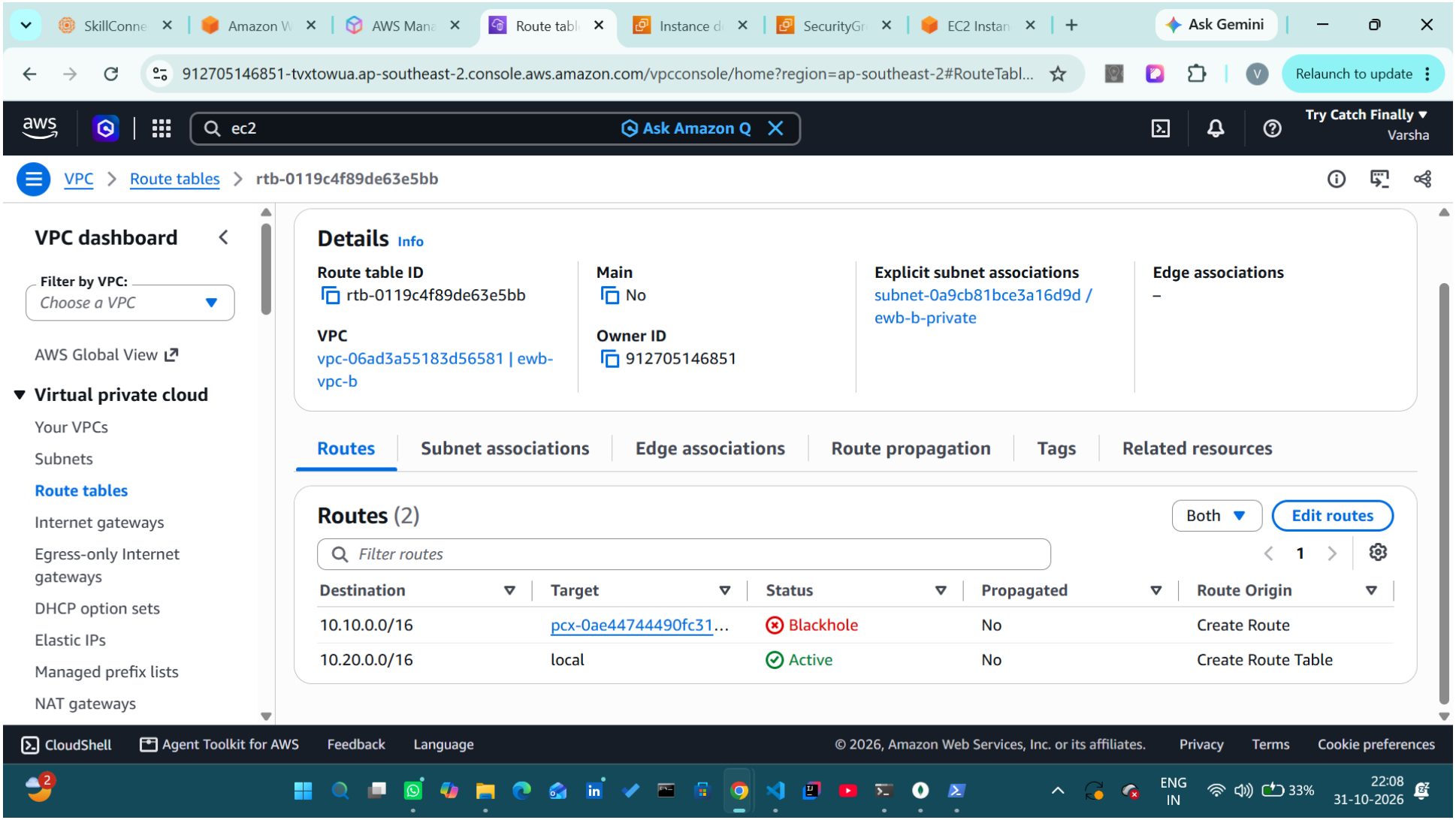
Evidence 35 — image(20261031-163331).png



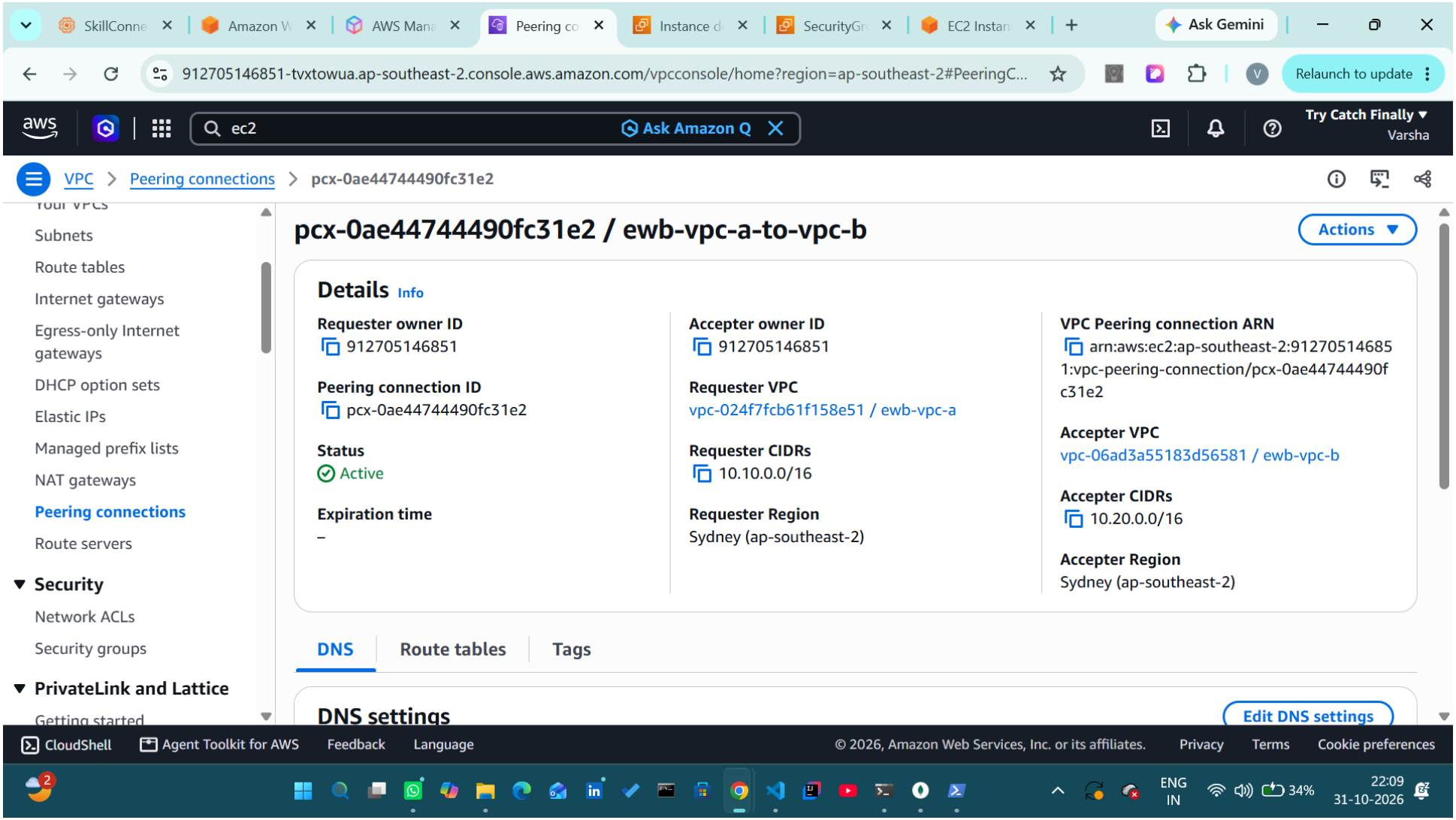
Evidence 36 — image(20261031-163815).png



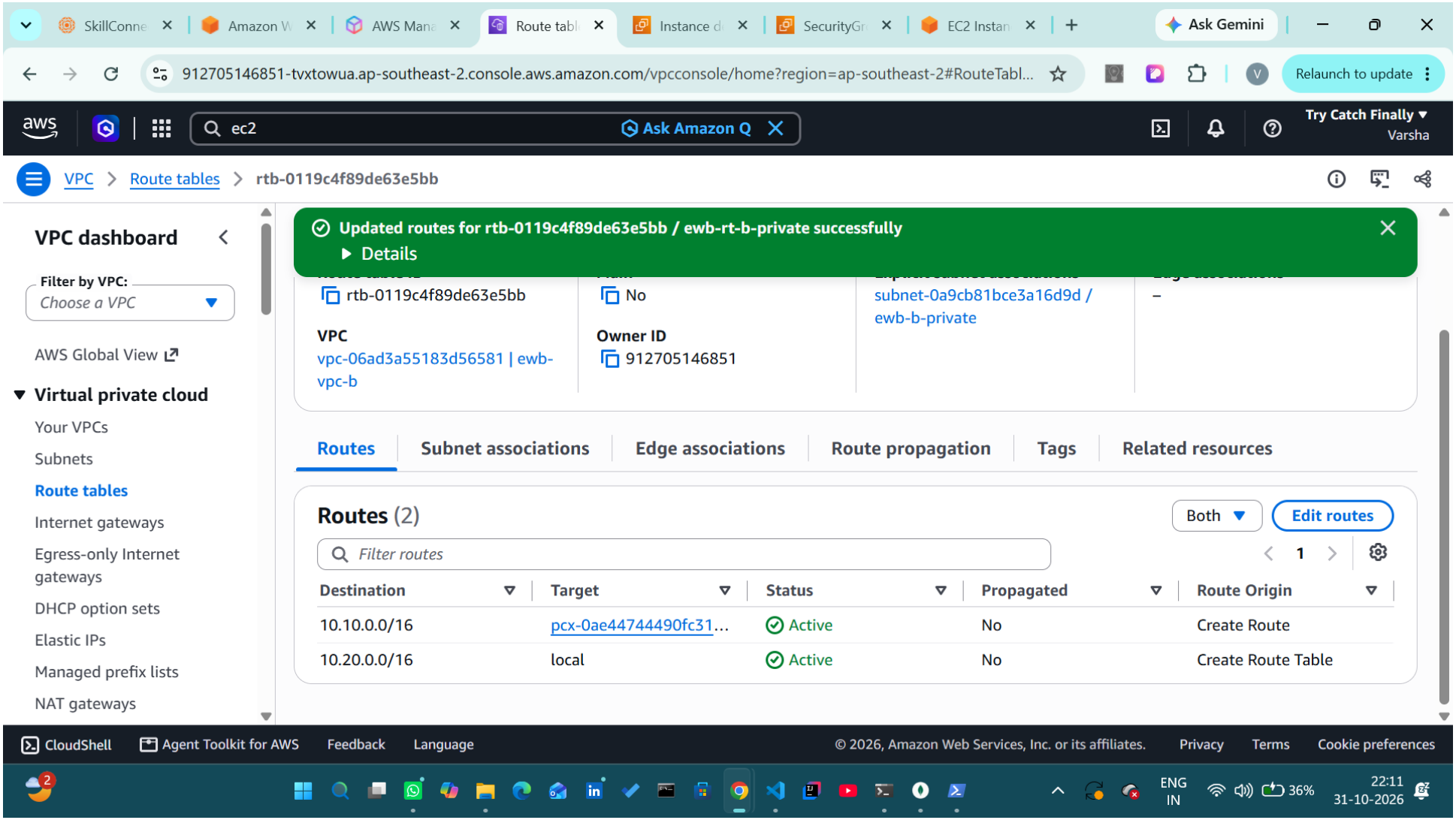
Evidence 37 — image(20261031-163905).png



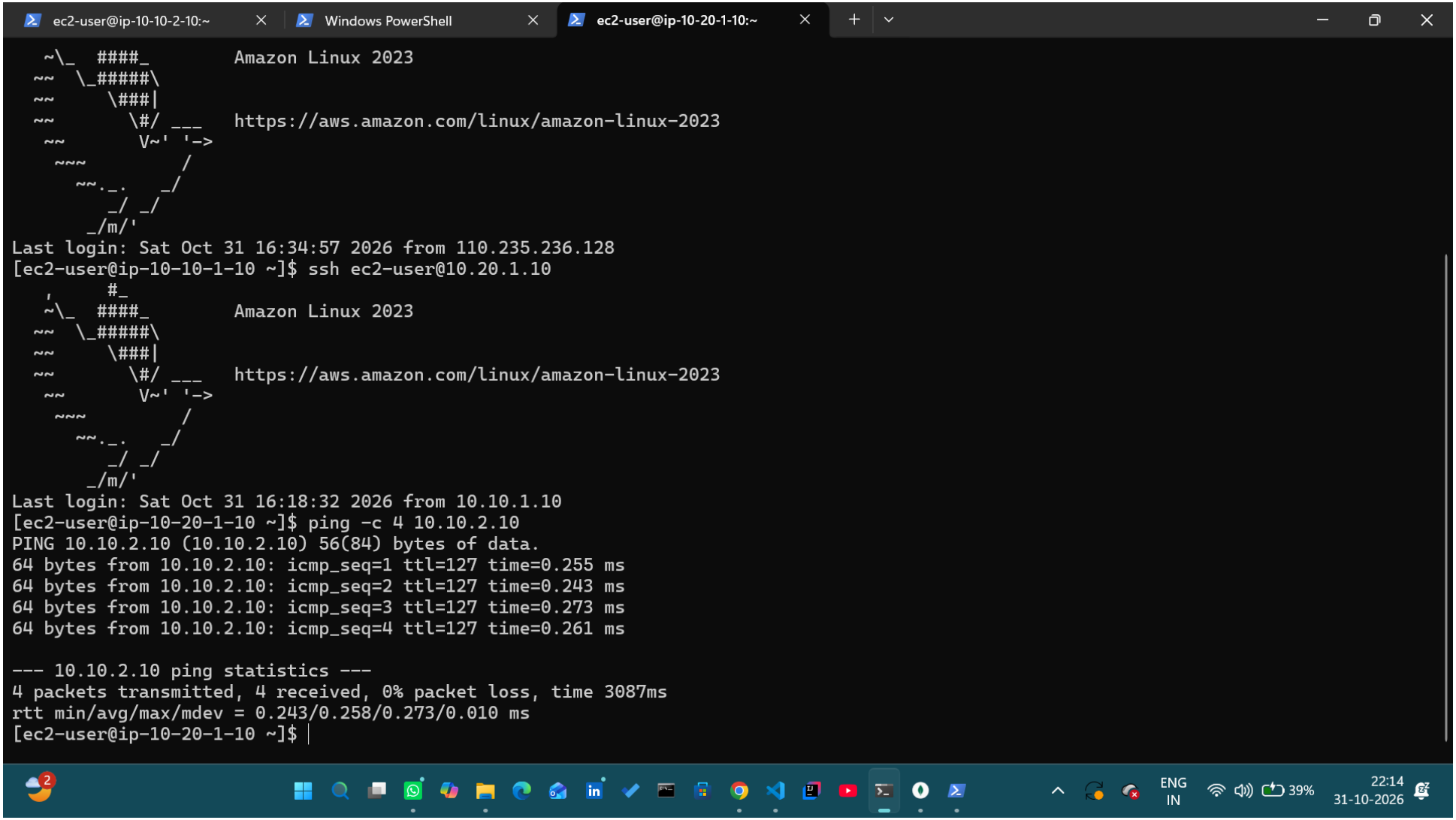
Evidence 38 — image(20261031-164004).png



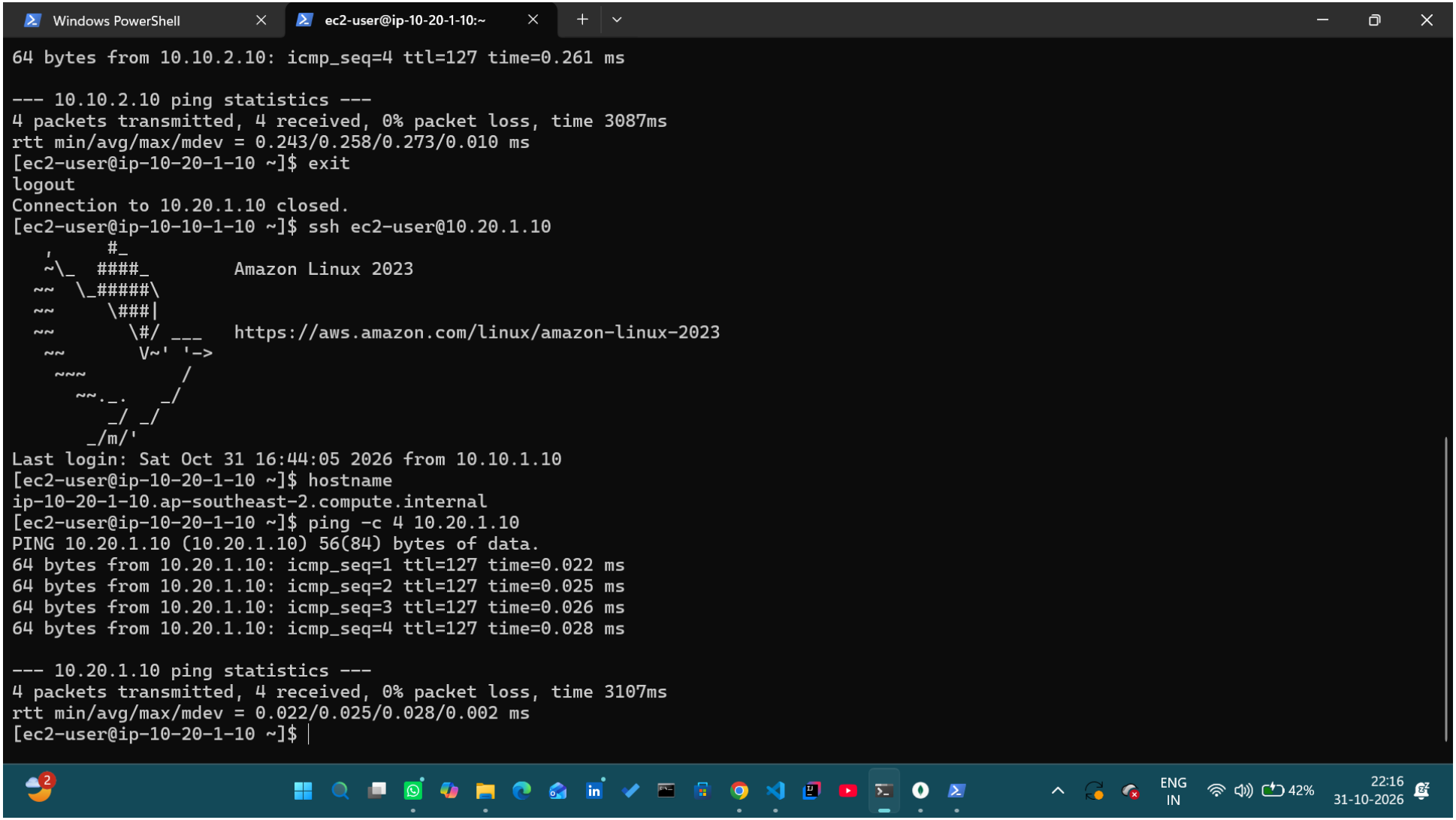
Evidence 39 — image(20261031-164152).png



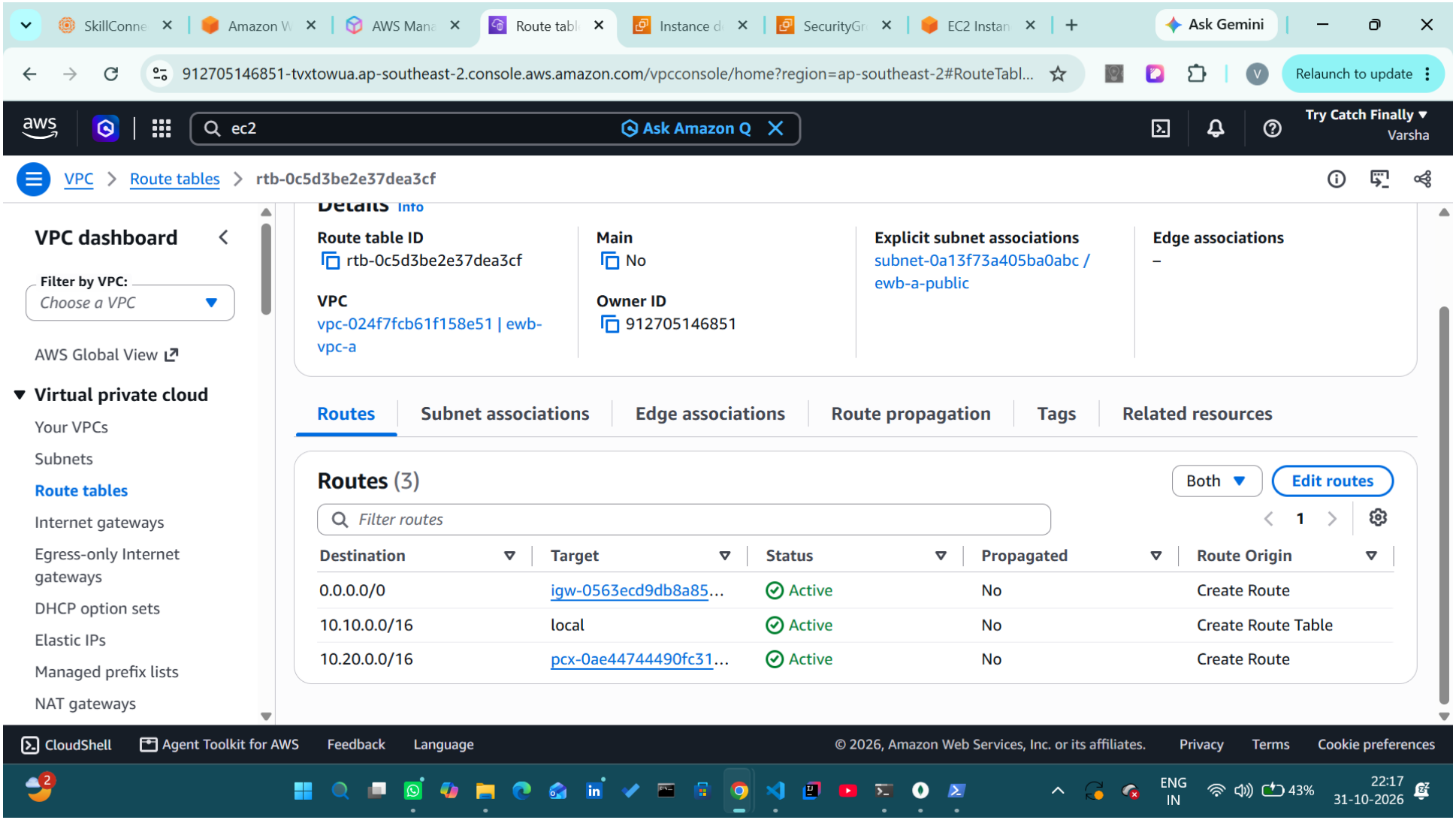
Evidence 40 — image(20261031-164421).png



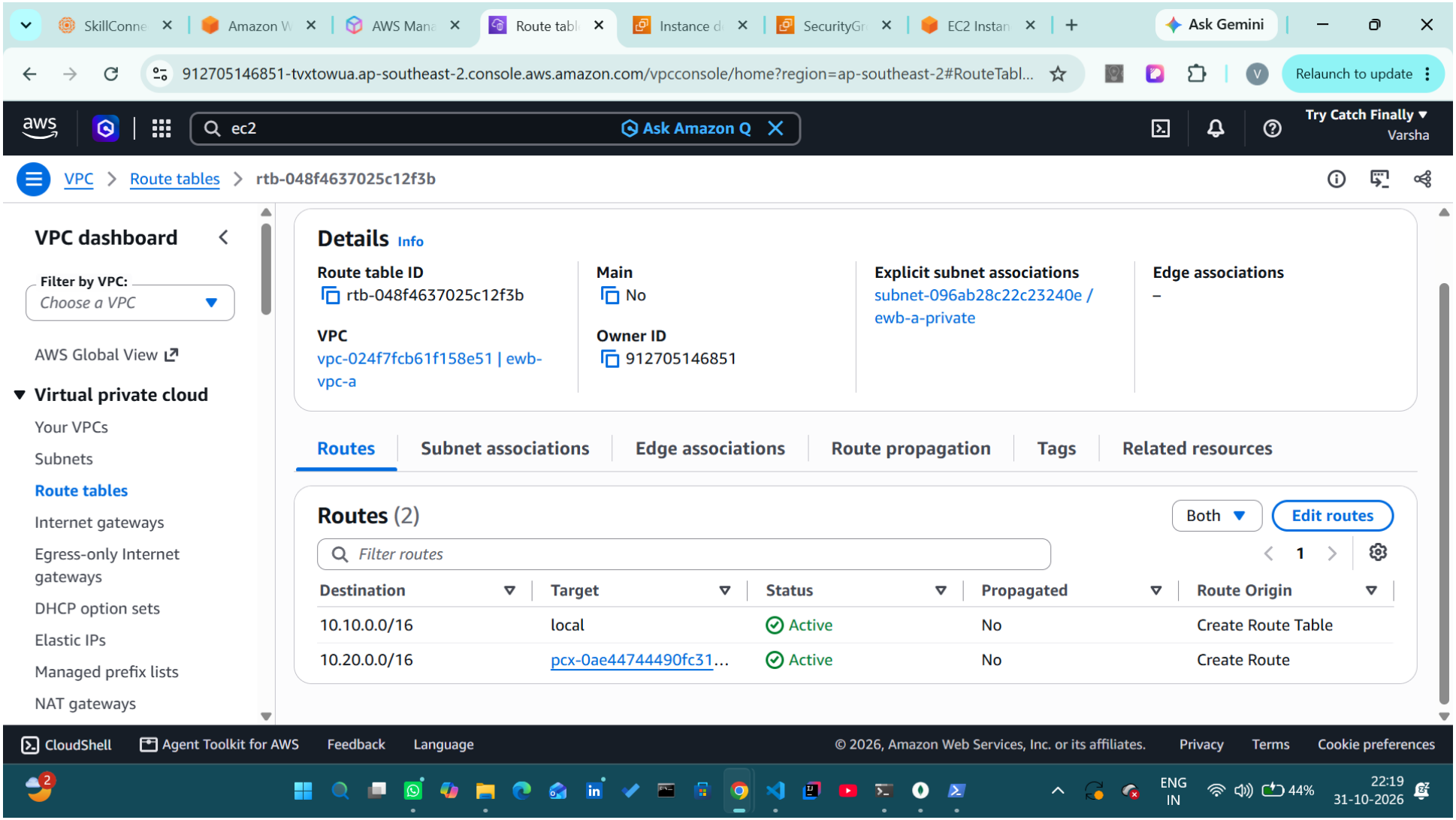
Evidence 41 — image(20261031-164700).png



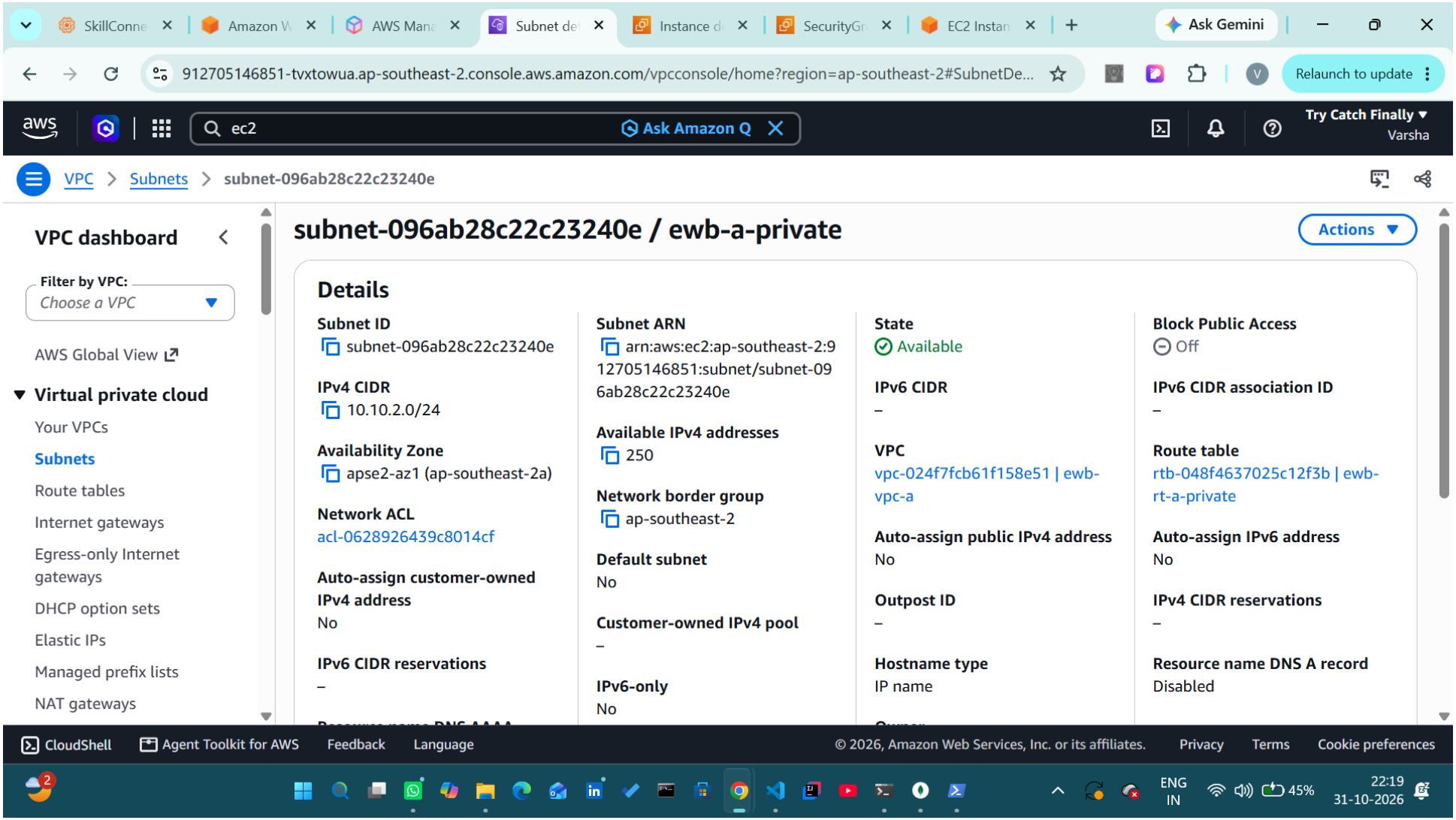
Evidence 42 — image(20261031-164806).png



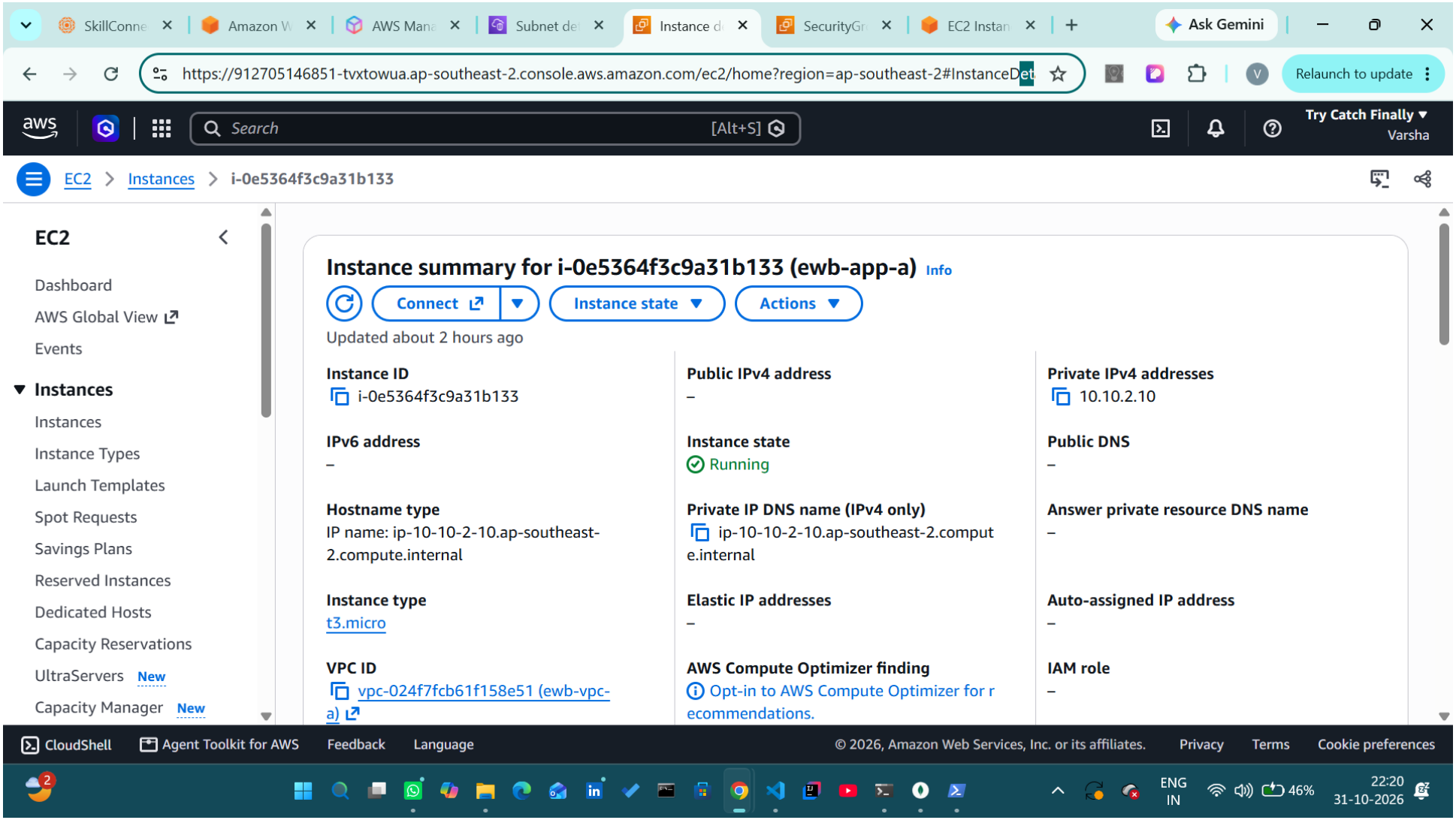
Evidence 43 — image(20261031-164937).png



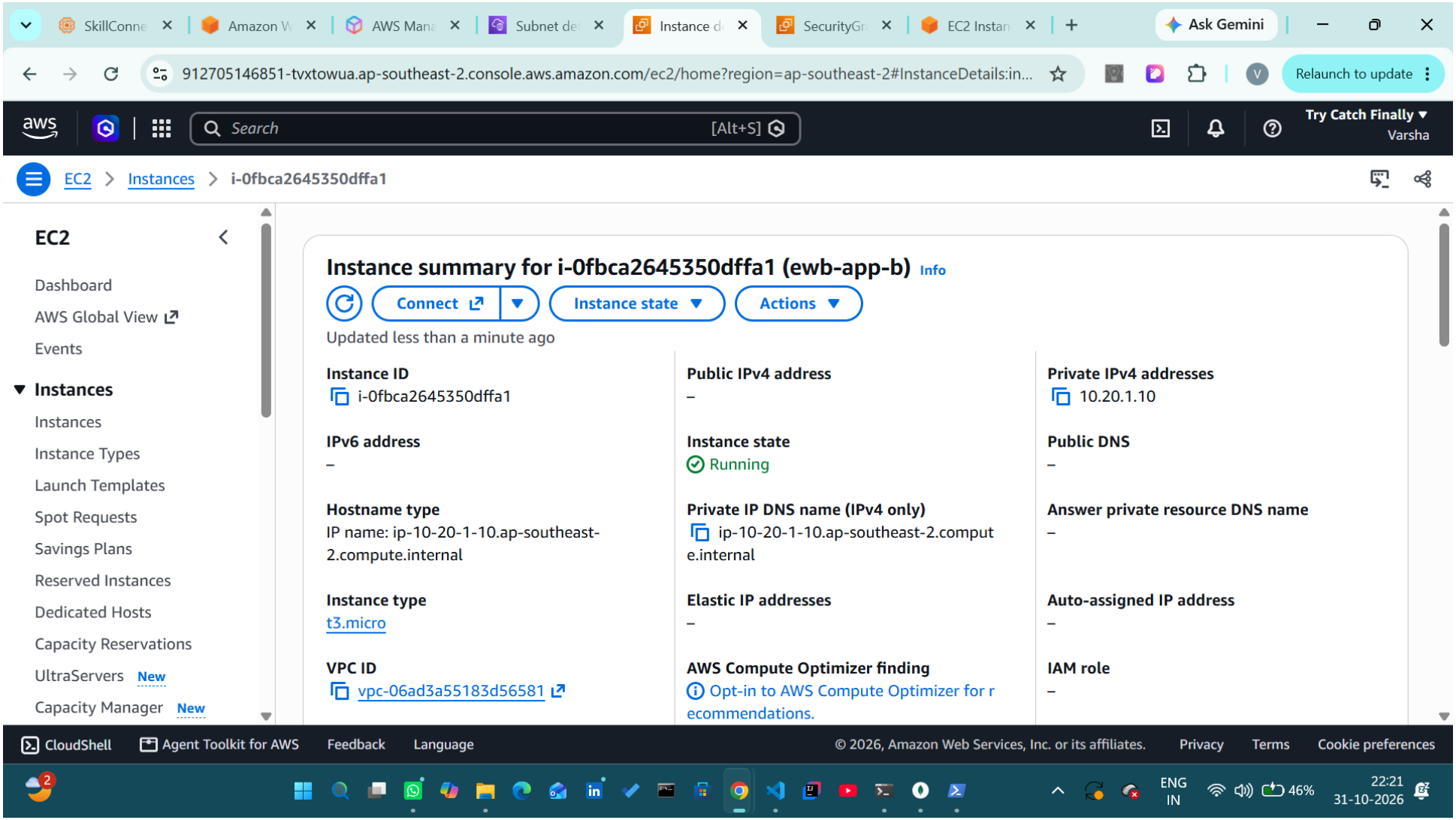
Evidence 44 — image(20261031-165007).png



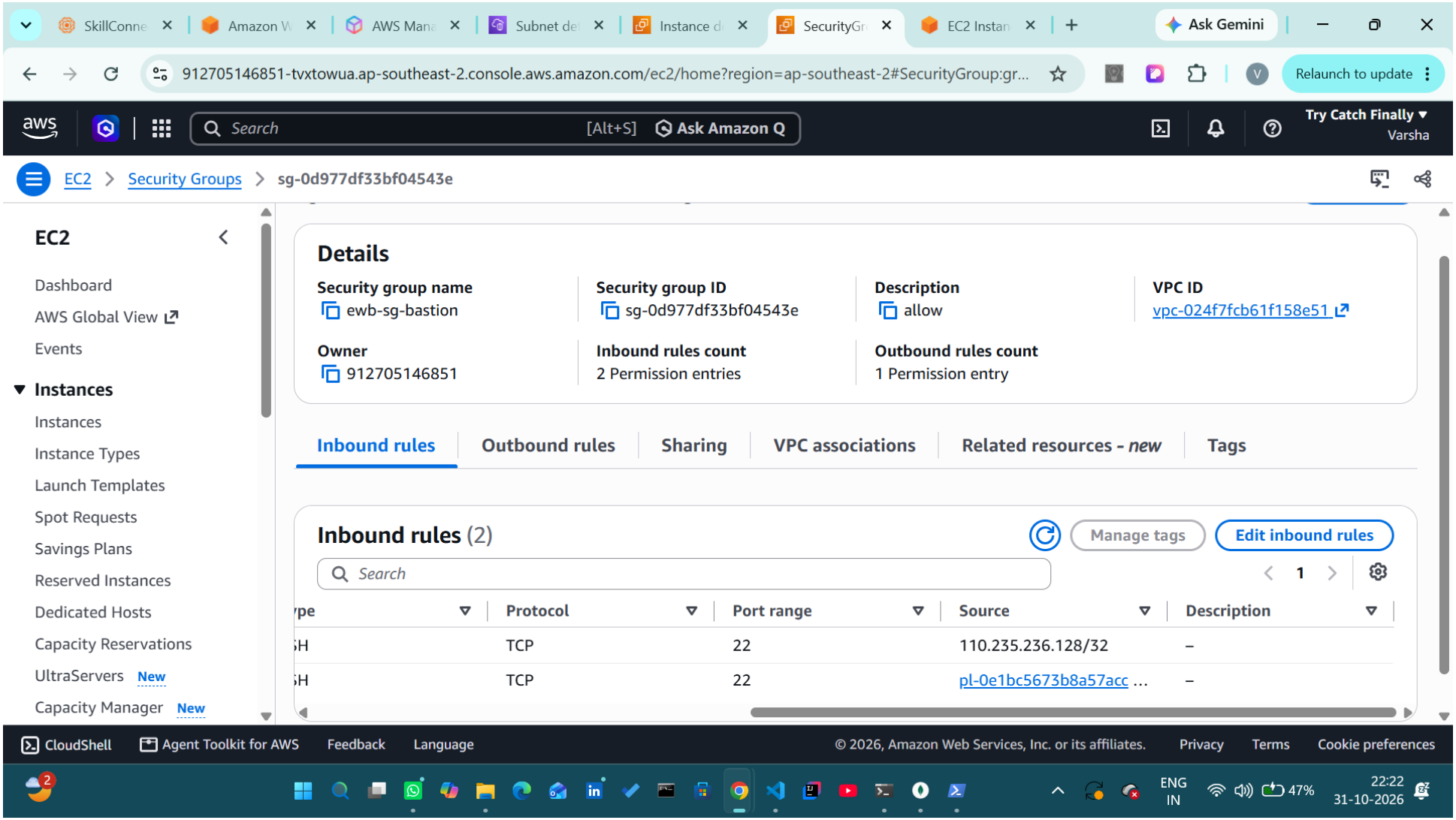
Evidence 45 — image(20261031-165107).png



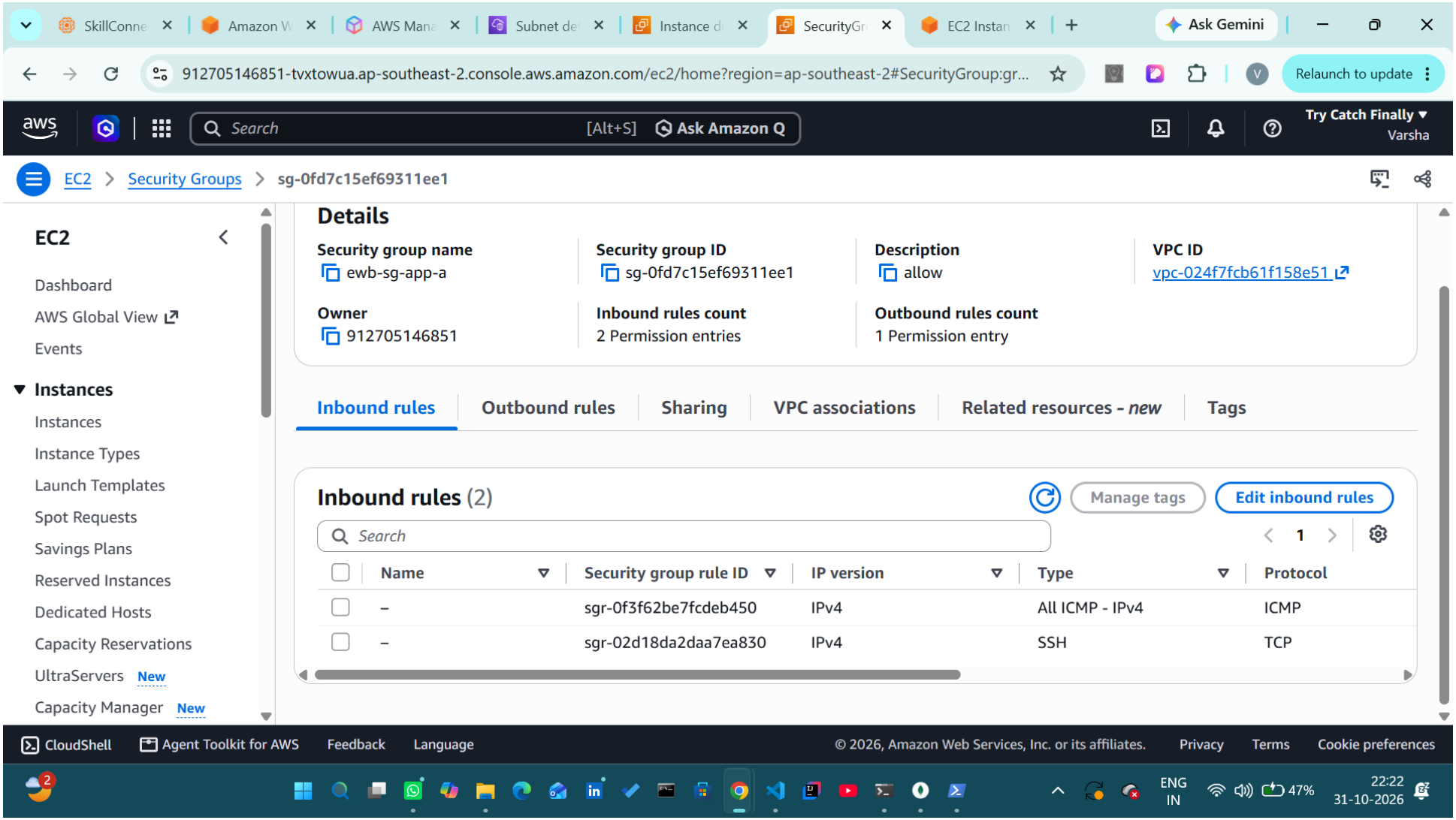
Evidence 46 — image(20261031-165131).png



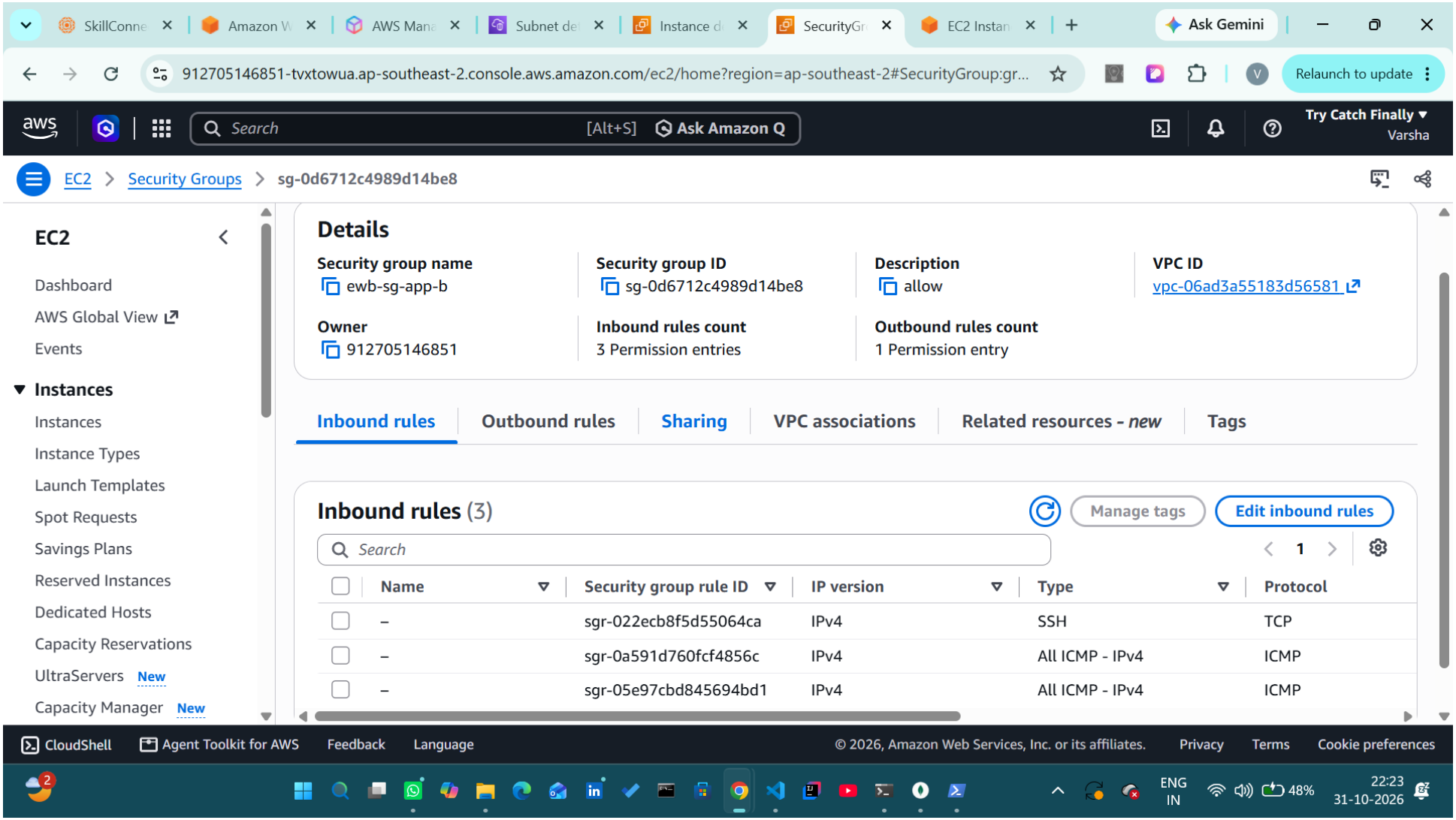
Evidence 47 — image(20261031-165243).png



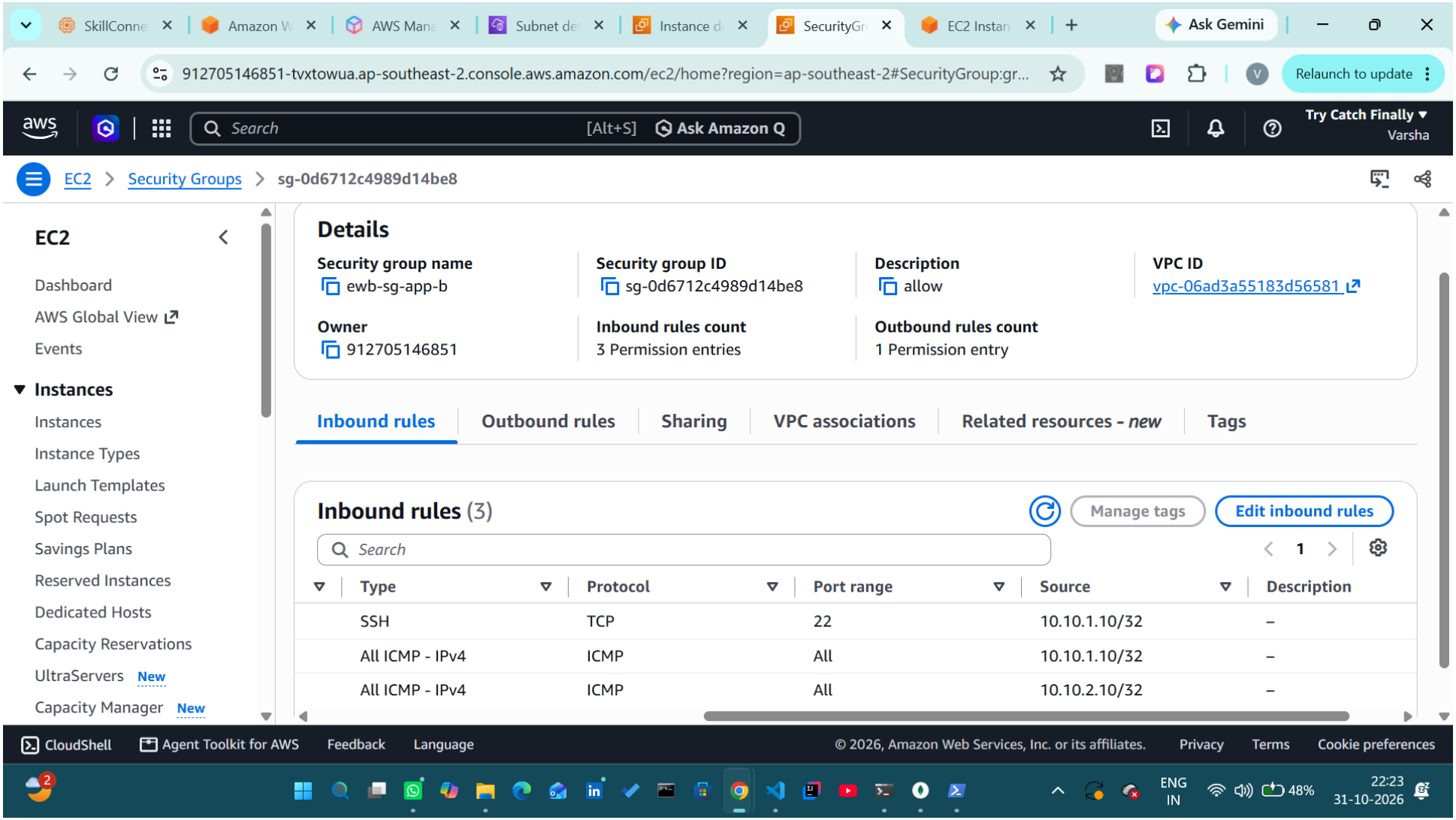
Evidence 48 — image(20261031-165300).png



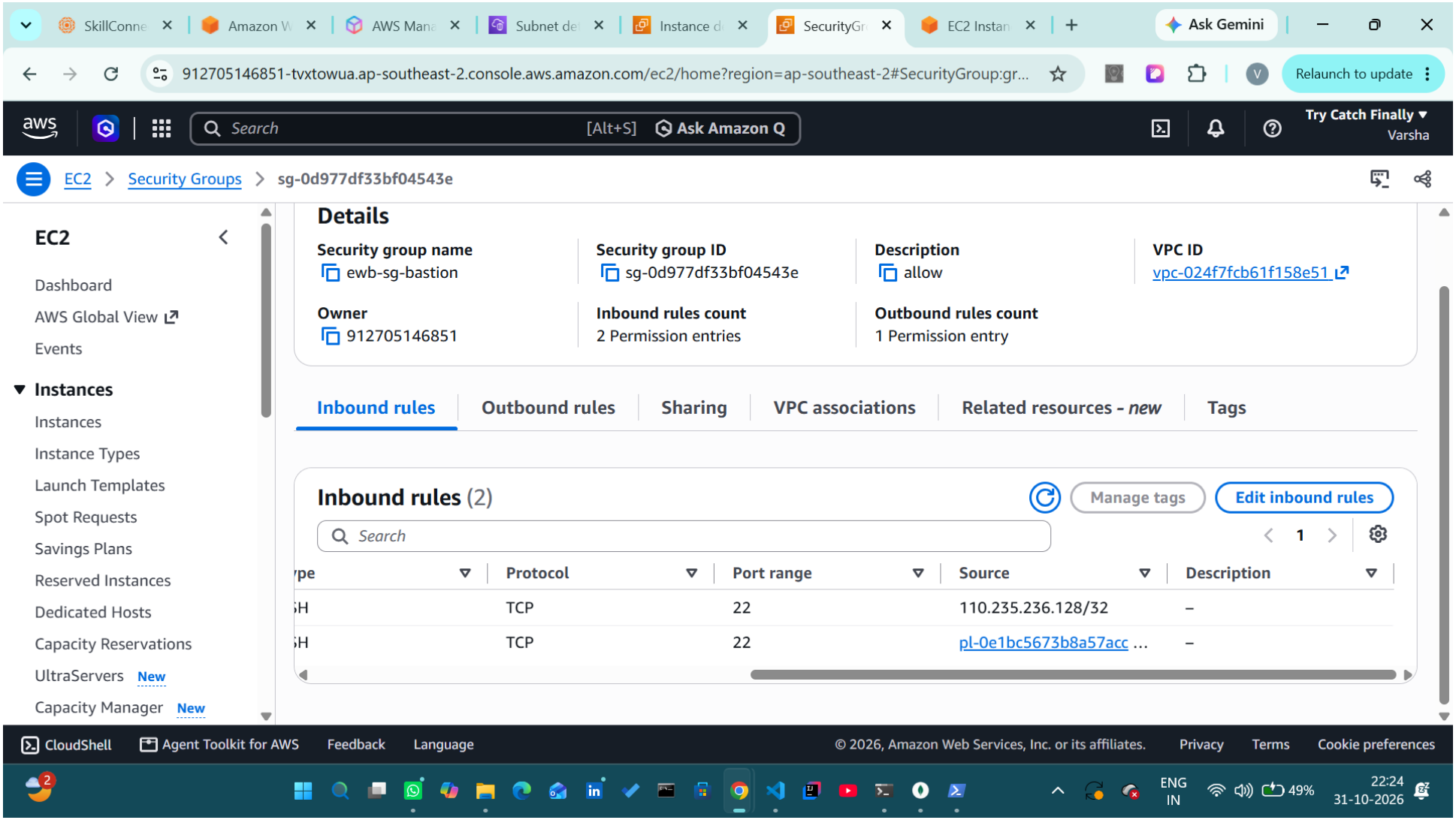
Evidence 49 — image(20261031-165325).png



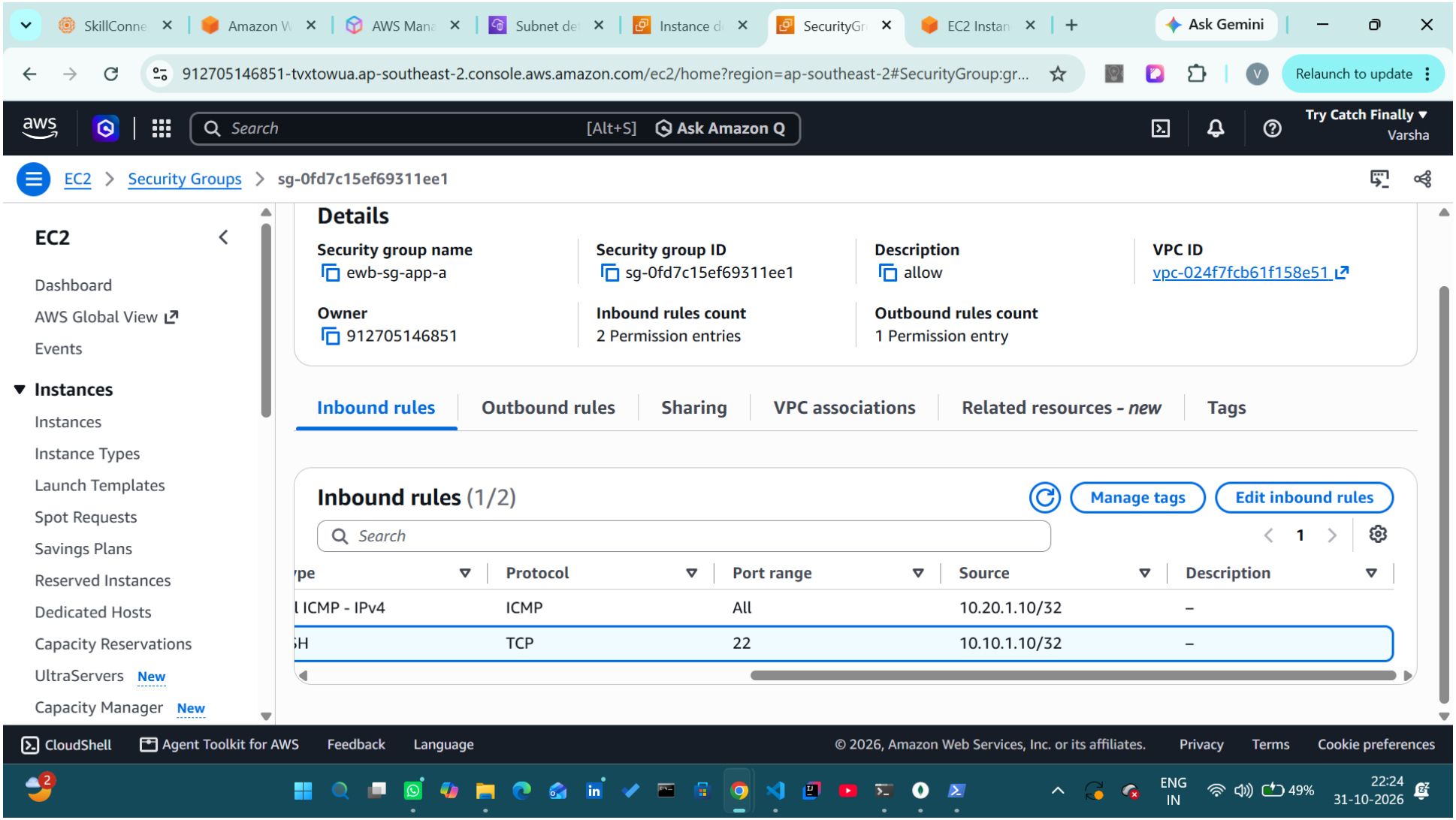
Evidence 50 — image(20261031-165408).png



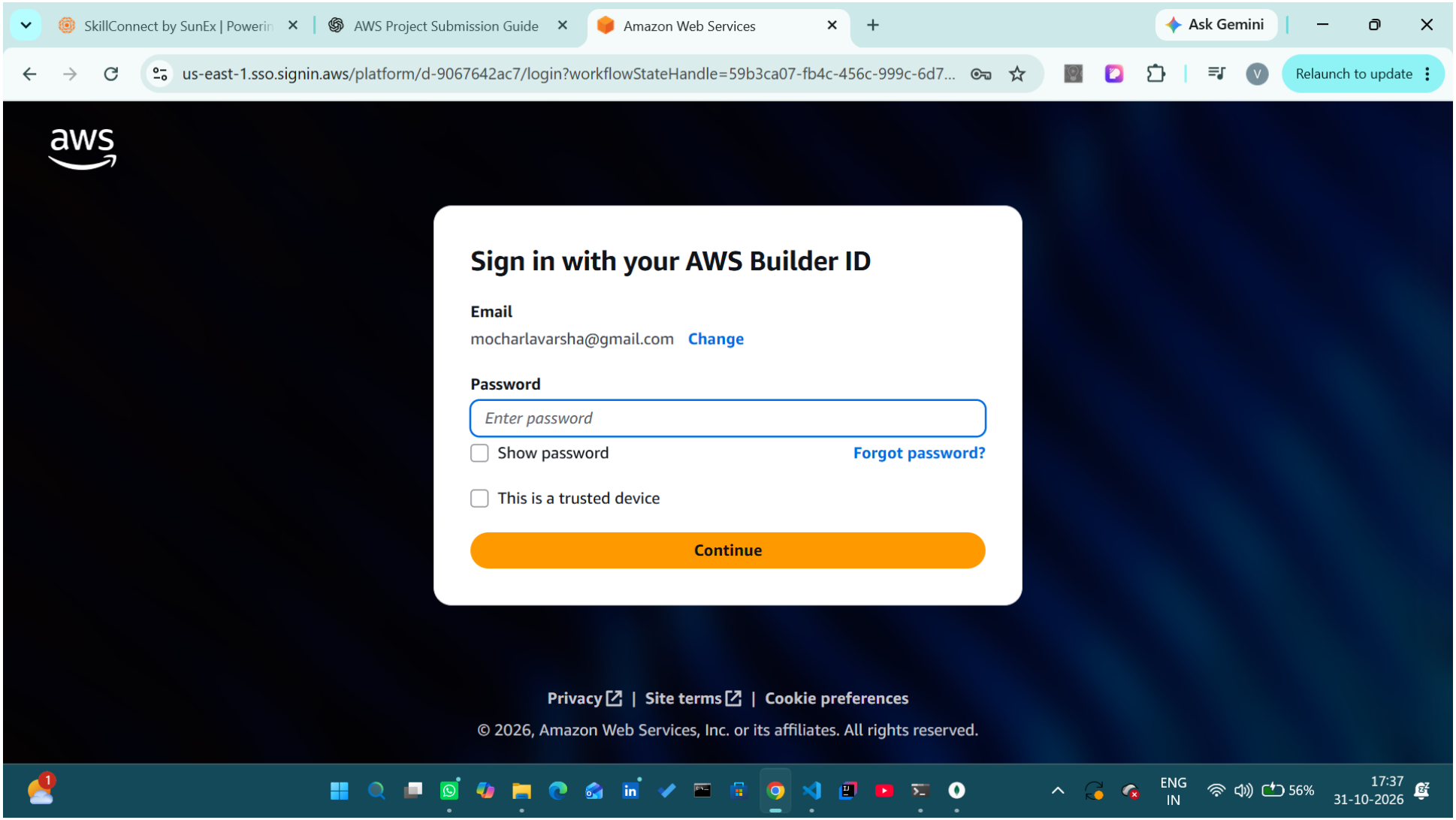
Evidence 51 — image(20261031-165423).png



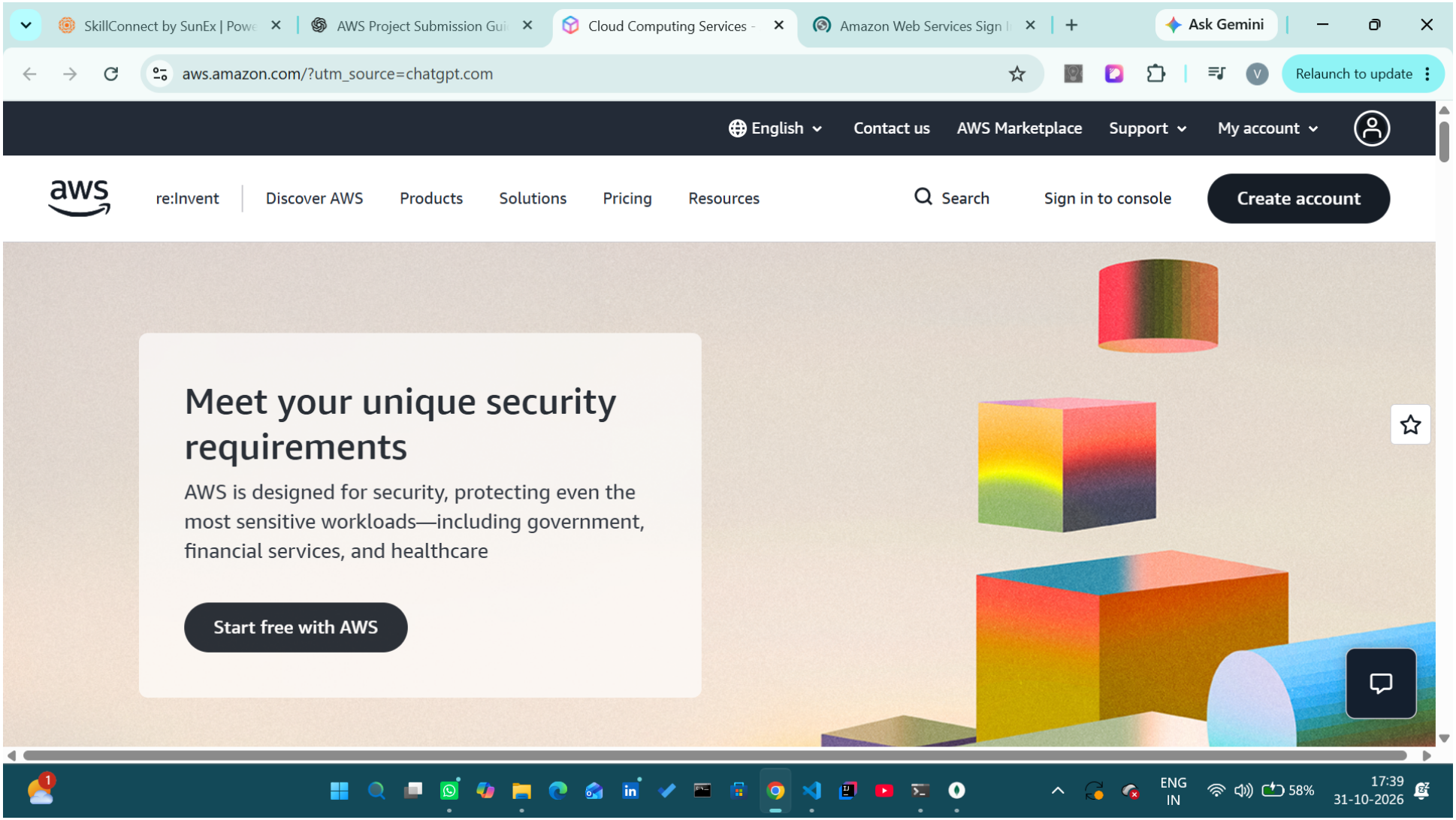
Evidence 52 — image(20261031-165441).png



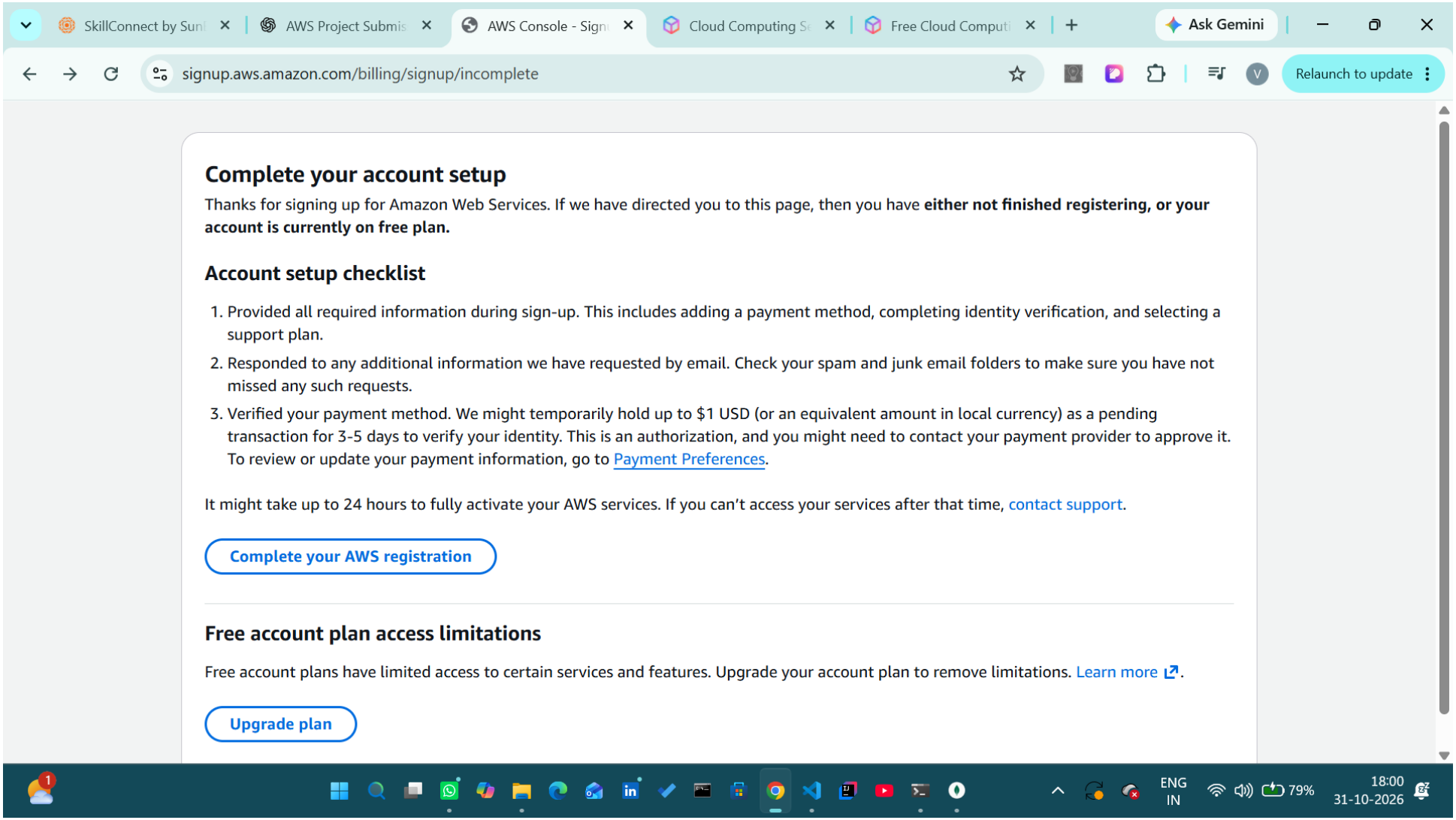
Evidence 53 — image(3).png



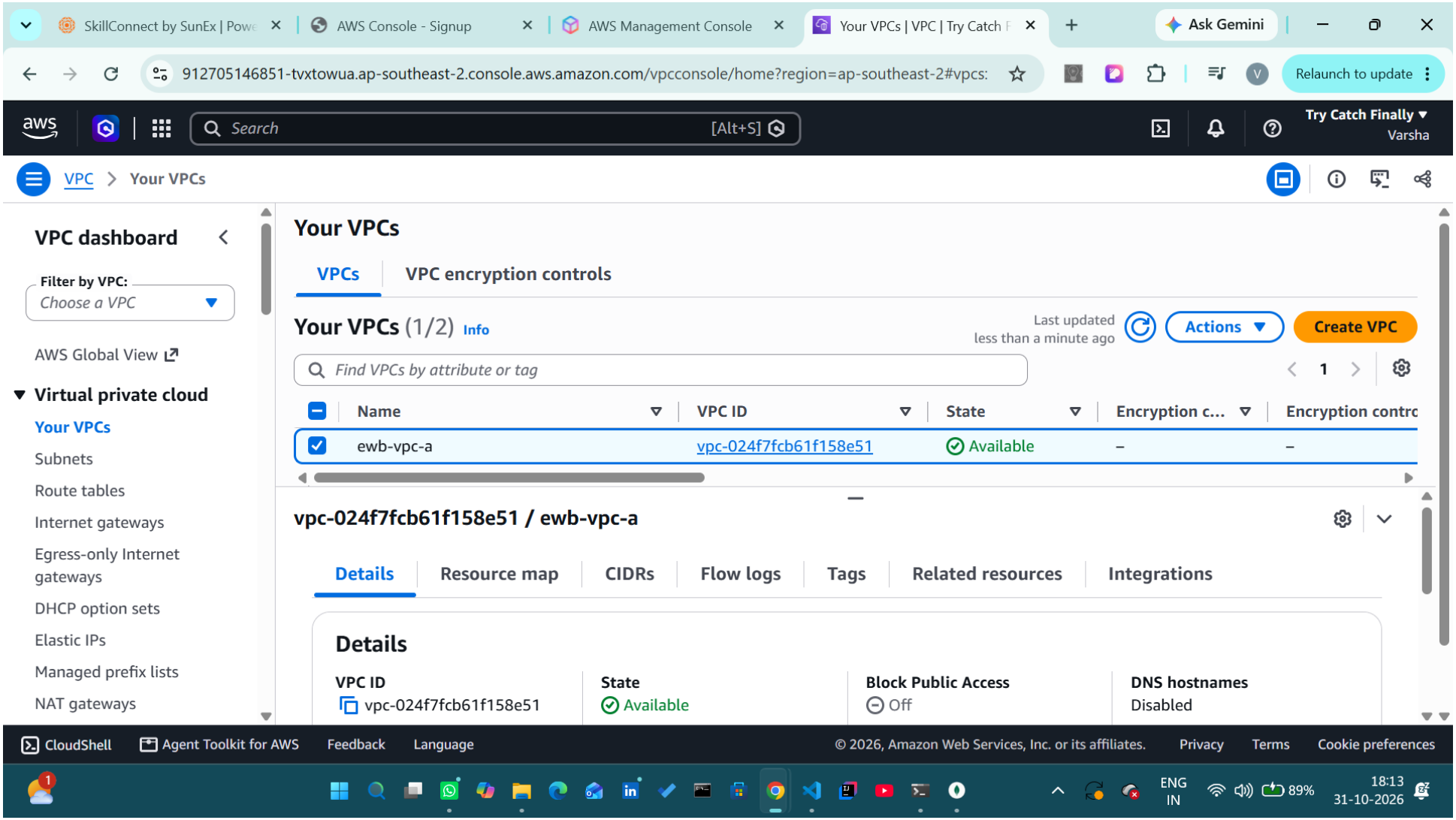
Evidence 54 — image(4).png



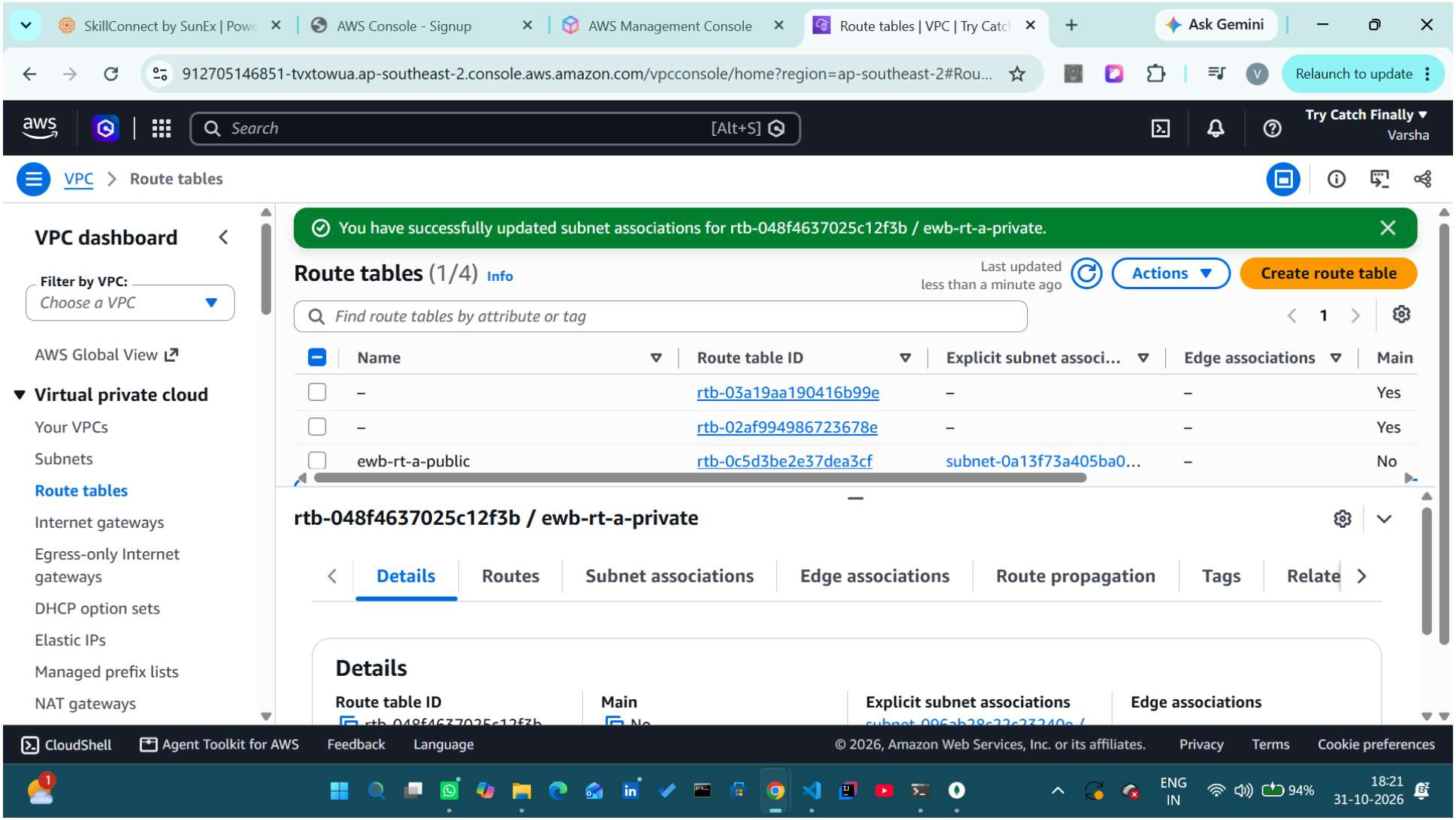
Evidence 55 — image(5).png



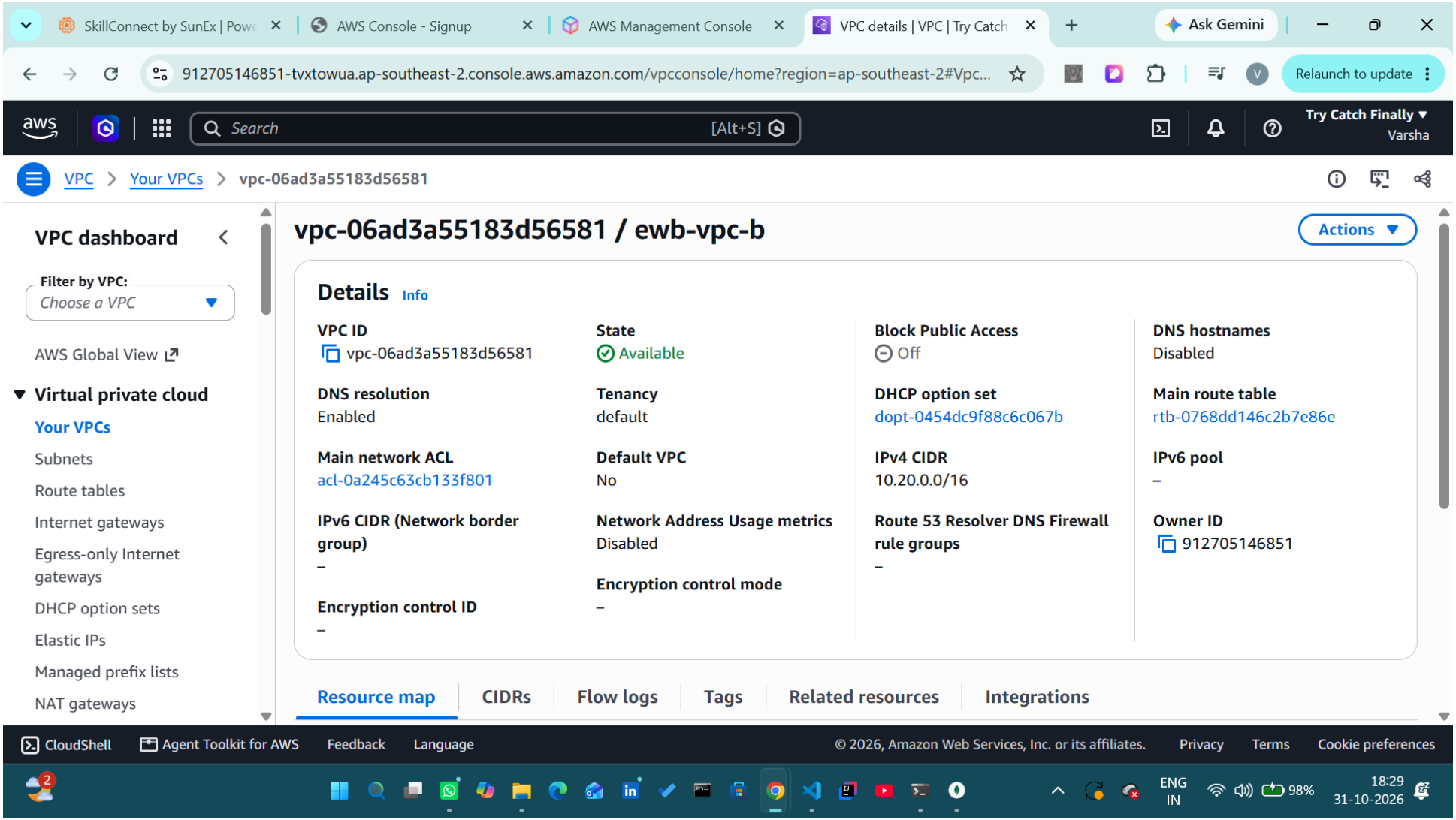
Evidence 56 — image(6).png



Evidence 57 — image(7).png



Evidence 58 — image(8).png



Evidence 59 — image(9).png

